SINKGRAB · BUILD PLAN

SinkGrab prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SKG-DEC-001 O2, O4, O5; SKG-DDR-006): crank slip clutch in place of the shear pin hub (3.4, joint 2, step 5), wider chain guard (3.7), dead-end load indicator in place of the shear link (3.13, joint 9, step 13, step 20), proof at 1.25 x the rated load (188 kg), clutch and indicator checks, four people at the T-bar, bentonite slurry for sinking; cost USD 3,063
0.4	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: pictures and sketches redrawn from the detailed D3 model (every bolt, pin and weld); the fixes of SKG-DEC-001 K1 to K22 written in (shear link sleeve, brake lever geometry, grab assembled on the line, spoil tub slid not lifted, carbon dioxide monitor read at depth, arm pivots at 60 mm, captive bearing bolts, set collars and others); pad and top plate heights corrected; cost USD 2,986
0.3	2026-10-04	Amish Chadha	Amish's decision 5A of 2026-10-04 (SKG-DDR-004): 340 mm slotted

			sleeve and 12 mm cross pin in place of the stop collar, so the scraper's arms fold to pass the ring; sketches SKG-DWG-112 and 113, joints 10 and 14, step 14 and the overview redrawn
0.2	2026-10-03	Amish Chadha	Amish's decisions 27A, 28B and 29B of 2026-10-03 (SKG-DDR-003): long two-hand crank handle, 3:1 closing tackle with an upper sheave in the head and the shear link on the crosshead, toe blades 40 mm past the ring; scraper arms carried unpinned
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (SKG-DDR-002)

SinkGrab prototype build plan

Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions are kept in the design decisions register (docs/06-design-decisions.md), not here.

1. What you are building

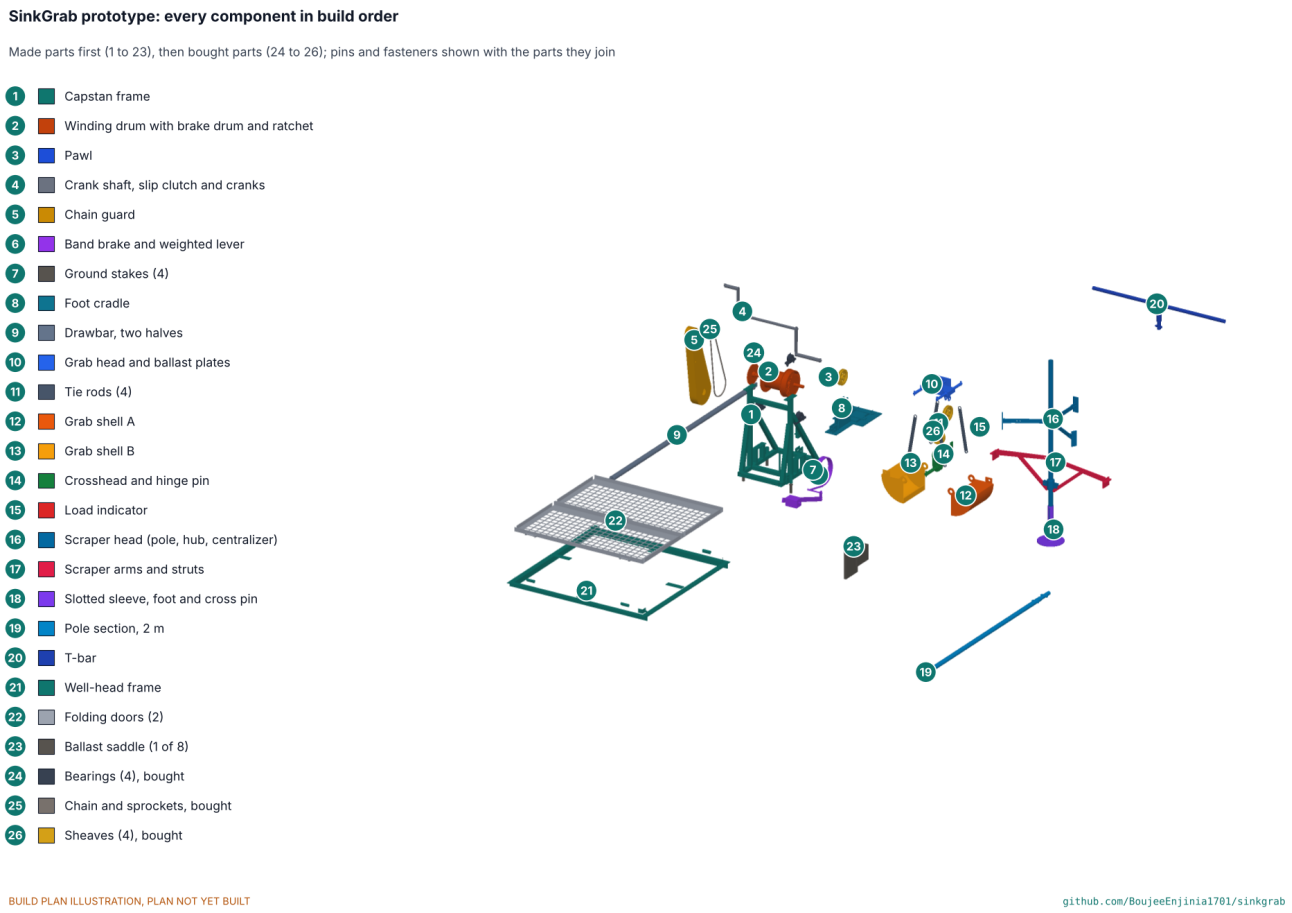


Figure 1. Every component pulled apart and numbered in build order: made parts 1 to 23, bought parts 24 to 26. Ropes, pins and fasteners are not shown.

The prototype is a kit for deepening a flooded caisson well from the surface. It has five groups: a hand capstan on the ground with a drawbar to a cradle under the HatchSide tripod's leg A foot; a clamshell grab worked on one rope; an under-curb scraper on a pole in 2 m sections; a well-head frame with two folding doors; and eight ballast saddles. You weld the frames, drum, cradle, grab, scraper and saddles from stock steel tube, plate and bar; have the ratchet wheel, pawl and shell end plates profile-cut and the shell skins rolled; and buy the bearings, chain drive, sheaves, ropes, rigging, dip tape, gas detector, carbon dioxide monitor and tubs. The parts cost about USD 2,986 from the bill of materials. The work needs a stick or MIG welder, a pillar drill, plate rolls (or a fabricator who has them) and hand tools. The HatchSide tripod is built to its own plan.

> **Safety:** SinkGrab is lifting equipment worked over an open well shaft, with a hand-cranked chain drive. A failed pin, rope or shackle can drop the grab; the chain, sprockets, drum, brake and cranks can trap fingers and clothing; the shaft can hold bad air; a person can fall in. The building work involves welding, grinding, rolling plate and lifting parts up to 25 kg. Nobody turns the cranks with the chain guard off, nobody enters the well, and nothing is lifted over the well before the safety stops in section 6 allow it. Load tests are TRL 4 work.

2. What changed to make it buildable

The concept showed what SinkGrab does; some of its parts could not be made or fitted as drawn. Each change keeps what SinkGrab does and is recorded in decision record SKG-DDR-002.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Hoist	A capstan clamped on one tripod leg	A ground capstan 3 m from a lead sheave in a cradle under the leg A foot, joined by a drawbar (Figures 15, 13)	The leg cannot carry it; the rope pull stays inside the kit
Grab lines	A closing line and a holding line	One line through a 3:1 tackle (over a sheave in the crosshead and one in the head); the head's weight opens the shells (Figures 19, 21)	One hoist, one rope; the 3:1 tackle closes the shells 50 % harder
Overload	None	A friction slip clutch on the crank shaft (Figure 6) and a peak-reading load indicator at the tackle's dead end on the crosshead (Figure 21), in place of the first design's shear pin and shear link (Amish's decision O4, 2026-10-25)	The tripod is rated 150 kg; a shear pin had no release window and fatigued
Lowering	Not defined	Cranks off, lowering on a band brake held on by a weight (Figure 8)	The brake sets if the operator lets go
Scraper	A scraper on a pole or a guided frame	Centred pole with a centralizer; two arms opened by struts when the pole's weight rests on the foot (Figures 24, 25)	Turns at depth; folds to pass the bore
Spoil chute	A chute and tipping hook	Well-head frame with folding doors; the grab dumps into a tub on the closed doors (Figures 29, 30)	The shaft is covered whenever the grab is up
Ballast frame	A frame on the top ring	U-saddles astride the ring wall on their own lines (Figure 32)	Placed and moved from the surface

The detailed model of 2026-10-24 (kit 1.9.1), which draws every bolt, pin and weld, found further details that could not be made, fitted or used safely as drawn. Each is now in the design and in this plan; the reasons are in the design decisions register (SKG-DEC-001 K1 to K21).

Table 1a. Changes found by the detailed model.

Component	Before	Now	Why
Shear link	The rope's thimble on the bare calibrated pin across a 12 mm gap	A hardened 16 mm sleeve filled the gap; since 2026-10-25 the link is replaced by the load indicator (decision O4)	Across a gap the pin bends and gives way near 0.15 kN
Band brake	Link pin below the lever pivot on the well side	Pivot 67 mm up on a plate outside the rail; link on the band's tangent to a pin on an upright arm 60 mm above the pivot; anchor end on an eye pinned between two posts (Figure 8)	The weight now pulls the band tight when the lever is let go
Grab handling	Two people lift the assembled grab	Assembled on the line under the tripod and moved only by the capstan; travels in pieces	At 52.7 kg it is beyond two or three people
Spoil tub	Not said how it leaves the doors	Slid off the closed doors on a board and tipped at the heap	A tub with one bite weighs about 50 kg
Air test	Four-gas detector	Four-gas detector and a carbon dioxide monitor	A four-gas detector does not read carbon dioxide, which collects in wells
Scraper arms	Pivots 45 mm from the pole axis	Pivots 60 mm out; closed eyes on the arms; folded radius 387 mm (Figure 25)	The arm's eye could not close round its pin without cutting into the hub
Drum bearings	Bolts with nuts under the pads	Bolts pushed up through the pads before welding, heads inside the pad posts (Figure 4)	The pad posts sit under the bolt holes
Crosshead	Free to slide along the hinge pin	Two set collars hold it central (Figure 19)	Keeps the tackle in line
Tie rods and struts	Ended at their hole centres	Closed eyes 25 and 15 mm beyond the holes	A hole cannot sit on the end of a bar
Crank handles	On M12 bolts	On 20 mm stub axles welded to the arms	A nut on the inner face would strike the chain guard as the crank turns
Foot cradle	Clamp bars level with the foot plate; rim 30 mm high	Clamp bars on top of the foot plate on four welded studs; rim 18 mm high (Figure 13)	The bars and rim overlapped the foot plate
Grab head	Box 15 mm below the head pin axis; ballast bolts at 75 mm	Box 19 mm below; ballast bolts at 40 mm through spacer tubes	The 25 mm pins cut the bottom plate; a bolt sat in the line's path
Rope clamp, guard, pawl pin, drawbar, toe blades, doors	U-bolt; flat tabs; welded pin; tubes against the clevis; lapped blades; a solid tray	Tapped boss and strap; clip angles with welded nuts; pressed pin; 6 mm weld gap; butt-welded blades; mesh on an angle frame with corner cleats	Each part can now be made and fitted

3. Making the components

Make and check each component before the step that needs it. Sizes are in millimetres. On the capstan, "+X" is the side with the brake and ratchet and "-X" the side with the chain; "front" faces the well. Workshop tolerance is 1 mm unless a step says otherwise. Weld 2 mm tube with MIG or 2.5 mm electrodes; fillets are 3 mm on tube and 5 mm on plate unless stated. Mark every part with its name in paint marker.

3.1 Capstan frame

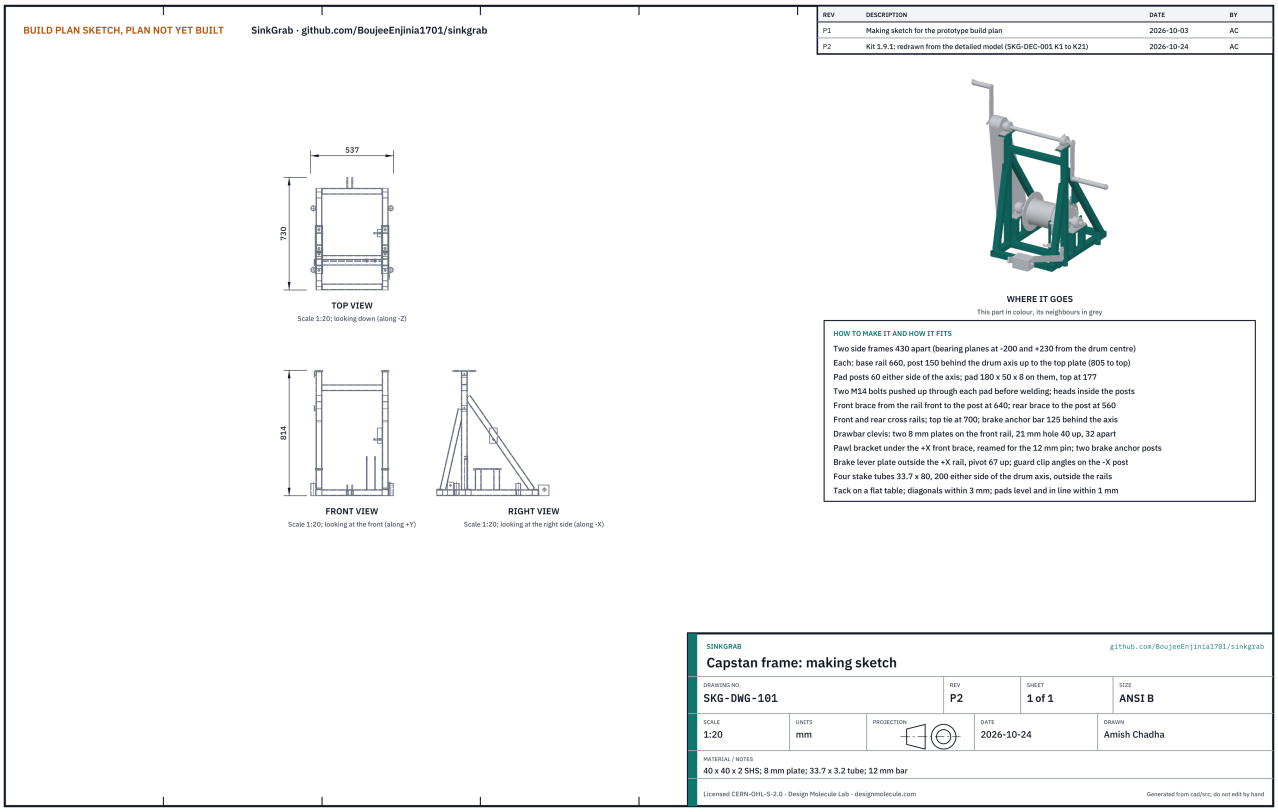


Figure 2. Capstan frame making sketch (SKG-DWG-101).

What it is and what it is made from. The welded frame that carries the drum, the crank shaft, the pawl and the brake. 40 x 40 x 2 mm steel square tube, about 7.4 m; 8 mm plate; 33.7 x 3.2 mm tube for the stake tubes; 12 mm bar for the pawl pin.

How to make it.

1. Cut two base rails 660 long, two posts 765 long, four pad posts 89 long, the cross rails, top tie and brake anchor bar 390 long. Cut the braces to fit in step 4.
2. On each base rail, weld a post upright 150 mm behind the drum axis line and two pad posts 60 mm either side of it.
3. Drill each bearing pad (180 x 50 x 8) for two M14 bolts 121 mm apart, push the bolts up through it from below and tack their heads to the underside; then weld the pad on its two pad posts (169 mm high), so its top is 177 mm above the floor and the bolt heads sit inside the posts. Weld a top plate (150 x 50 x 8, drilled for two M12 bolts 105 mm apart) on the post (805 mm high), its top 813 mm up.

4. Fit the front brace from the front of the rail to the post near 640 mm up and the rear brace from the rear of the rail to the post near 560 mm up; weld.
5. Stand the two side frames 430 mm apart, centre to centre, on a flat floor. Weld in the front and rear cross rails, the top tie between the posts 700 mm up, and the brake anchor bar 125 mm behind the drum axis.
6. Weld the two drawbar clevis plates (8 mm, 70 x 70, 21 mm hole 40 mm up) to the front cross rail, 32 mm apart and centred.
7. Weld the pawl bracket under the +X front brace, with a reamed 12 mm hole for the pawl pin; two brake anchor posts (6 mm plate) on the anchor bar either side of the brake drum, each with a 10.5 mm hole 240 mm up and 140 mm behind the drum axis; the lever pivot plate (8 mm) on the outside face of the +X rail, 240 mm behind the axis, with a 12.5 mm hole 67 mm up.
8. Weld a stake tube (33.7 x 80) upright on the outside of each rail, 200 mm either side of the drum axis, and two guard clip angles (3 mm) on the -X post at 560 and 780 mm up.
9. Check the pad and top plate holes against the bearings bought.

How it fits the parts next to it. The drum bearings bolt on the pads (Figure 4), the crank bearings on the top plates, the pawl hangs on its pin (Figure 7), the brake anchors to the post (Figure 8) and the drawbar pins into the clevis (Figure 15).

Check before moving on. Diagonals of the base within 3 mm; both pads level and in line within 1 mm; both top plates likewise; about 23 kg.

3.2 Winding drum

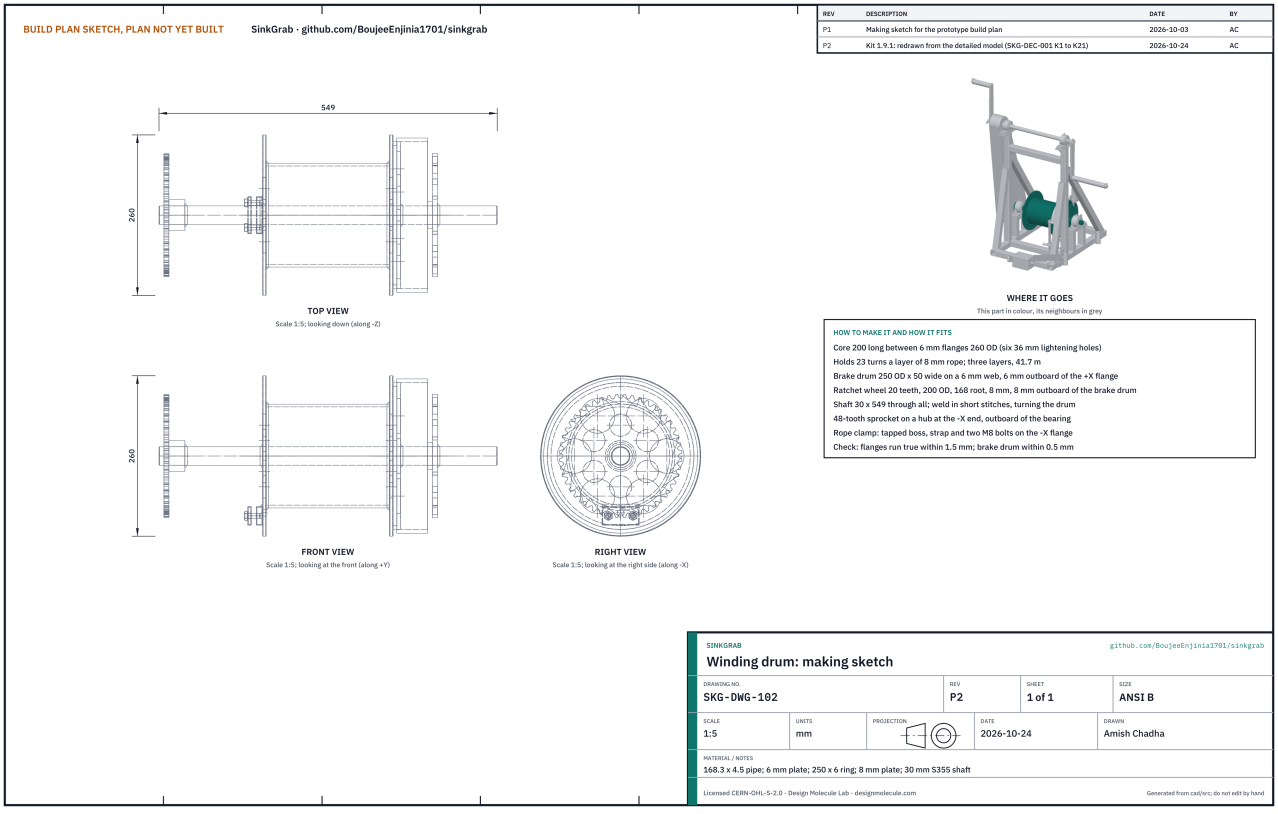


Figure 3. Winding drum making sketch (SKG-DWG-102).

What it is and what it is made from. The drum that winds the line, with its brake drum, ratchet wheel and big sprocket on one shaft. 168.3 x 4.5 mm pipe; 6 mm plate; a 250 x 6 mm ring 50 wide; 8 mm plate for the ratchet; 30 mm bright steel bar (S355 or EN8); the bought 48-tooth sprocket.

How to make it.

1. Cut the core 200 mm long with square ends. Cut two flanges 260 mm across from 6 mm plate with a 30 mm centre hole and six 36 mm lightening holes on a 100 mm circle.
2. Push the 549 mm shaft through both flanges; set the flanges on the core ends, the shaft central within 0.5 mm; weld in short stitches, turning as you go.
3. Weld the brake drum web (6 mm, with six 56 mm holes) and ring on the shaft 6 mm outboard of the +X flange, with a spacer ring between.
4. Weld the ratchet wheel (20 teeth, 200 mm outside, 168 mm at the roots) 8 mm outboard of the brake drum, its teeth as in Figure 7.
5. At the -X end, weld the big sprocket on a short hub, outboard of where the bearing will sit.
6. Weld the rope clamp boss (10 mm plate, two M8 tapped holes) to the -X flange's outer face beside the 11 mm rope hole; the line's dead end comes through the hole and is clamped under a 6 mm strap with two M8 bolts.

How it fits the parts next to it. The shaft runs in the two drum bearings (Figure 4); the band brake wraps the brake drum (Figure 8); the pawl drops into the ratchet (Figure 7). The line is wound on from below.

Check before moving on. Turned in V-blocks, the flanges run true within 1.5 mm and the brake drum within 0.5 mm; about 18 kg.

3.3 Drum bearings (bought)

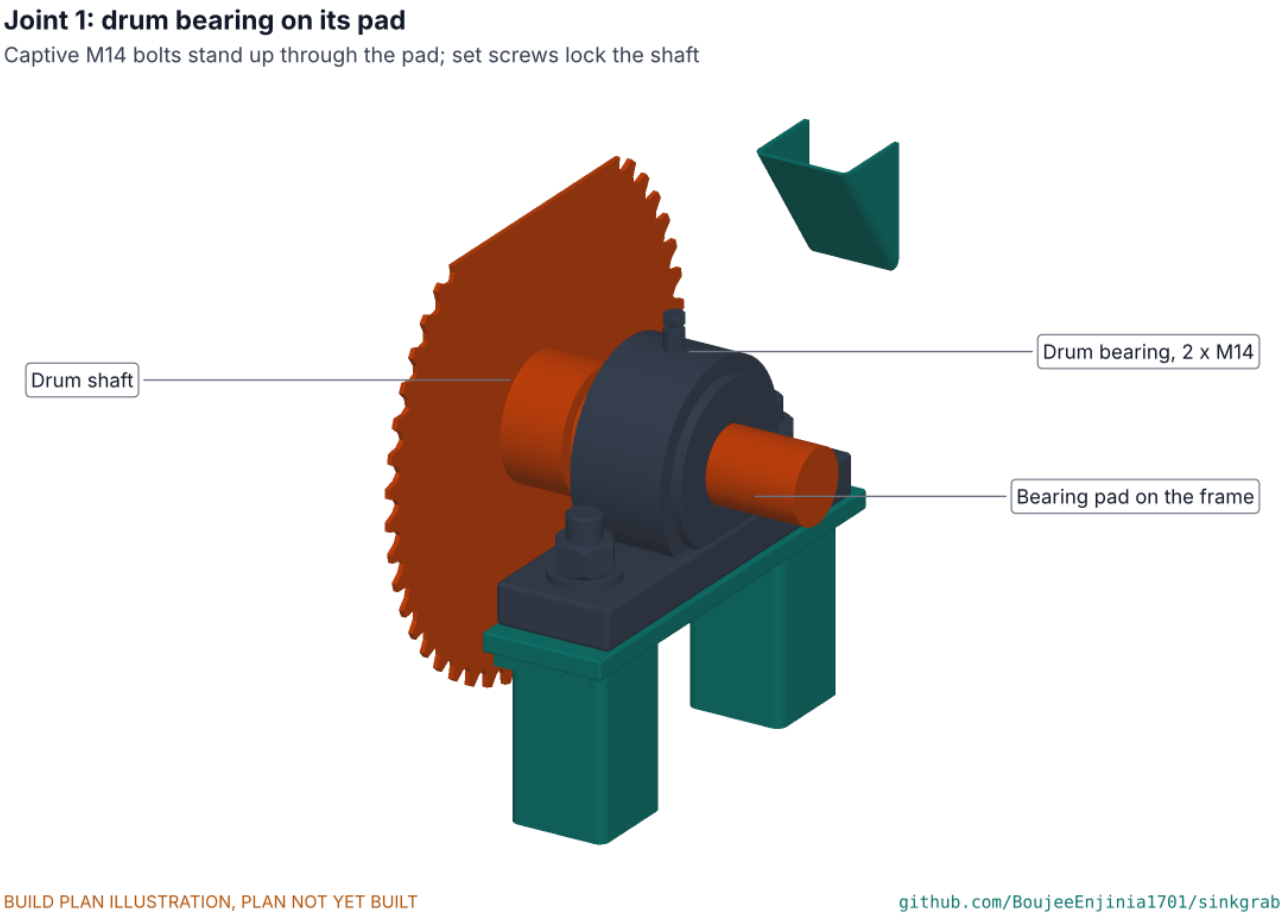


Figure 4. Joint 1. Each UCP206 bearing drops over the two M14 bolts standing up from its pad (their heads tacked under the pad, inside the pad posts) and is held by washers and nuts; its set screws lock the shaft.

Buy two UCP206 pillow block bearings with 30 mm bores. Nothing to make.

3.4 Crank shaft, slip clutch and cranks

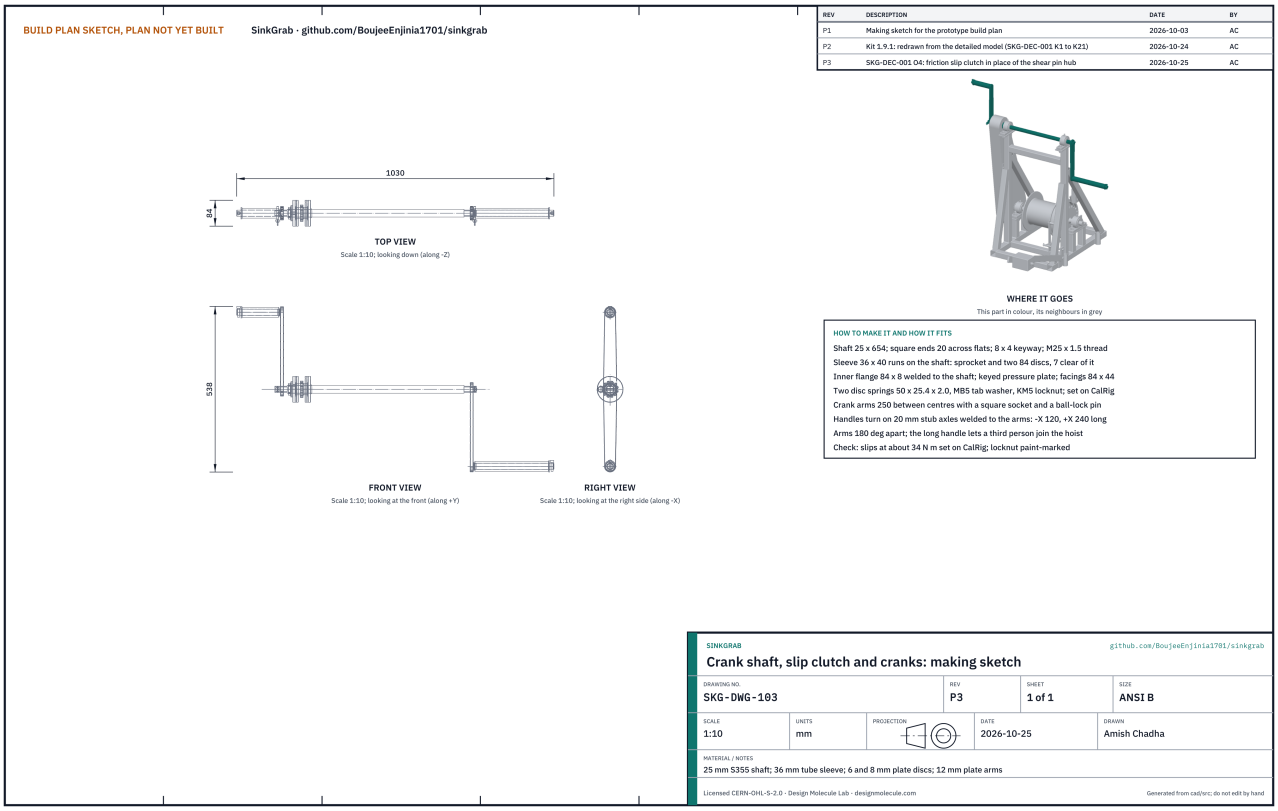


Figure 5. Crank shaft, slip clutch and cranks making sketch (SKG-DWG-103).

What it is and what it is made from. The shaft the people turn, and the friction slip clutch that drives the chain and slips if the line pulls more than the tripod's 150 kg rating (Amish's decision O4, 2026-10-25). 25 mm bright steel bar; 36 mm tube for the driven sleeve; the bought 12-tooth sprocket; 6 and 8 mm plate for the clutch discs, flange and pressure plate; two woven friction facings 84 x 44 x 4 mm; an 8 x 7 x 8 mm key; two disc springs 50 x 25.4 x 2.0; a KM5 locknut and MB5 tab washer; 12 mm plate; 32 mm tube; two ball-lock pins. One crank has a long handle so a third person can join the hoist.

> **Safety:** the clutch is the only thing that keeps the line inside the tripod's rating. It is set on CalRig and the locknut is paint-marked; nobody tightens it on site. If it slips during a hoist, stop and find out why.

How to make it.

1. Cut the shaft 654 mm long; mill or file a 20 mm square, 40 mm long, on each end for the cranks. Cut an 8 x 4 mm keyway 18 mm long where the sketch shows it, under the pressure plate, and an M25 x 1.5 thread outboard of it for the locknut.
2. Cut the driven sleeve 40 mm long from 36 mm tube bored to run on the shaft. Weld the 12-tooth sprocket round its middle. Profile-cut two driven discs 84 mm across from 6 mm plate, bored 36 mm, grind both faces flat, and weld one to each end of the sleeve, 7 mm clear of the sprocket on each side so the chain's side plates pass.

3. Profile-cut the inner flange and the pressure plate, 84 mm across, from 8 mm plate; bore the flange 25 mm and the pressure plate 25.1 mm with an 8 mm keyway. Have a friction facing 84 x 44 mm bonded to one face of each (a brake and clutch reliner does this).
4. Slide the inner flange onto the -X end, facing outboard, to its place on the sketch and weld it to the shaft on its inboard face.
5. Make two crank arms 250 mm between centres, each with a square socket to fit the shaft end and a hole for its ball-lock pin. Weld a 20 mm stub axle to the outer face of each arm's end; slide on the handle (32 mm tube bored 20.5 mm), 120 mm long on the -X crank and 240 mm long on the +X crank for two people's hands side by side, and retain it with a washer and split pin. Nothing projects from the arm's inner face.

How it fits the parts next to it.

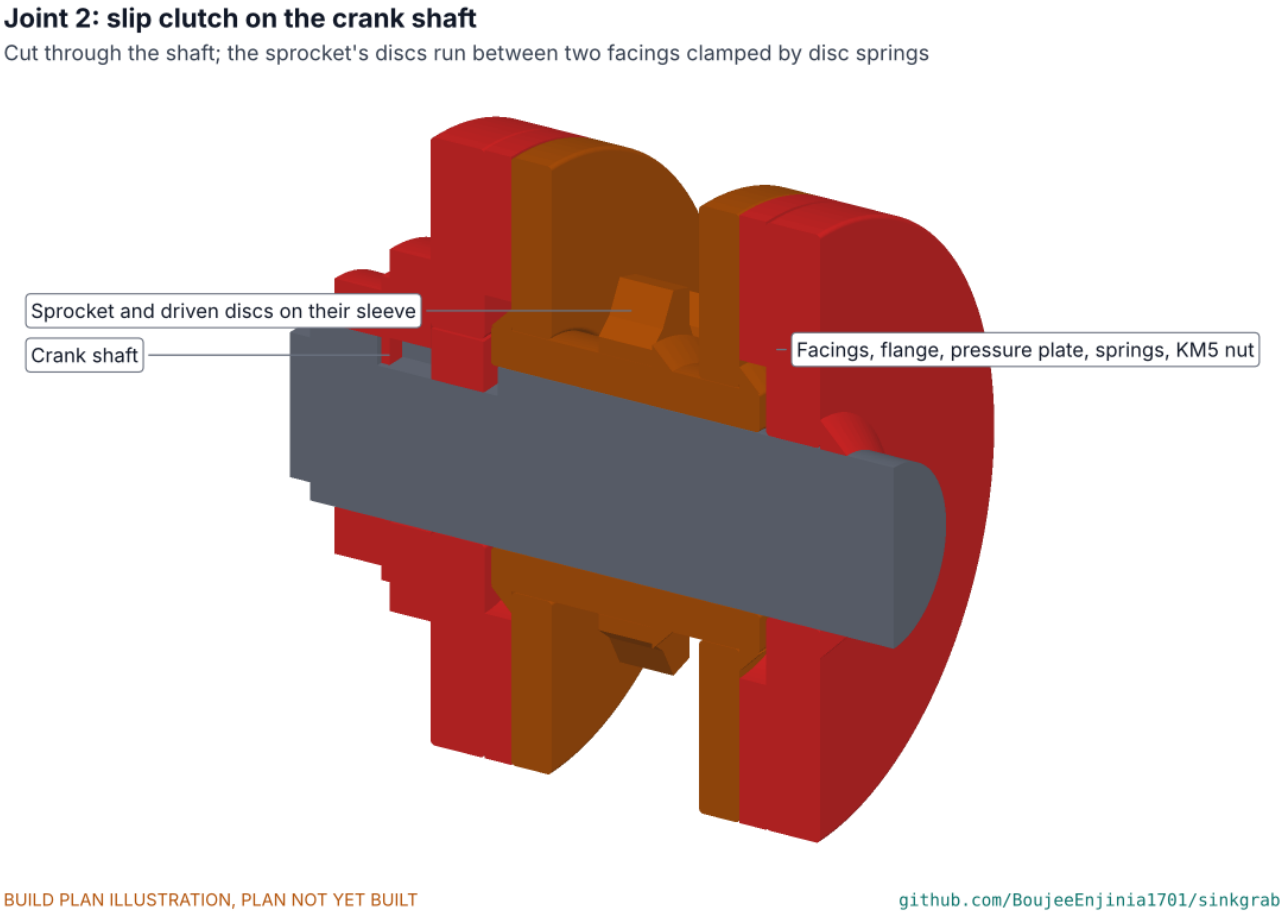


Figure 6. Joint 2. The sprocket and its two driven discs run free on the shaft between two friction facings: one on the flange welded to the shaft, one on the keyed pressure plate. Two disc springs and the KM5 locknut clamp them. Set on CalRig to slip at about 34 N m: if the capstan pull passes about 1.1 to 1.5 kN (it depends on the rope layer), the clutch slips, the cranks turn without lifting and the pawl holds the drum. Nothing breaks; the hoist carries on once the snag is cleared.

Check before moving on. With the locknut slack the sprocket turns freely by hand; set on CalRig, the clutch slips at the set torque and not below the steady pull of the full grab on the third layer. Paint-mark the locknut across the tab washer. Each crank slides on and locks with its pin.

3.5 Pawl

The pawl is drawn on the brake's making sketch (Figure 9). Cut it from 8 mm plate: a 12 mm pivot hole and a nose with a 3 mm radius that sits 94 mm from the drum axis. Deburr; keep the nose edge square. Press the 12 mm pin into the bracket's reamed hole; the pawl goes on it with a washer and a split pin.

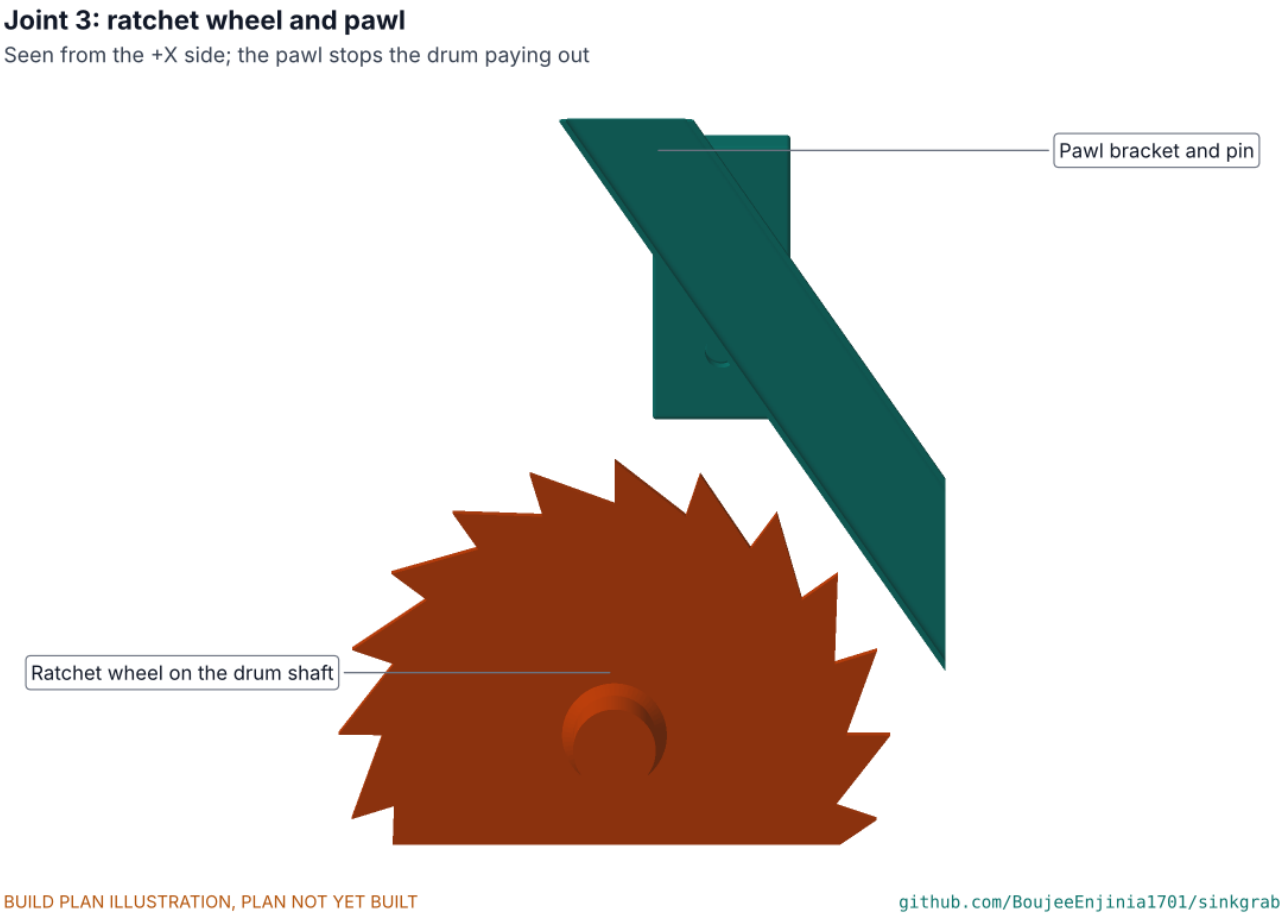


Figure 7. Joint 3, seen from the +X side. The pawl drops into the ratchet by its own weight and stops the drum turning to pay out line. To lower, it is flipped up onto the bracket.

Check before moving on. The pawl falls into every tooth gap as the drum is turned slowly by hand.

3.6 Band brake and weighted lever

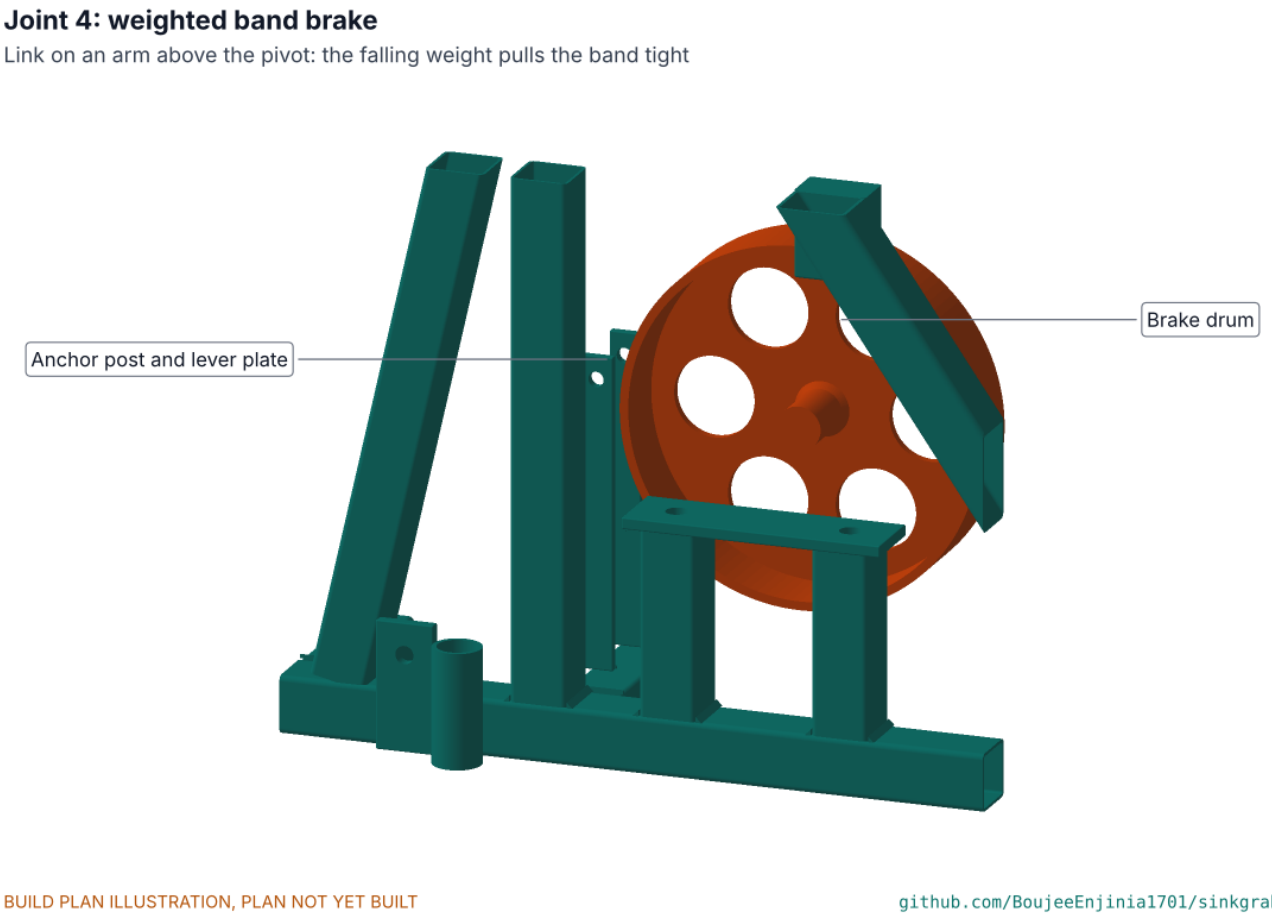


Figure 8. Joint 4. The lined band wraps 270 degrees round the brake drum. Its anchor end is an eye pinned between two posts; its live end pulls a link that runs along the band's tangent to a pin on the lever's upright arm, 60 mm above the lever pivot. As the 6 kg weight at the lever's end falls, that pin moves away from the band and pulls it tight; lifting the lever about 30 mm frees the drum.

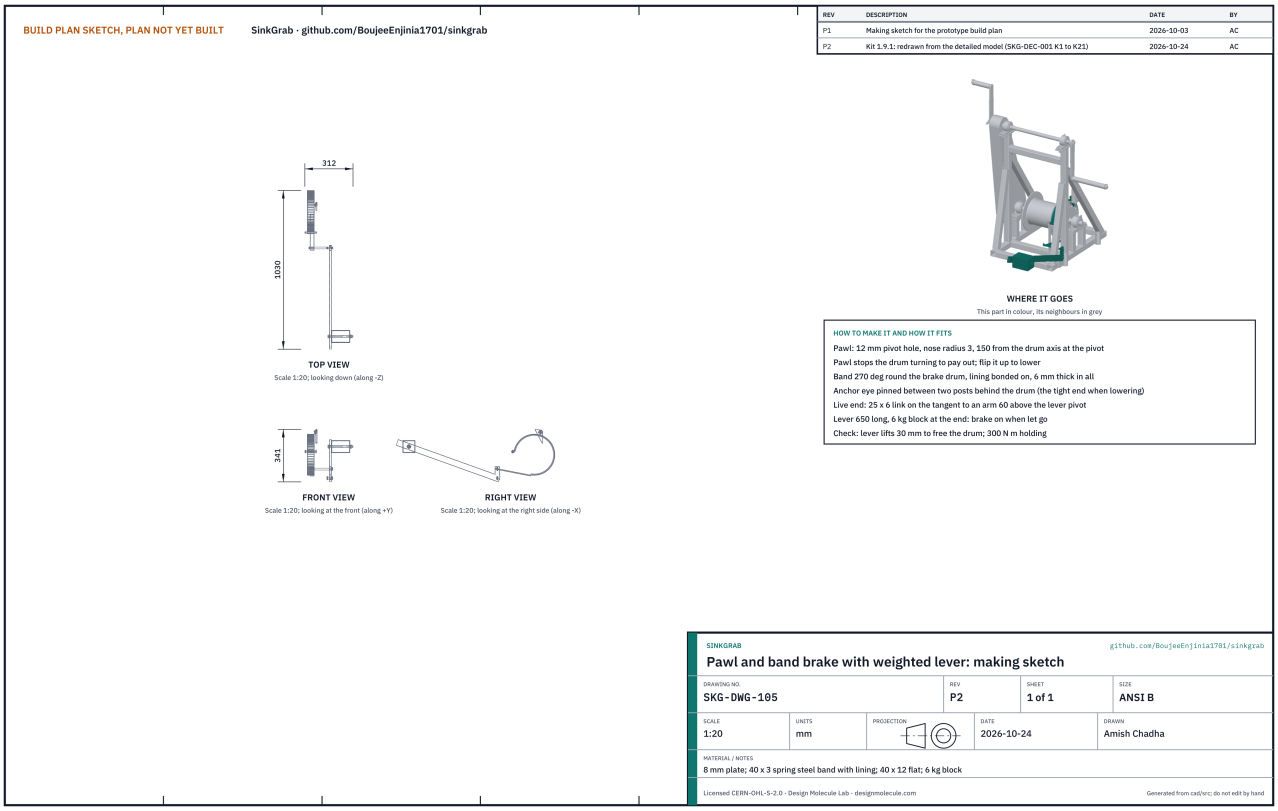


Figure 9. Pawl and band brake with weighted lever making sketch (SKG-DWG-105).

What it is and what it is made from. A brake that is on unless someone holds it off. 40 x 3 mm spring steel strip about 1.25 m with bonded lining; 25 x 6 mm flat for the link; 40 x 12 mm flat 650 mm long for the lever; a 6 kg steel block; 12 mm pins.

How to make it.

1. Bond or rivet the lining to the band; form the band round a 250 mm former.
2. Weld a 16 mm tube eye across the anchor end of the band and pin it with a 10 mm pin between the two anchor posts, so the band leaves the drum toward the back.
3. Weld the 25 x 6 link to the band's live end along its tangent; drill its free end 12.5 mm, 220 mm from the band.
4. Weld a 12 mm upright arm to the lever at its pivot end, with a 12.5 mm hole 60 mm above the pivot hole. Pin the lever to the pivot plate (pivot 67 mm above the floor) and the link to the upright arm; bolt the 6 kg block to the lever's far end with one M12 bolt.

Check before moving on. With the weight on, two people cannot turn the drum by the flanges; lifting the lever 30 mm frees it.

3.7 Chain guard

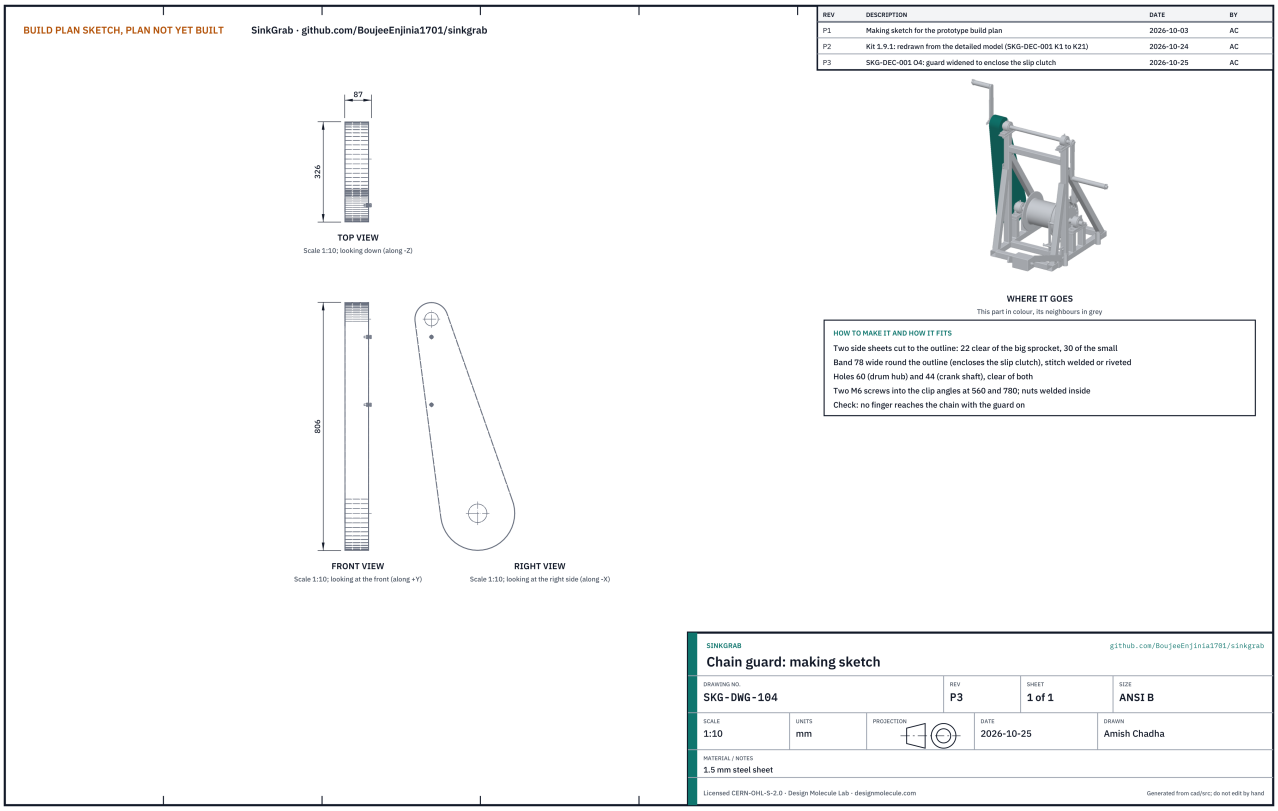


Figure 10. Chain guard making sketch (SKG-DWG-104).

What it is and what it is made from. A closed box round the chain and both sprockets. 1.5 mm steel sheet.

How to make it. Cut both side sheets to the outline (22 mm clear of the big sprocket, 30 mm of the small one), cut the shaft holes, bend a 78 mm band round the outline (wide enough to enclose the slip clutch) and stitch weld or rivet it to both sheets. Drill the +X sheet for two M6 screws at the clip angles and weld an M6 nut inside each hole.

Check before moving on. With the guard on, a finger cannot reach the chain, sprockets or clutch. Drill a 6 mm drain hole at the guard's lowest point so rain does not stand on the clutch.

3.8 Ground stakes (make 4)

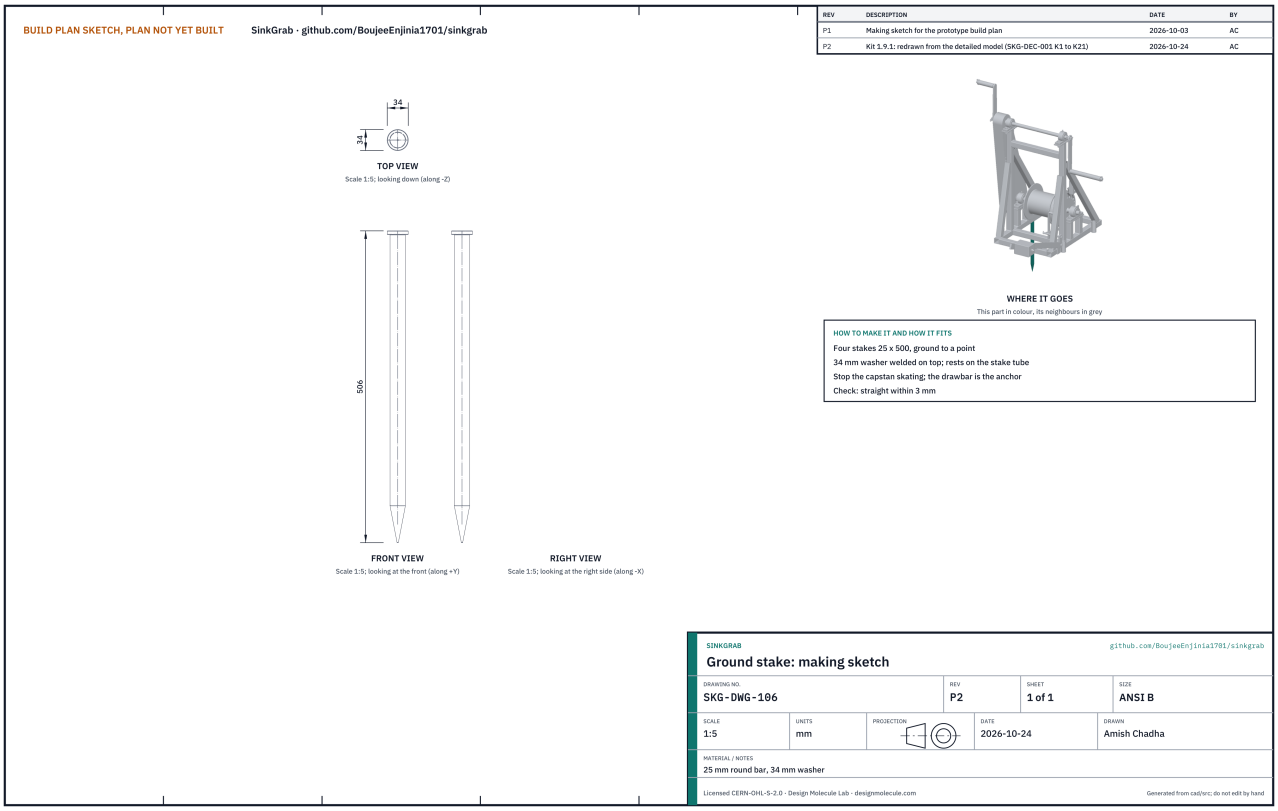


Figure 11. Ground stake making sketch (SKG-DWG-106).

25 mm round bar 500 mm long, ground to a point, with a 34 mm washer welded on top. Each is driven through a stake tube until its head rests on it. The stakes stop the capstan skating; the drawbar carries the pull. Check: straight within 3 mm.

3.9 Foot cradle and lead sheave

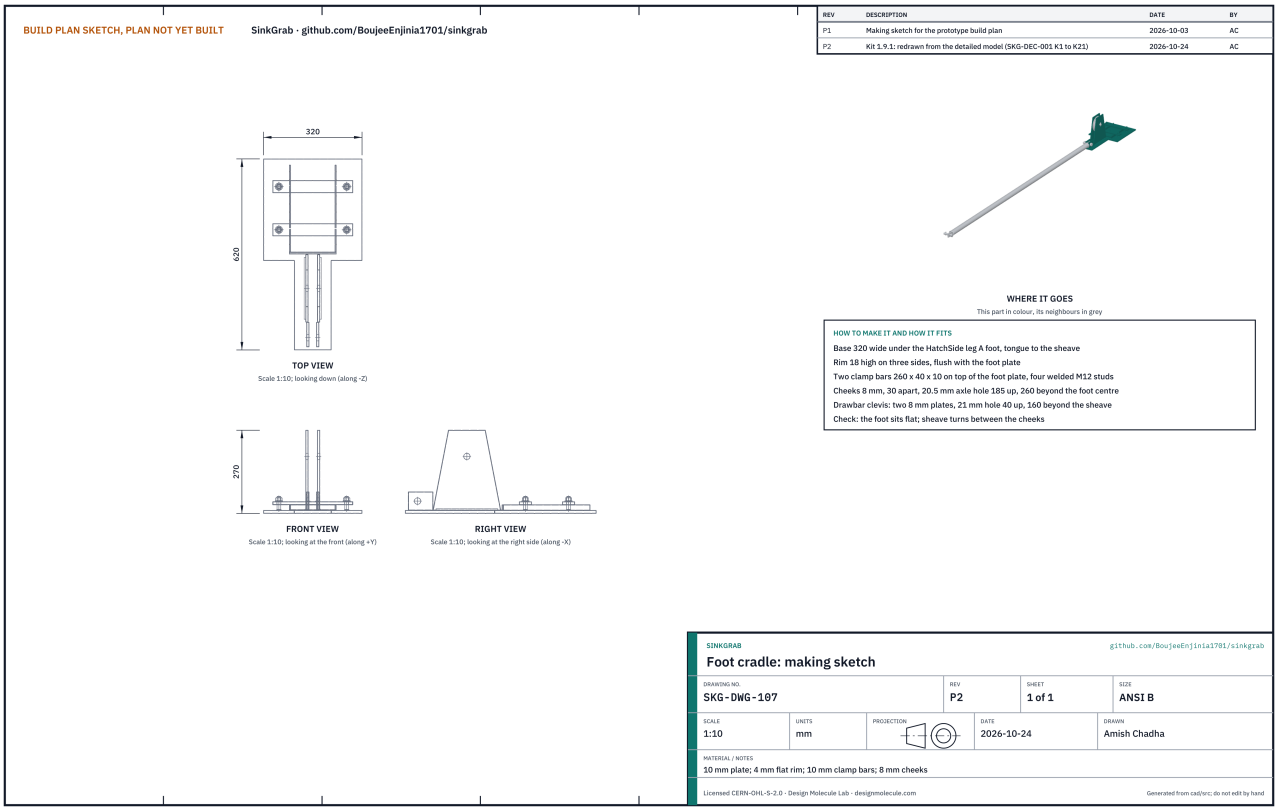


Figure 12. Foot cradle making sketch (SKG-DWG-107).

What it is and what it is made from. A plate the tripod's leg A foot stands in, carrying the lead sheave and the drawbar clevis. 10 mm plate; 4 mm flat; 10 mm bar for the clamp bars; 8 mm plate; four M12 studs with nuts; the bought sheave on a 20 mm axle.

How to make it.

1. Cut the base 320 x 330 with a 120 mm wide tongue running out to the sheave and the clevis.
2. Weld an 18 mm rim of 4 mm flat on three sides of the foot position, open toward the well, 2 mm clear of the foot plate and no higher than it.
3. Weld four M12 studs 45 mm long upright on the base for the clamp bars, 220 mm apart across and 140 mm apart along.
4. Weld the two sheave cheeks (8 mm, 30 mm apart) on the tongue, their 20.5 mm axle holes 185 mm up and 260 mm beyond the foot centre.
5. Weld the two drawbar clevis plates at the end of the tongue, 21 mm holes 40 mm up, 160 mm beyond the sheave.

How it fits the parts next to it.

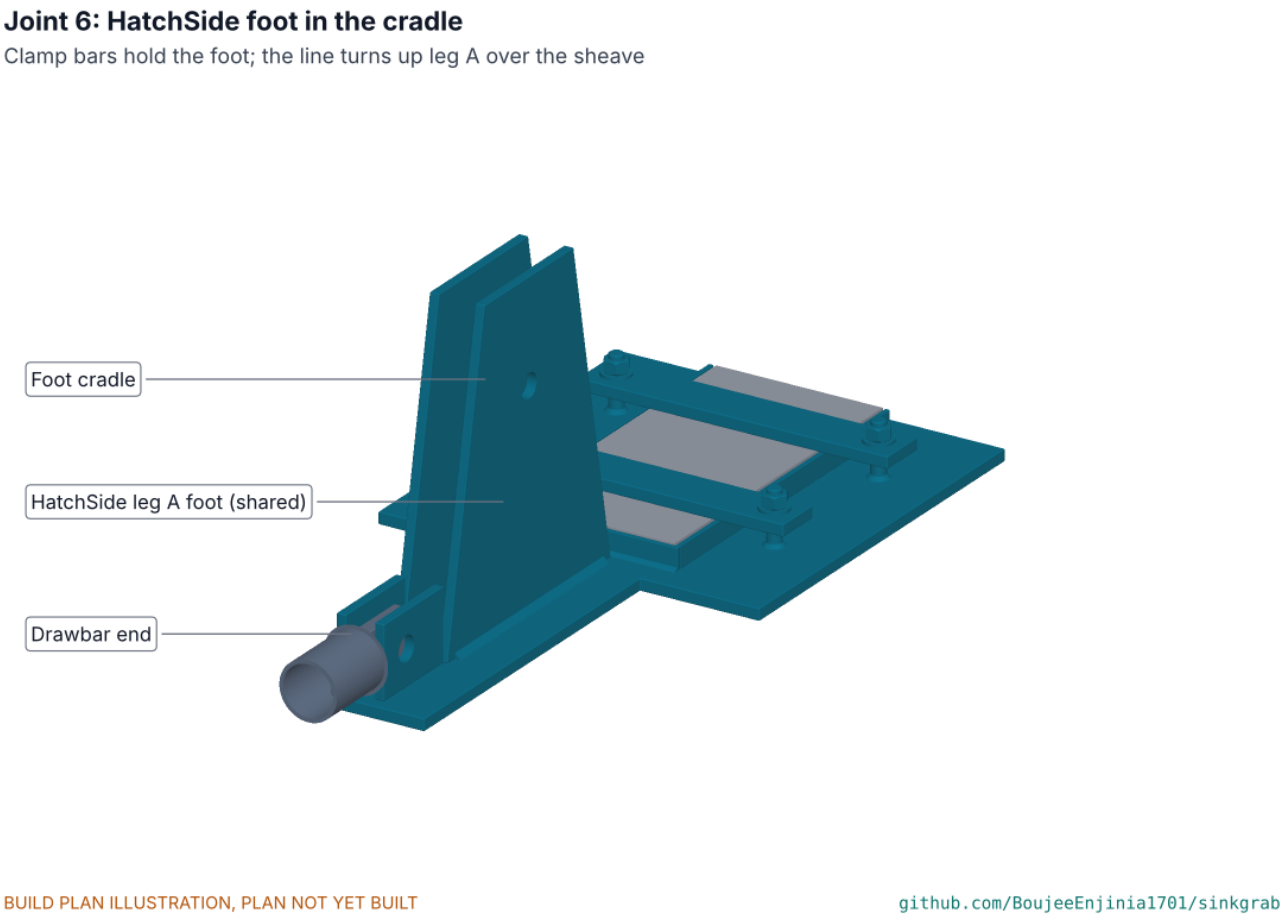


Figure 13. Joint 6. The HatchSide foot sits inside the rim; two clamp bars lie across the top of it on the four studs and are pulled down with nuts. The line comes in under the sheave from the capstan and leaves upward beside leg A.

Check before moving on. The foot drops in and sits flat; the sheave turns freely between the cheeks.

3.10 Drawbar

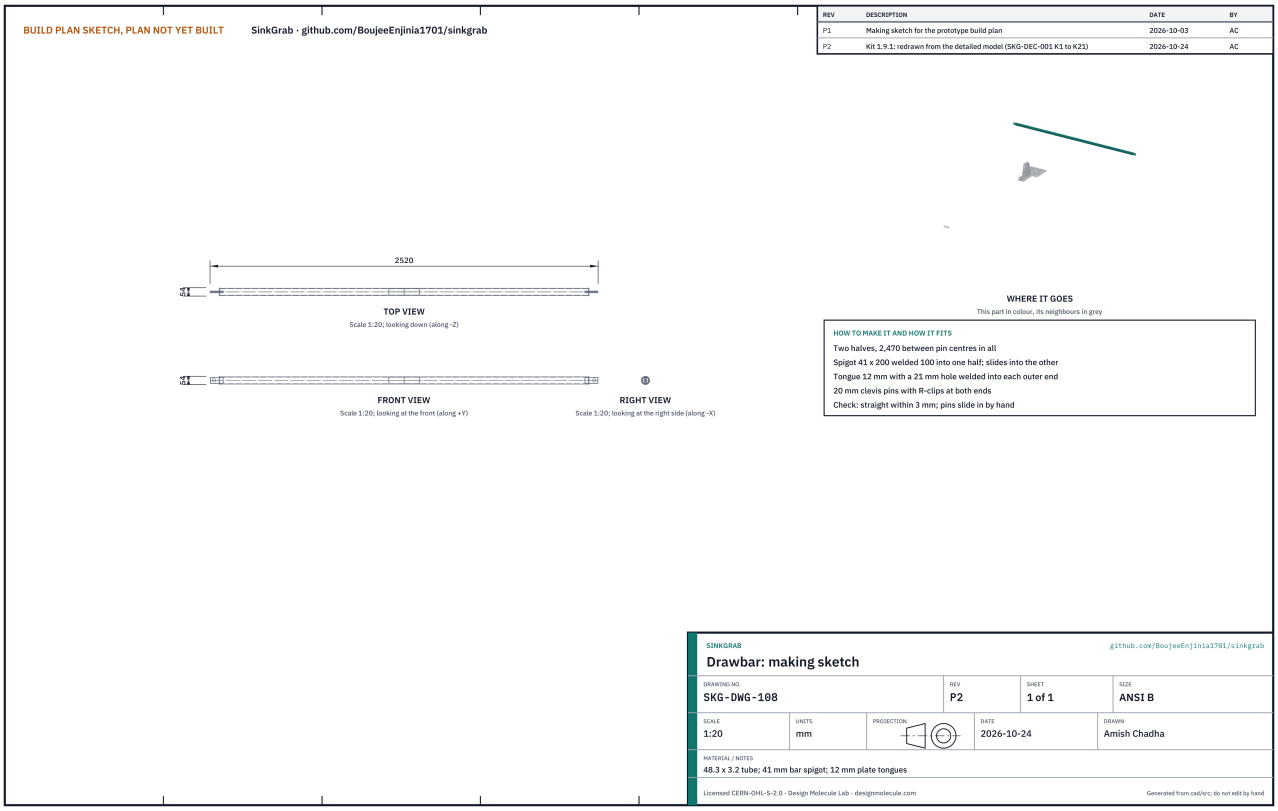


Figure 14. Drawbar making sketch (SKG-DWG-108).

What it is and what it is made from. The strut that carries the line's pull from the capstan back to the cradle. 48.3 x 3.2 mm tube in two halves; a 41 mm bar spigot 200 long; 12 mm plate tongues; two 20 mm clevis pins with R-clips.

How to make it. Cut both halves so the pins are 2,470 mm apart when joined, the tube ends stopping 6 mm short of the clevis plates. Weld the spigot 100 mm into one half. Slot a 12 mm tongue with a 21 mm hole 25 mm into each outer end and weld round the tube end.

How it fits the parts next to it.

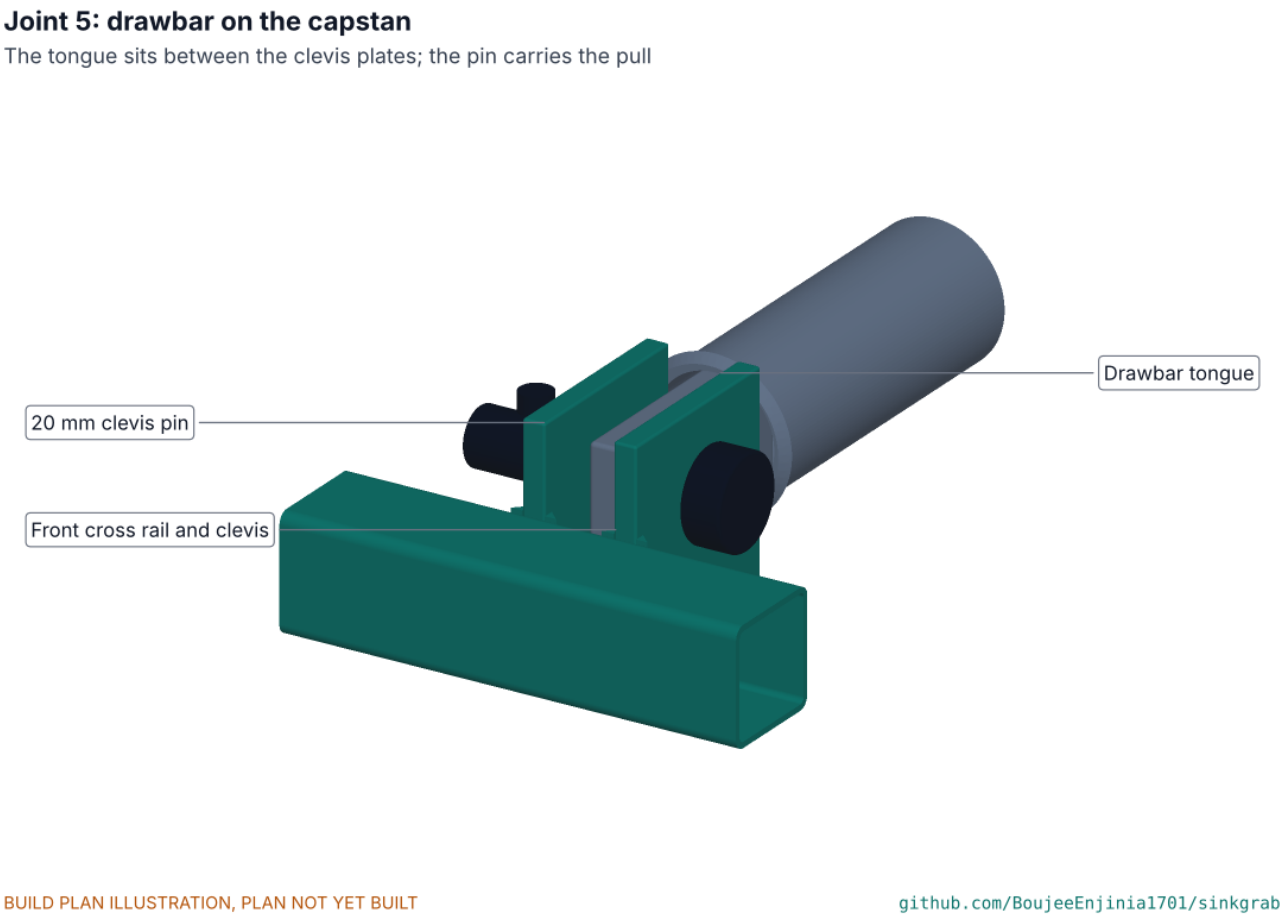


Figure 15. Joint 5. The tongue sits between the clevis plates on the capstan's front rail; the 20 mm pin carries the pull. The other end is the same at the cradle.

Check before moving on. Joined, it is straight within 3 mm; both pins slide in by hand.

3.11 Grab shells (make 2)

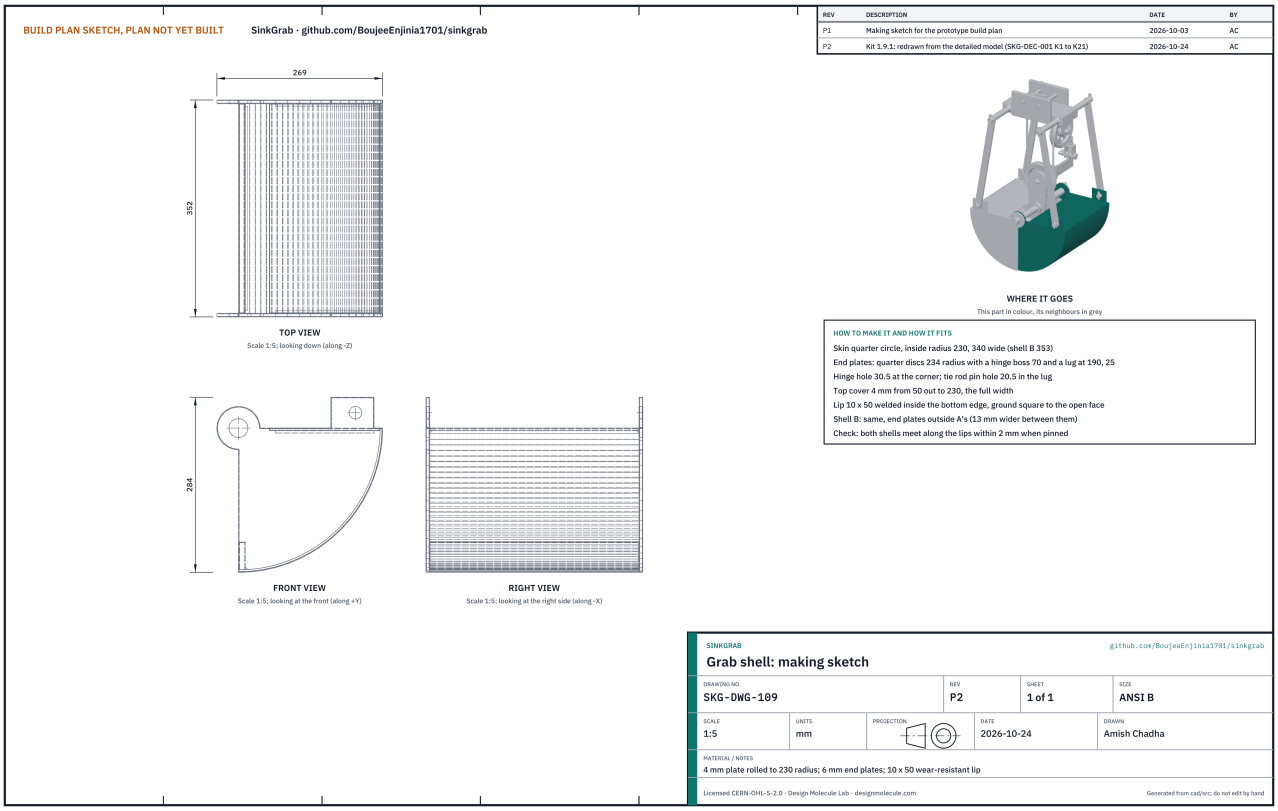


Figure 16. Grab shell making sketch (SKG-DWG-109).

What it is and what it is made from. The two scoops that bite the sand. 4 mm plate rolled to a 230 mm inside radius; 6 mm plate end plates (profile cut); 4 mm top cover; a 10 x 50 mm wear-resistant lip.

How to make it.

1. Roll the skin to a quarter circle of 230 mm inside radius, 340 mm wide for shell A and 353 mm for shell B.
2. Profile-cut the four end plates: a quarter disc of 234 mm radius with a 70 mm boss round the hinge corner and a lug at the outer top corner; a 30.5 mm hinge hole and a 20.5 mm tie rod hole.
3. Tack the end plates to the skin, shell B's end plates 13 mm further apart than shell A's so they pass outside A's at the hinge.
4. Weld the 4 mm top cover from 50 mm off the hinge out to the skin.
5. Weld the lip inside the bottom edge and grind its face square to the open side.

How it fits the parts next to it. Both shells hang on one 30 mm pin through their end plates and the crosshead (Figure 19); each is closed by two tie rods from its lugs (Figure 20).

Check before moving on. Pinned together, the lips meet within 2 mm along their length.

3.12 Grab head with ballast plates

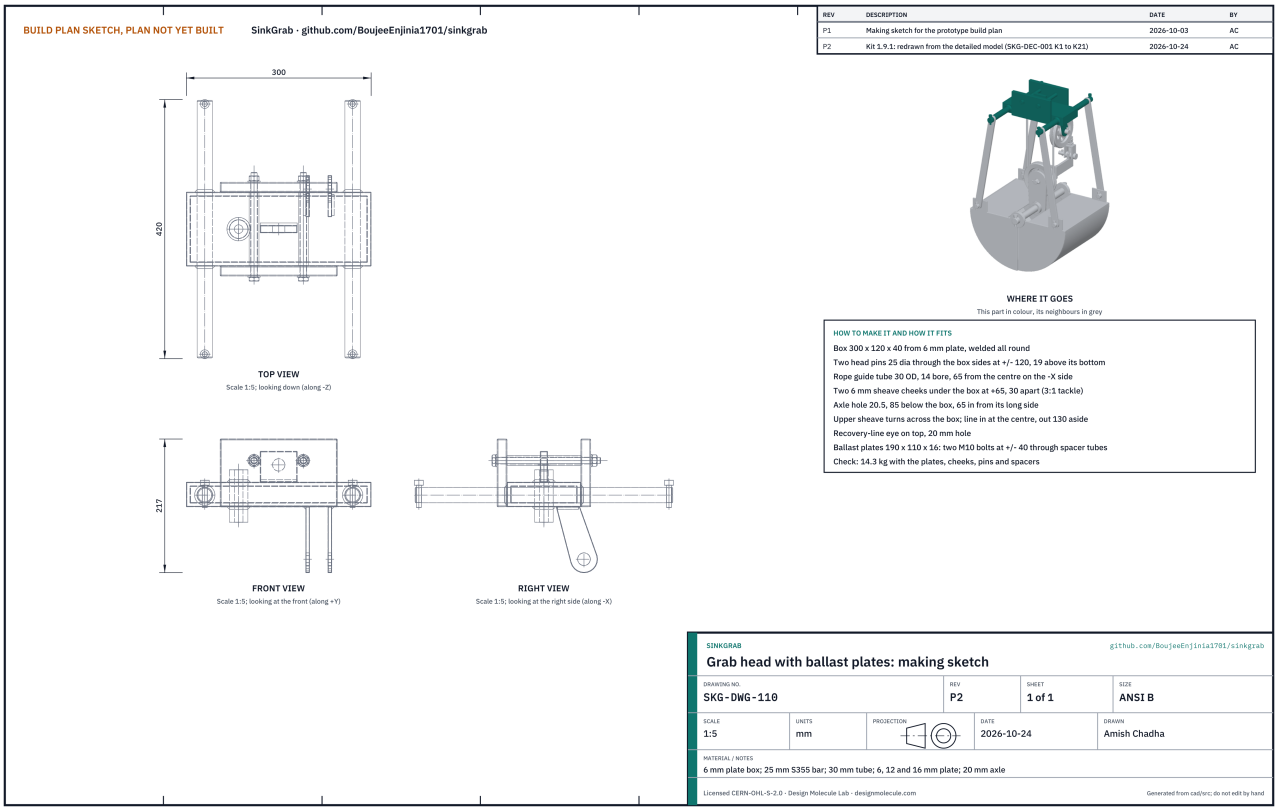


Figure 17. Grab head with ballast plates making sketch (SKG-DWG-110).

What it is and what it is made from. The heavy top block that holds the tie rods, guides the line and carries the upper sheave of the tackle. 6 mm plate; 25 mm bright bar; 30 mm tube; 12 mm plate; two 16 mm ballast plates; M10 bolts; the bought upper sheave on a 20 mm axle.

How to make it.

1. Weld a box 300 x 120 x 40 from 6 mm plate.
2. Pass the two 25 mm head pins (418 long) through the box sides, 240 mm apart and 19 mm above the box's bottom face so they clear the 6 mm bottom plate, and weld; drill each end for a split pin.
3. Weld the rope guide tube (30 mm outside, 14 mm bore) through the box 65 mm from the centre; flare its bottom end.
4. Weld two 6 mm cheek plates under the box, 65 mm the other side of the centre and 30 mm apart, hanging down to a 20.5 mm axle hole 85 mm below the box and 65 mm in from the box's long side. Weld the recovery-line eye on top.
5. Fit the upper sheave between the cheeks on its 20 mm axle with R-clips. It stands across the box, turned 90 degrees to the crosshead sheave, so the line comes up into it on the centre line and goes down again 130 mm to one side.
6. Drill both ballast plates for two M10 bolts, 80 mm apart and 75 mm above the box's bottom face; the bolts pass through 16 mm spacer tubes above the box, clear of the line rising from the guide tube.

Check before moving on. About 14.3 kg with both plates, cheeks, pins and spacers, and 2 kg more with the sheave and axle; the 8 mm line runs freely through the guide and over the upper sheave.

3.13 Crosshead, tie rods and load indicator

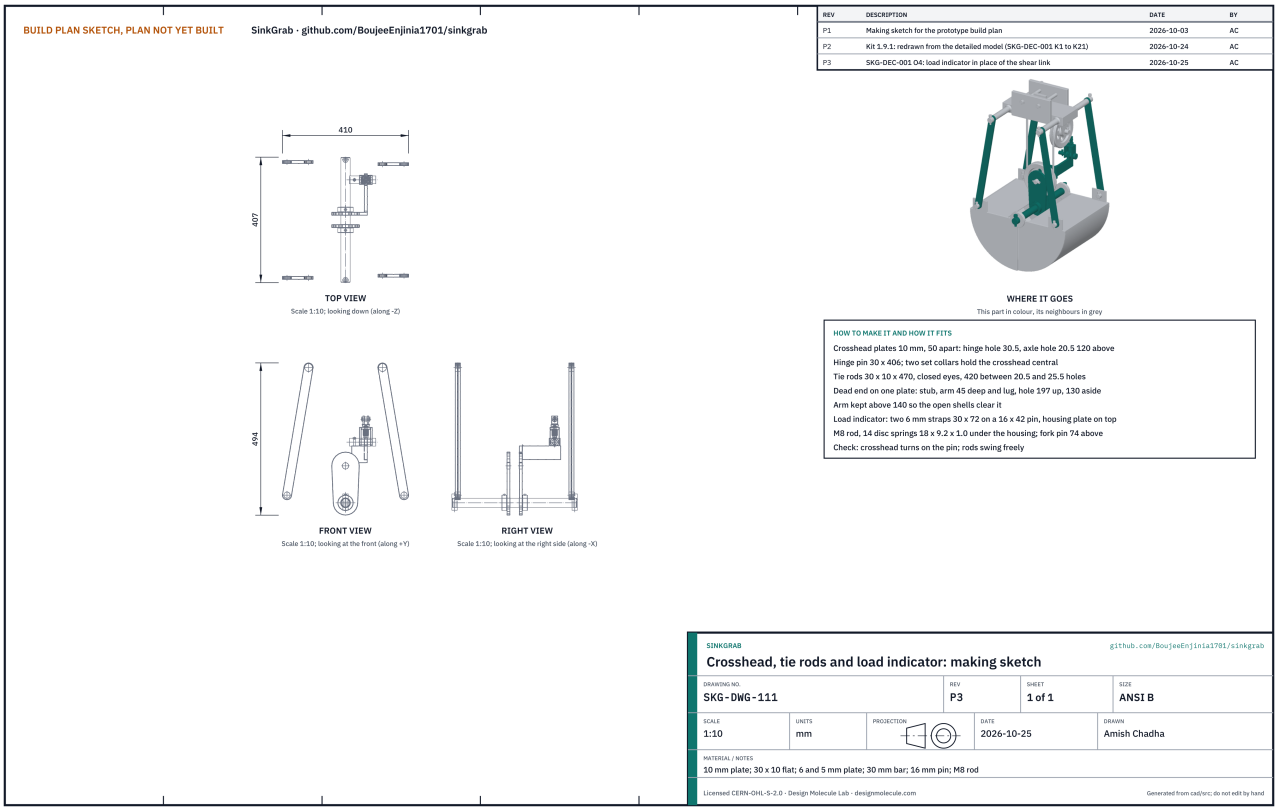


Figure 18. Crosshead, tie rods and load indicator making sketch (SKG-DWG-111).

What it is and what it is made from. The crosshead carries the hinge pin, the lower sheave of the tackle and the dead end of the line; the four tie rods join the head to the shells; the load indicator at the dead end shows the highest pull of each lift (decision O4, 2026-10-25; it replaces the shear link and does not release). 10 mm plate; 30 mm bar; 30 x 10 mm flat; 6 and 5 mm plate; 26 mm tube; a 16 x 42 mm clevis pin; an M8 stainless rod; 14 disc springs 18 x 9.2 x 1.0; an 8 mm pin; an 8 x 2 mm O-ring.

How to make it.

1. Cut the two crosshead plates with a 30.5 mm hinge hole and a 20.5 mm axle hole 120 mm above it.
2. On one plate weld the dead end: a 10 mm stub plate running out sideways from the plate, a 10 mm arm plate 45 mm deep along the hinge direction to 150 mm from the centre, and a 10 mm lug on its end with a 16.5 mm hole 197 mm above the hinge and 130 mm from the centre. Keep the arm above 140 mm so the open shells clear it.
3. Cut the hinge pin 406 mm long from 30 mm bar; drill both ends for split pins. Two 30 mm set collars on the pin, one each side of the crosshead, hold it central.
4. Cut four tie rods 470 mm long from 30 x 10 flat with rounded ends and 20.5 and 25.5 mm holes 420 mm apart, so each hole sits in a closed eye.
5. Load indicator: cut two straps 30 x 72 from 6 mm plate with a 16.5 mm hole 12 mm from the bottom, and a housing plate 30 x 30 from 6 mm plate with an 8.5 mm centre hole; weld the housing across the strap tops so

the straps stand 30 mm apart inside. Cut two spacer tubes from 26 mm tube to fill the gaps between the straps and the dead-end lug. Cut a fork base 23 x 24 from 6 mm plate, drill and tap it M8, and weld two cheeks 20 x 28 from 5 mm plate to it 9 mm apart, each with an 8.5 mm hole 17 mm above the base. Thread an M8 stainless rod 40 mm long into the fork base and scribe rings on it every 0.5 mm above the housing. Stack the 14 disc springs in series on the rod under the housing, then the washer and nut; fit the O-ring on the rod above the housing. Set the rings against CalRig in 0.25 kN steps up to 1.5 kN.

How it fits the parts next to it.

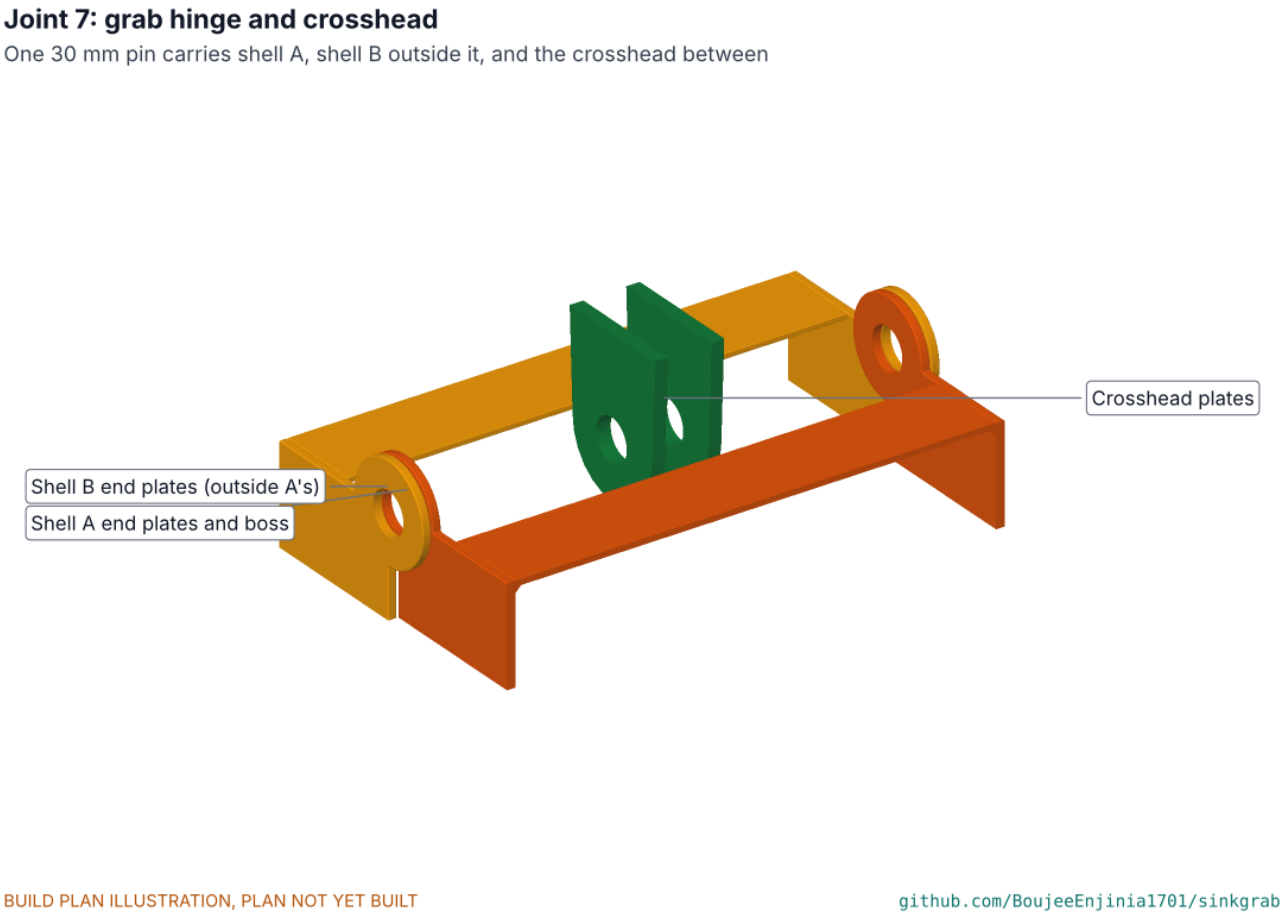


Figure 19. Joint 7. One 30 mm pin carries shell A's end plates, shell B's end plates outside them, and the two crosshead plates between. The sheave (not shown) turns on its own axle 120 mm above the pin.

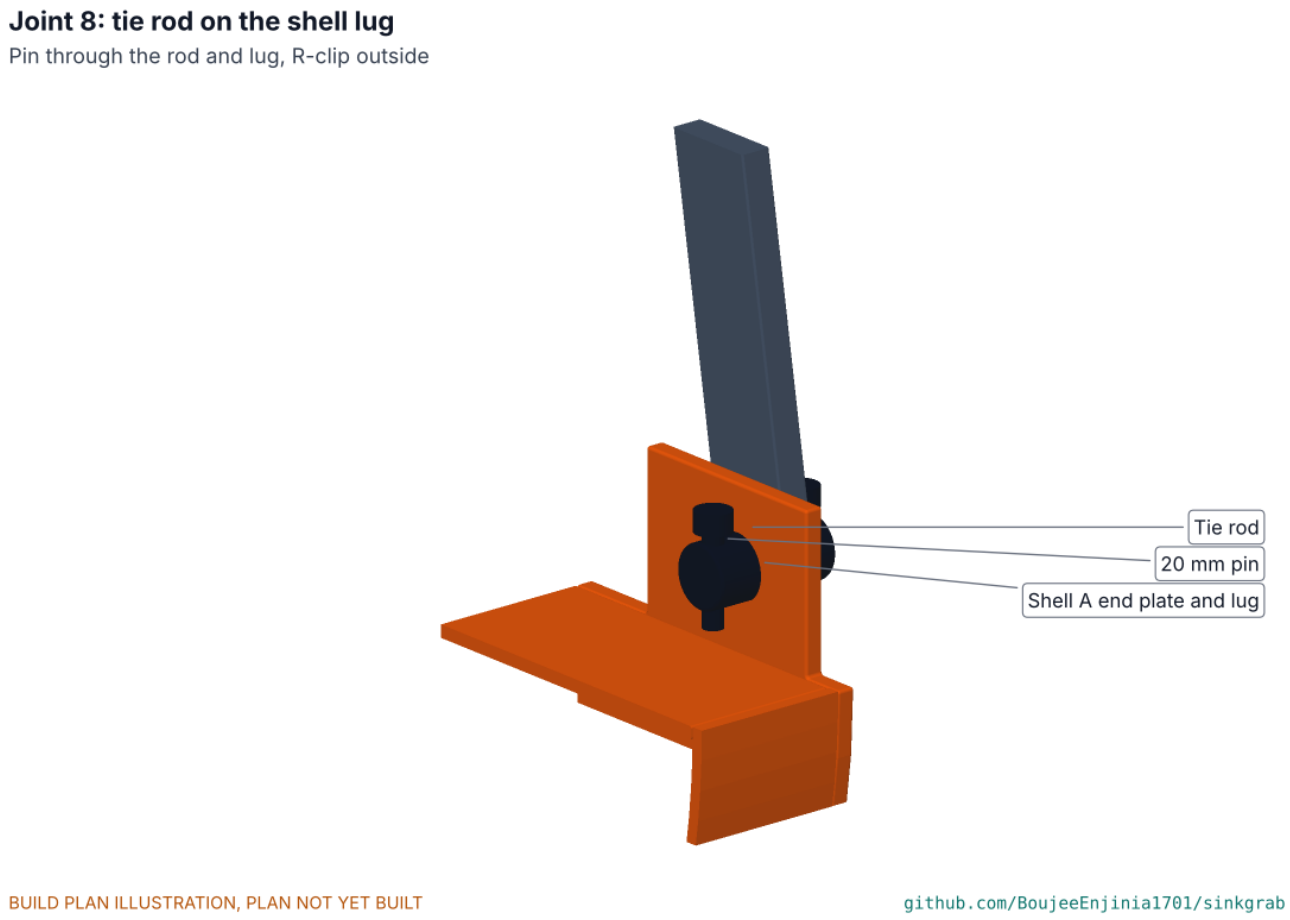


Figure 20. Joint 8. Each tie rod is pinned to its shell's lug with a 20 mm pin and R-clip; shell A's rods run outside A's end plates, shell B's outside B's.

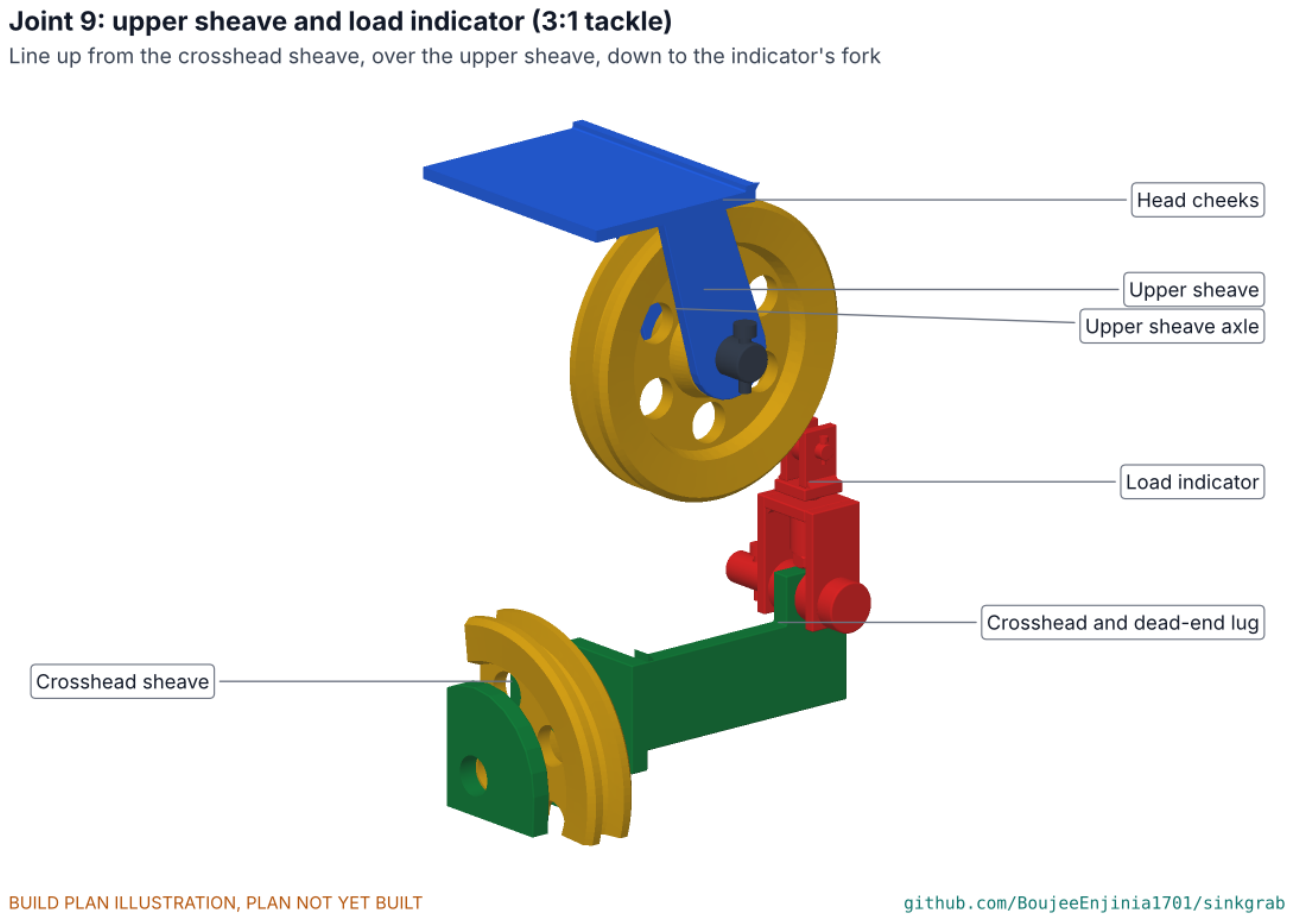


Figure 21. Joint 9. The line comes up from the crosshead sheave, over the upper sheave in the head and down to the load indicator on the crosshead's dead-end lug; its thimble sits on a sleeve on the fork pin, and the line's pull draws the rod up against the disc springs. The O-ring tell-tale stays at the highest travel, read against the rings at the doors after each lift and pushed back down. Three parts of line between head and crosshead close the shells 50 % harder than two. A snag is held by the clutch at the capstan; nothing at the grab releases.

Check before moving on. The crosshead turns on the pin between its collars; the rods swing freely; the indicator's rod slides freely and its rings have been set on CalRig (see the register).

3.14 Scraper head

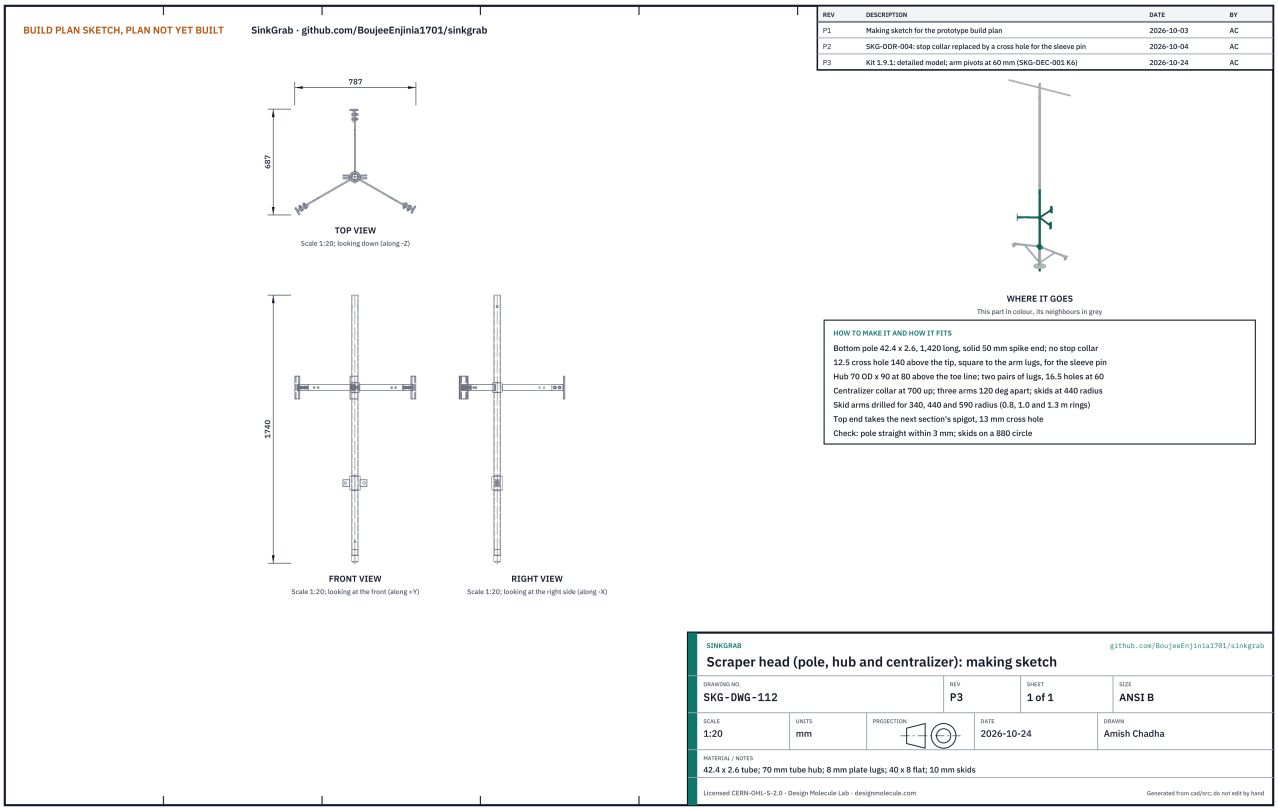


Figure 22. Scraper head (pole, hub and centralizer) making sketch (SKG-DWG-112).

What it is and what it is made from. The bottom pole section that carries the arms and centres itself in the ring. 42.4 x 2.6 mm tube; 50 mm bar for the spike; 70 mm tube for the hub; 8 mm plate lugs; 40 x 8 flat; 10 mm skid plates.

How to make it.

1. Cut the pole 1,420 mm and weld a 50 mm solid spike in the bottom end. Drill a 12.5 mm cross hole through the pole 140 mm above the tip, square to the arm lugs, for the sleeve's cross pin. There is no stop collar: the pin stops the sleeve.
2. Weld the hub 80 mm above the toe line, with two pairs of 8 mm lugs either side, 16.5 mm holes 60 mm off the pole axis, so each arm's eye clears the hub.
3. Weld the centralizer collar 700 mm up with three arms 120 degrees apart; drill each arm for the skids at 340, 440 and 590 mm radius; bolt the skids at 440 for a 1.0 m ring.
4. Drill a 13 mm cross hole 75 mm below the top end for the next section's bolt.

Check before moving on. Straight within 3 mm; the skids lie on an 880 mm circle.

3.15 Scraper arms, struts and sleeve

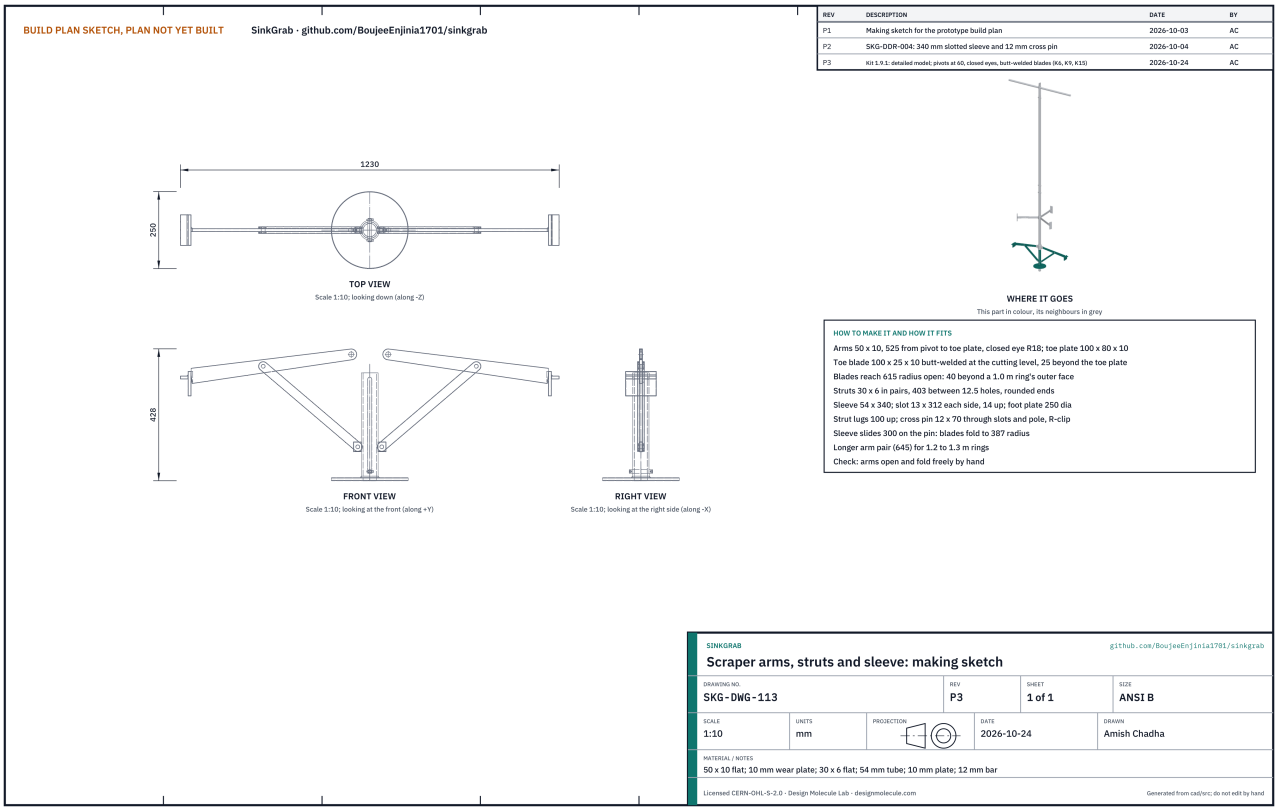


Figure 23. Scraper arms, struts and sleeve making sketch (SKG-DWG-113).

What it is and what it is made from. Two arms whose toes cut under the ring and 40 mm beyond its outer face, opened by the pole's weight, and the slotted sleeve that lets them fold to pass the ring. 50 x 10 mm flat; 10 mm wear-resistant plate for the toes and toe blades; 30 x 6 mm flat for the struts; 54 mm tube and a 250 mm, 10 mm disc for the sleeve and foot; 12 mm bright bar for the cross pin; 12 and 16 mm pins.

How to make it.

1. Cut two arms from 50 x 10 flat, 525 mm from the pivot hole to the toe plate, narrowing to a closed eye of 18 mm radius round the pivot hole; weld a toe plate 100 x 80 x 10 square across each end.
2. Cut two toe blades 100 x 25 from 10 mm wear-resistant plate. Butt-weld one to the outer face of each toe plate at the cutting level, standing 25 mm out, so the blade's edge is 615 mm from the pole axis.
3. Drill each arm for its 16 mm pivot pin and, 289 mm out, a 12.5 mm strut pin.
4. Cut four struts from 30 x 6 flat with 12.5 mm holes 403 mm apart and rounded ends 15 mm beyond each hole.
5. Cut the sleeve 340 mm long from 54 mm tube. Cut a slot 13 mm wide and 312 mm long through each side, starting 14 mm above the bottom end, with the top of each slot rounded; the two slots face each other. Weld the foot disc under the sleeve, and a 10 mm strut lug on each of the other two sides, 100 mm above the bottom, with a 12.5 mm hole.
6. Cut the cross pin 70 mm long from 12 mm bright bar and drill a small hole near one end for an R-clip.
7. For 1.2 to 1.3 m rings, make a second pair of arms 645 mm long, with the same blades.

How it fits the parts next to it.

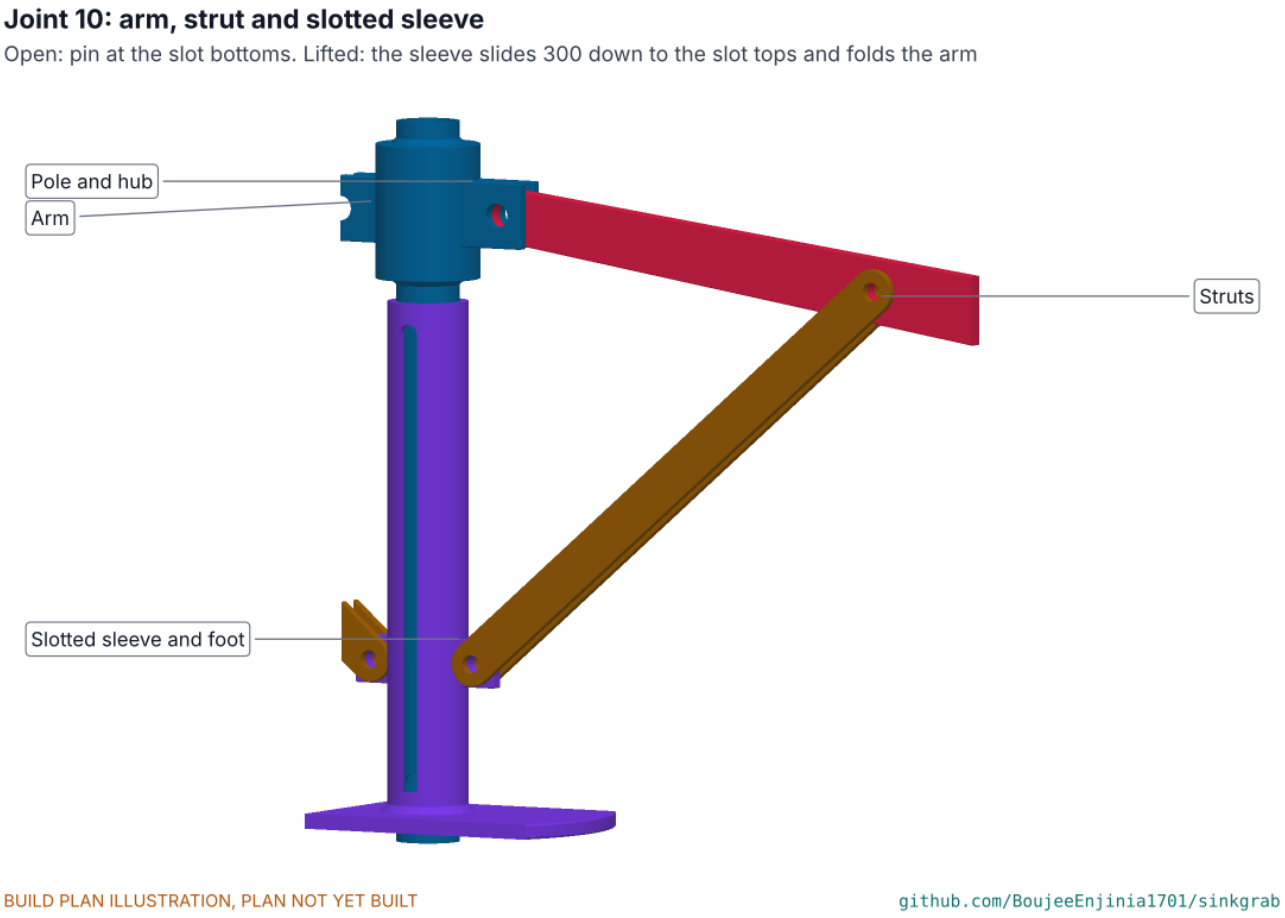


Figure 24. Joint 10. The sleeve slides on the pole, held by a cross pin through the pole that runs in a slot on each side of the sleeve. When the foot rests on the sump floor and the line is slacked, the pole slides down through the sleeve until the pin reaches the bottom of the slots, and the struts push the arms out.

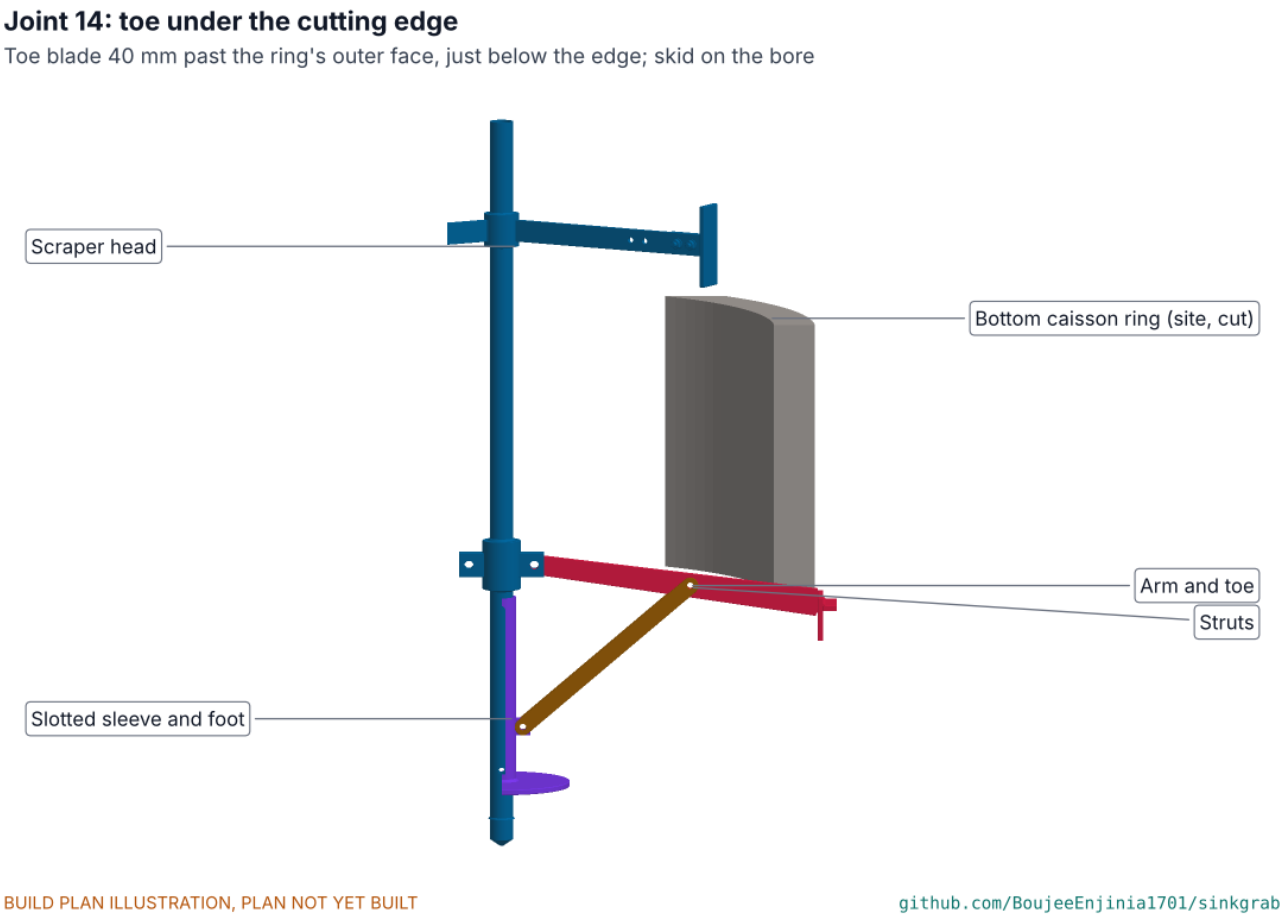


Figure 25. Joint 14, the ring cut away. Open, the toe blade reaches 40 mm past the ring's outer face, just below its cutting edge, so the ring slides down through loosened soil; the skid bears on the bore above. Lifting the pole lets the sleeve slide 300 mm down it until the pin reaches the top of the slots; that folds the blades to 387 mm radius so the head passes the ring.

Check before moving on. The arms open and fold freely by hand by sliding the sleeve the full length of its slots; folded, the blades sit inside an 800 mm circle. The head weighs about 27.3 kg with the arms on, so unpin the two arms for carrying: about 21.9 kg for the head and 5.5 kg for the arms.

3.16 Pole sections and T-bar

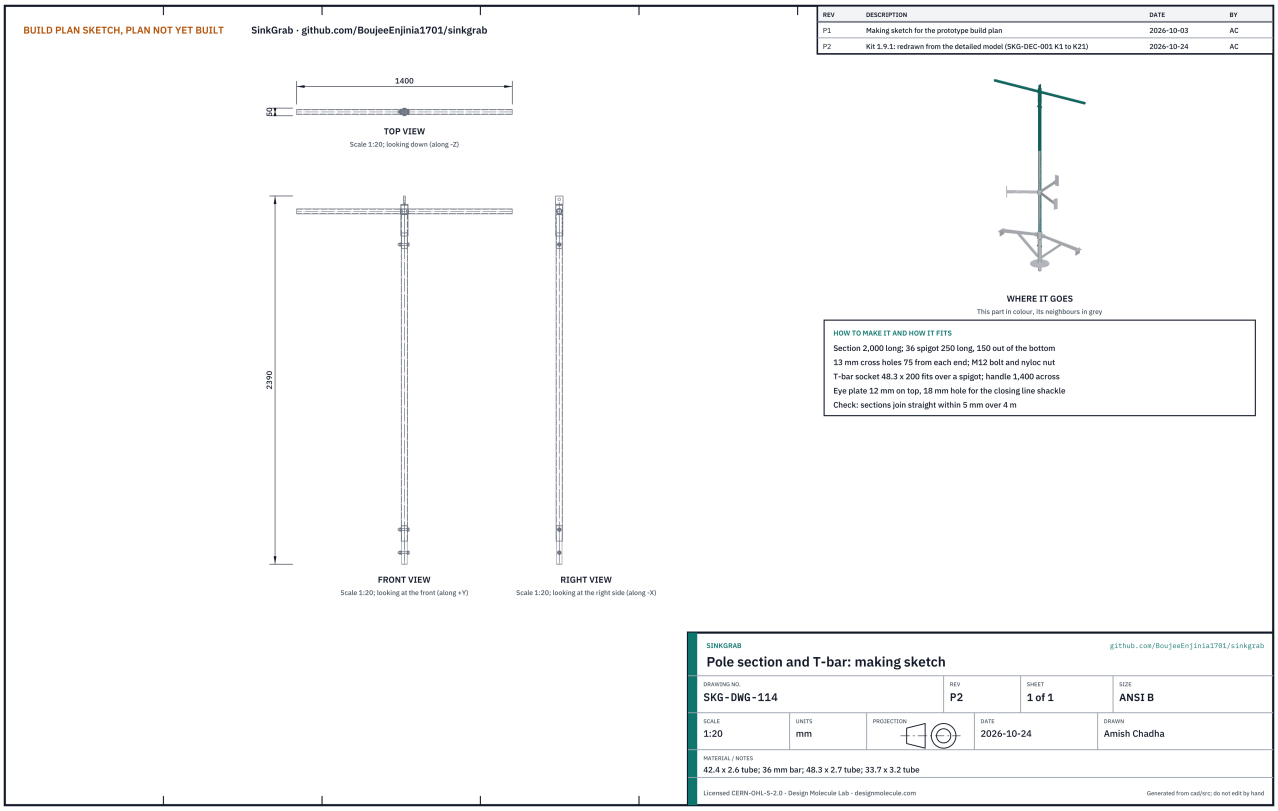


Figure 26. Pole section and T-bar making sketch (SKG-DWG-114).

What it is and what it is made from. 2 m extensions and the handle. 42.4 x 2.6 mm tube; 36 mm bar; 48.3 x 2.7 mm tube; 33.7 x 3.2 mm tube; 12 mm plate; M12 bolts.

How to make it. Cut each section 2,000 mm; weld a 36 mm spigot 250 mm long into the bottom end, 150 mm out; drill 13 mm cross holes 75 mm from each end. For the T-bar, weld the 1,400 mm handle across a 200 mm socket of 48.3 tube and the eye plate on top.

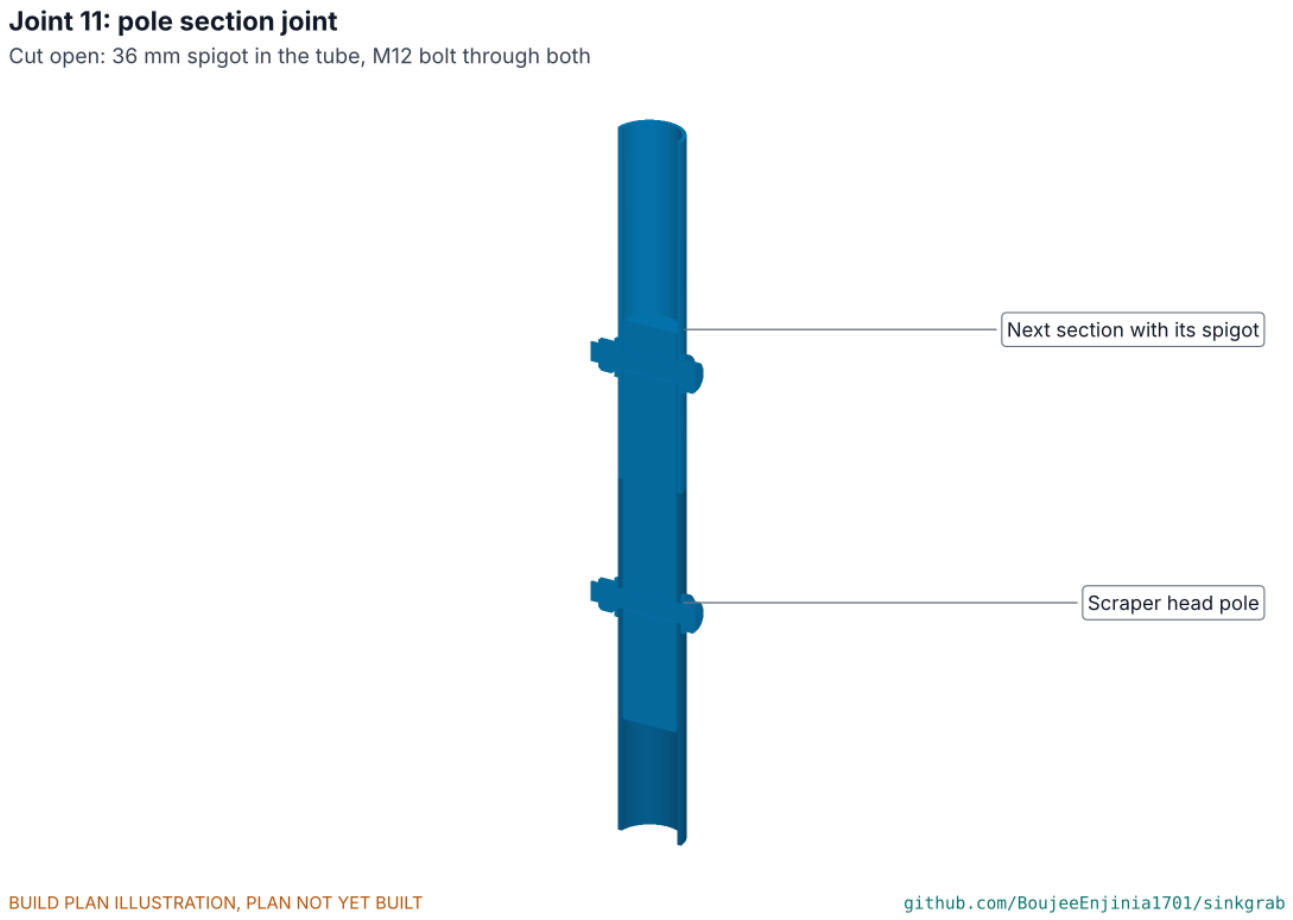


Figure 27. Joint 11, cut open. The spigot slides into the tube below; one M12 bolt through both carries the weight and the turning.

Check before moving on. Two sections bolted together are straight within 5 mm over 4 m.

3.17 Well-head frame

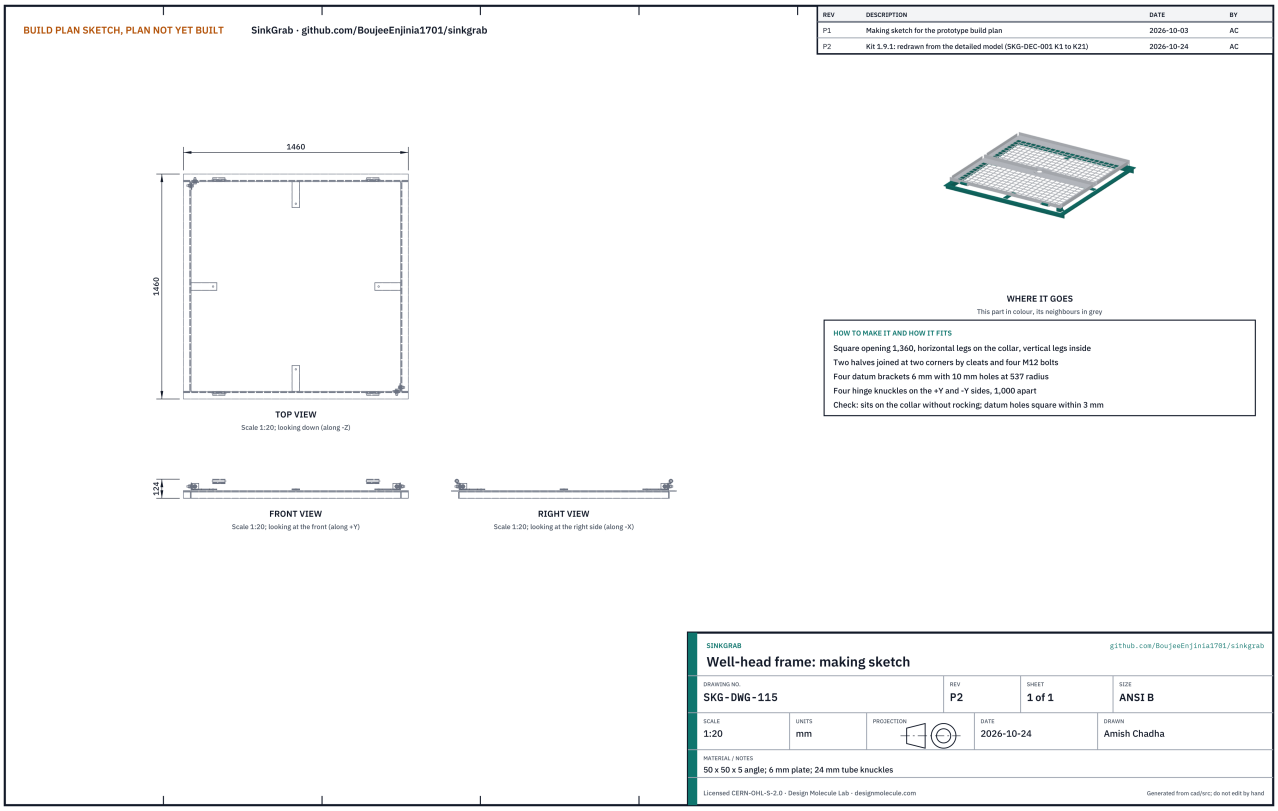


Figure 28. Well-head frame making sketch (SKG-DWG-115).

What it is and what it is made from. A square frame that sits on the well collar and carries the doors and the tilt datum. 50 x 50 x 5 mm angle; 6 mm plate; 24 mm tube for the hinge knuckles; M12 bolts.

How to make it.

1. Cut and weld the angle into two L-shaped halves that join at two corners, by a 50 x 50 x 5 cleat inside each corner and four M12 bolts, into a frame with a 1,360 mm square opening, horizontal legs out and down onto the collar, vertical legs up at the inner edge.
2. Weld four datum brackets (6 mm plate) under the frame at the middle of each side, each with a 10 mm hole 537 mm from the centre, over the caisson ring wall.
3. Weld two hinge knuckles on each of the two hinged sides, 1,000 mm apart.

Check before moving on. It sits on the collar without rocking; the four datum holes lie on a square within 3 mm.

3.18 Folding doors (make 2)

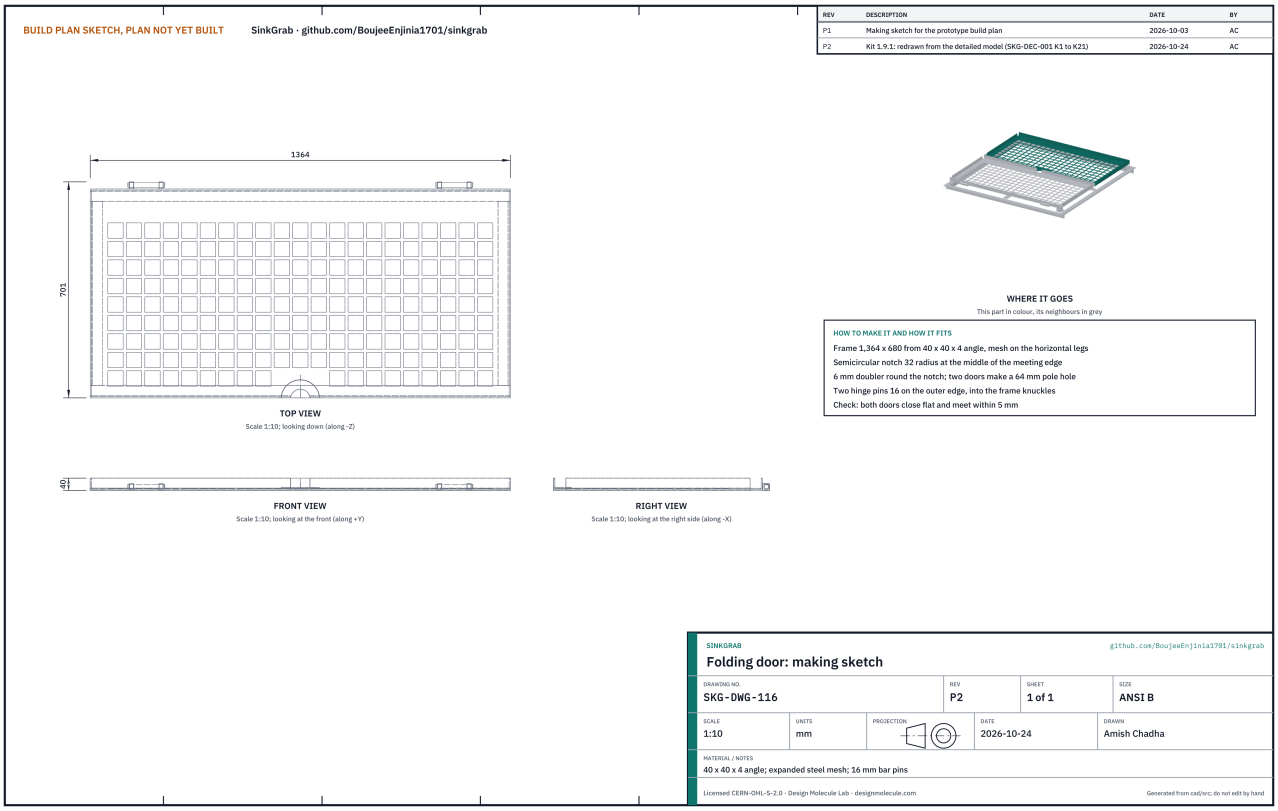


Figure 29. Folding door making sketch (SKG-DWG-116).

What it is and what it is made from. Two mesh doors that close over the shaft. 40 x 40 x 4 mm angle; expanded steel mesh; 6 mm plate doubler; 16 mm bar.

How to make it. Weld a frame 1,364 x 680 from angle, horizontal legs inward at the bottom, with the mesh welded on the horizontal legs. Cut a 32 mm radius notch at the middle of the meeting edge and weld a 6 mm doubler round it. Weld two hinge pins on the outer edge to match the knuckles.

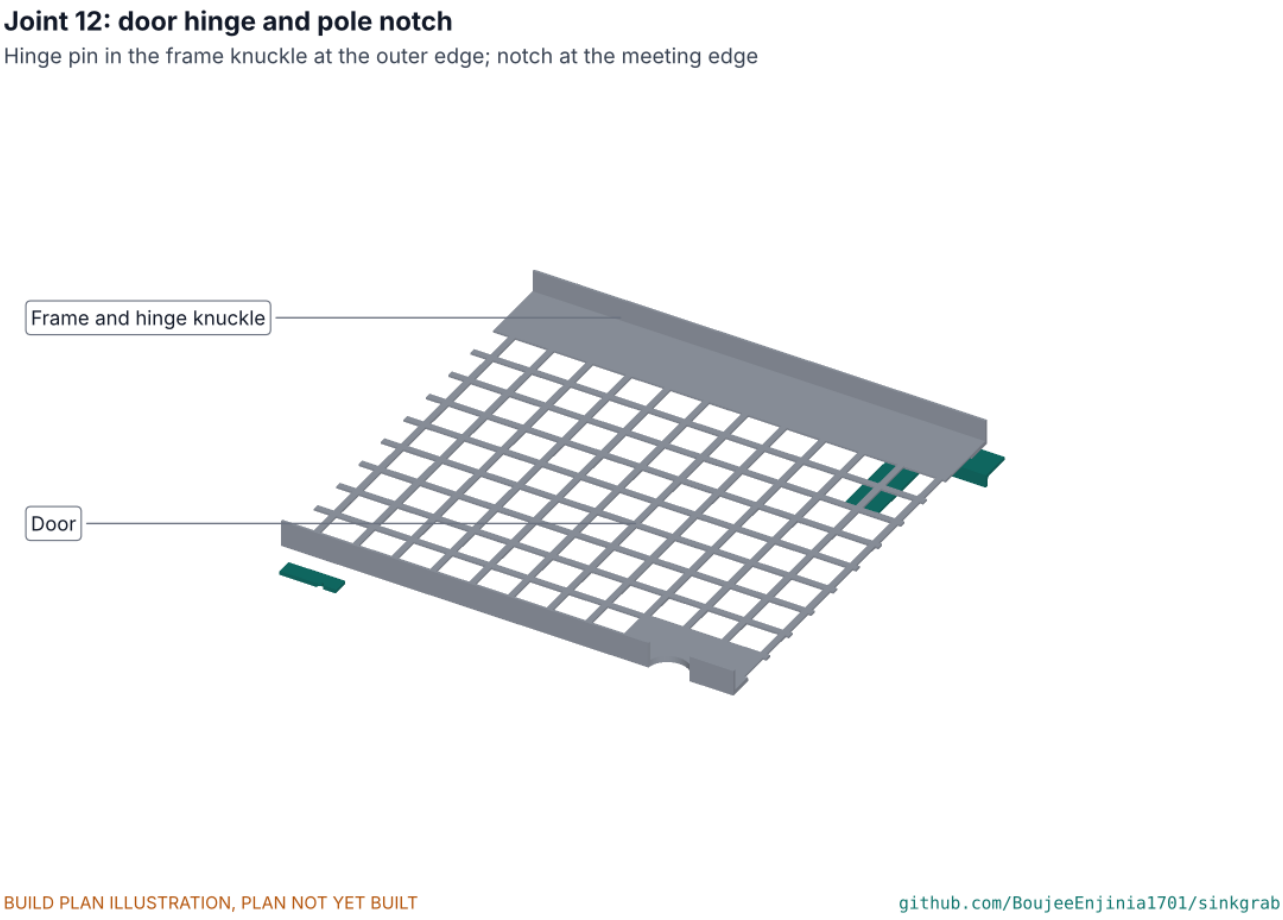


Figure 30. Joint 12. Each door hinges on two pins in the frame knuckles at its outer edge and rests on the frame's vertical legs. Closed, the two notches make a 64 mm hole for the pole.

Check before moving on. Both doors close flat and meet within 5 mm; each is about 20 kg.

3.19 Ballast saddles (make 8)

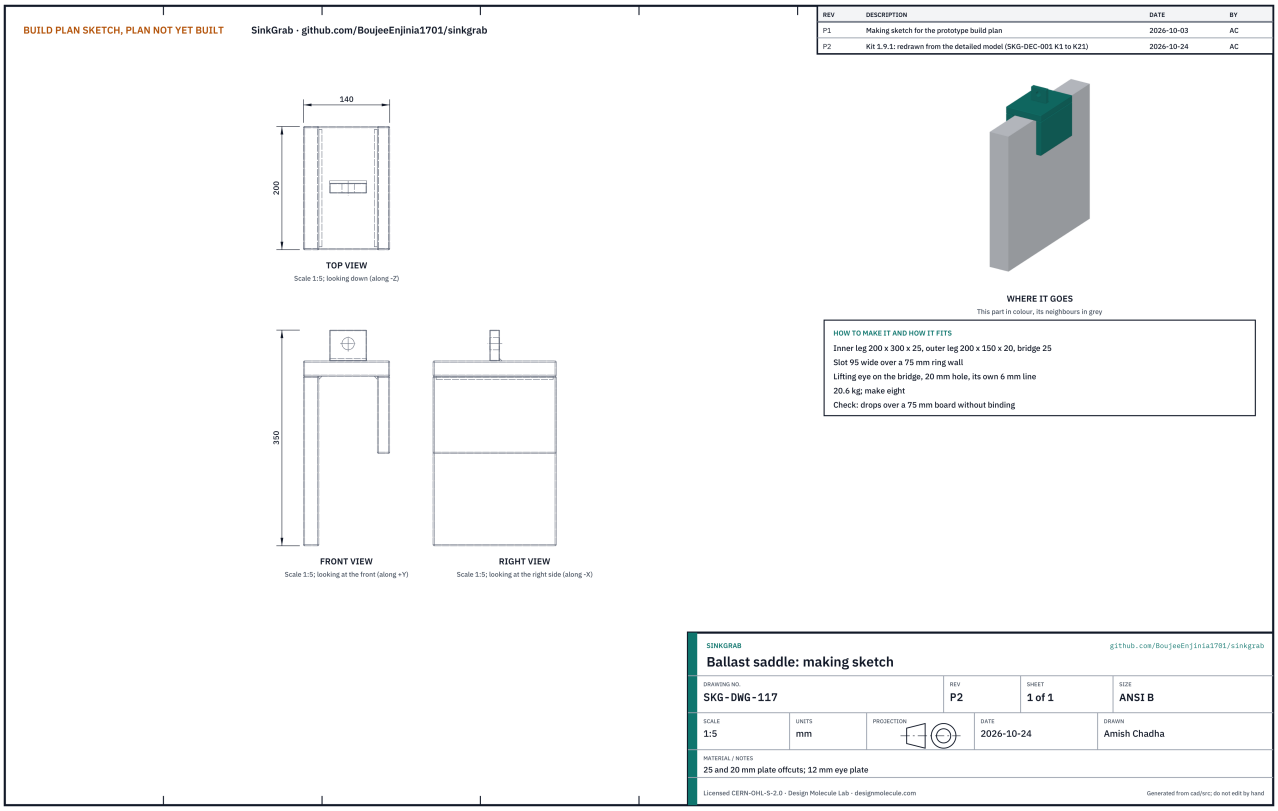


Figure 31. Ballast saddle making sketch (SKG-DWG-117).

25 and 20 mm plate offcuts welded into a U: inner leg 200 x 300 x 25, outer leg 200 x 150 x 20, bridge 25 mm, a 95 mm slot, and a lifting eye with a 20 mm hole for its own 6 mm line. About 20.6 kg each.

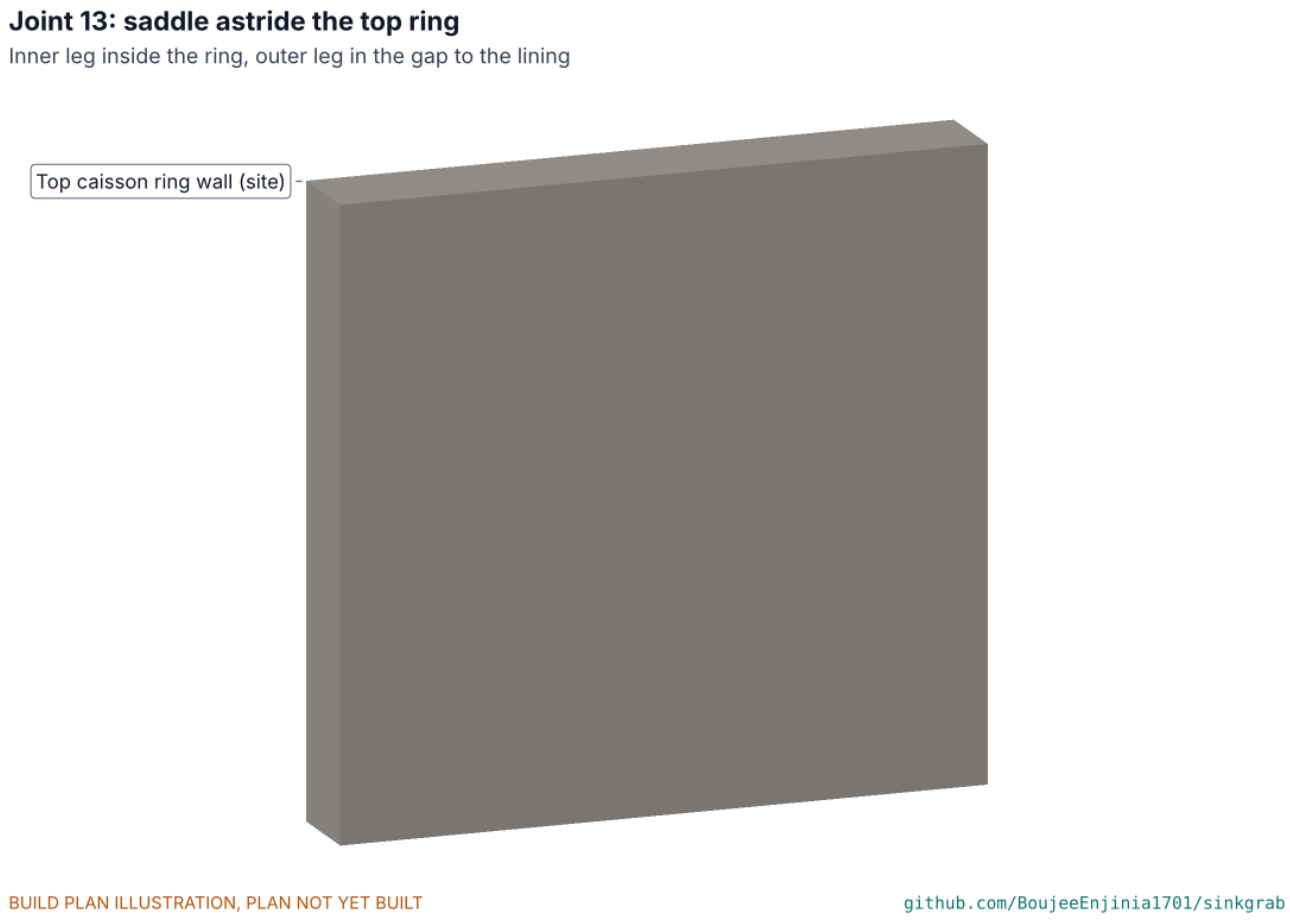


Figure 32. Joint 13. The saddle drops over the ring wall: the inner leg inside the ring, the outer leg in the gap to the lining.

Check before moving on. It drops over a 75 mm board without binding.

3.20 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

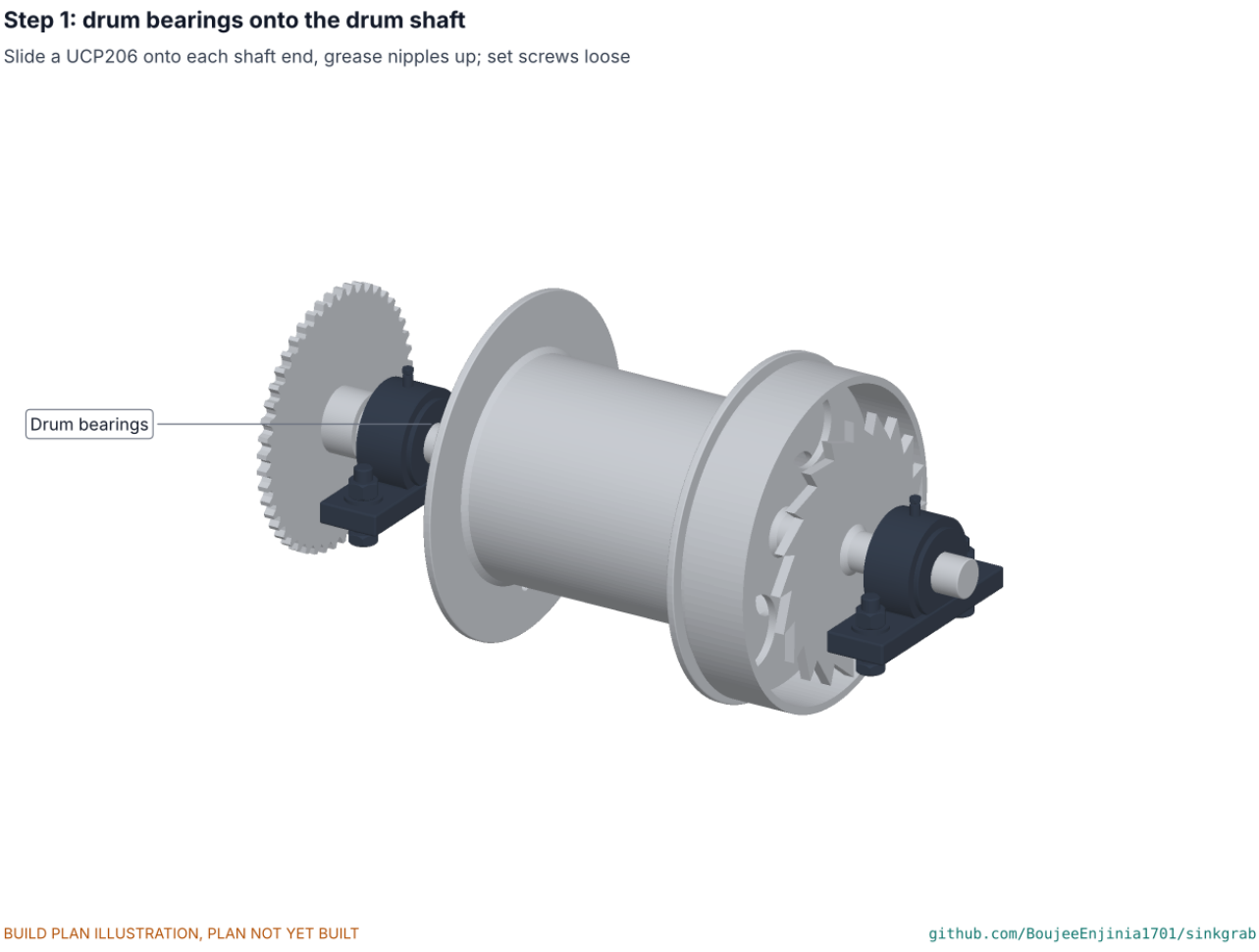
- **Bearings (lines 21 and 22).** Two UCP206 (30 mm) and two UCP205 (25 mm) pillow block ball bearings.
- **Sprockets and chain (line 23).** ISO 08B-1: a 12-tooth sprocket bored to suit the driven sleeve and a 48-tooth plate wheel bored 30 mm; 1.6 m of chain, breaking load at least 17.8 kN, with one connecting link.
- **Crank slip clutch parts (line 24).** Two woven friction facings 84 x 44 x 4 mm bonded to the flange and pressure plate; two disc springs 50 x 25.4 x 2.0 (DIN EN 16983, series B); a KM5 locknut and MB5 tab washer; a parallel key 8 x 7 x 8.
- **Load indicator parts (line 14).** Fourteen disc springs 18 x 9.2 x 1.0 (DIN EN 16983, series A), an M8 stainless nut and washer, spare 8 x 2 mm O-rings.
- **Sheaves (line 25).** Four sheaves 150 mm outside, 130 mm at the rope, grooved for 8 mm fibre rope, 28 mm wide, 20 mm bore with a ball bearing, rated at least 500 kg: one for the cradle, one for the crosshead, one for the grab head (the 3:1 tackle), one to fit in the HatchSide head in place of its wire-rope sheave.
- **Ropes (lines 26 to 28).** 45 m of 8 mm polyester double braid (at least 12 kN) with a thimble eye splice for the closing line; 40 m of 8 mm for the recovery line; eight 32 m lengths of 6 mm braid (at least 5 kN) for the saddles.

- **Rigging (line 29).** A ball-bearing swivel and bow shackles, WLL 500 kg, and one WLL 1,000 kg shackle for the T-bar eye.
- **Dip tape and plumb line (line 30), gas detector (line 31), spoil tubs (line 32), barrier kit and briefing card (line 33), fasteners and consumables with lead-free primer and paint (line 34).**
- **Carbon dioxide monitor (line 35).** A portable safety monitor reading 0 to 5 % carbon dioxide, with alarms near 0.5 % and 3 %, and a calibration certificate; used beside the four-gas detector, which does not read carbon dioxide.

4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in. Steps 1 to 15 are done in the workshop; steps 16 to 20 at the well. Two people throughout.

Step 1: drum bearings onto the drum shaft

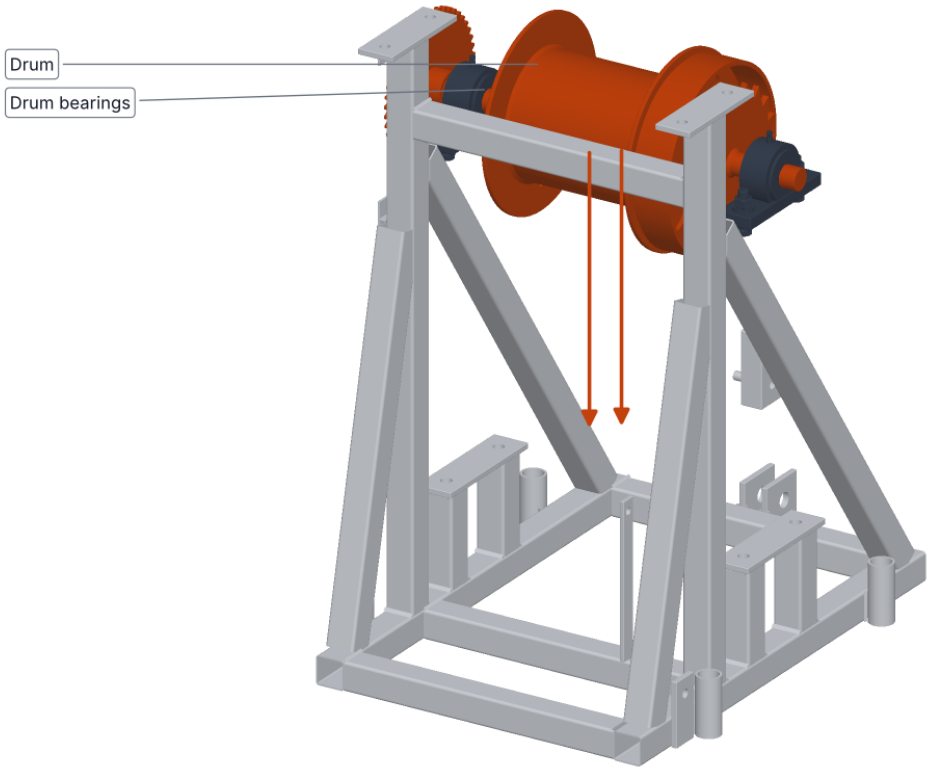


Slide a UCP206 onto each end of the drum shaft, grease nipples up, set screws loose.

Step 2: lower the drum onto the pads

Step 2: lower the drum onto the pads

Two people; over the captive M14 bolts; centre the drum; set screws

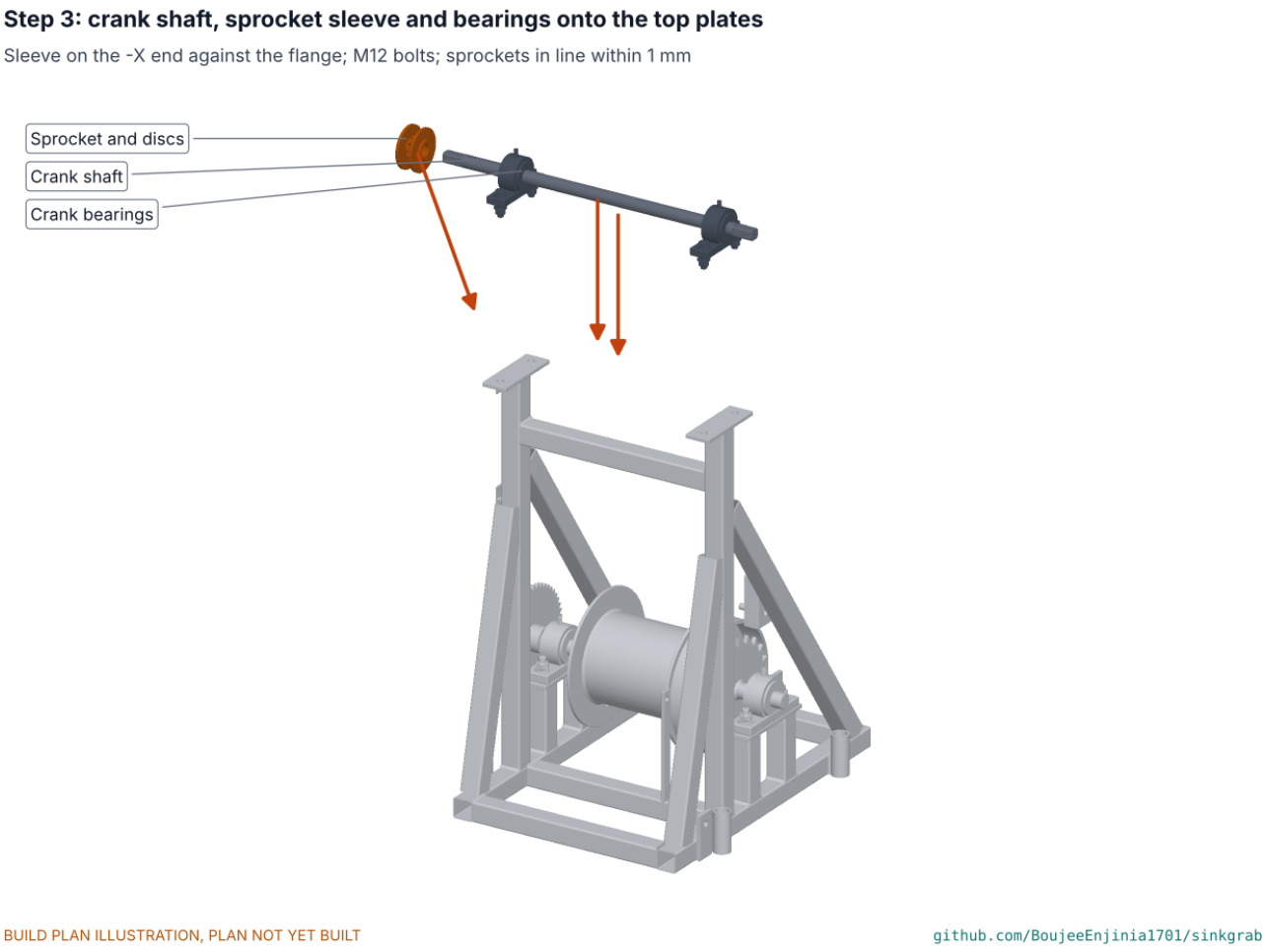


BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/sinkgrab

Two people lower the drum and bearings onto the pads, sprocket to the -X side, each bearing dropping over the two M14 bolts standing up from its pad. Fit washers and nuts, centre the drum and tighten the set screws.

Step 3: crank shaft, hub and bearings onto the top plates

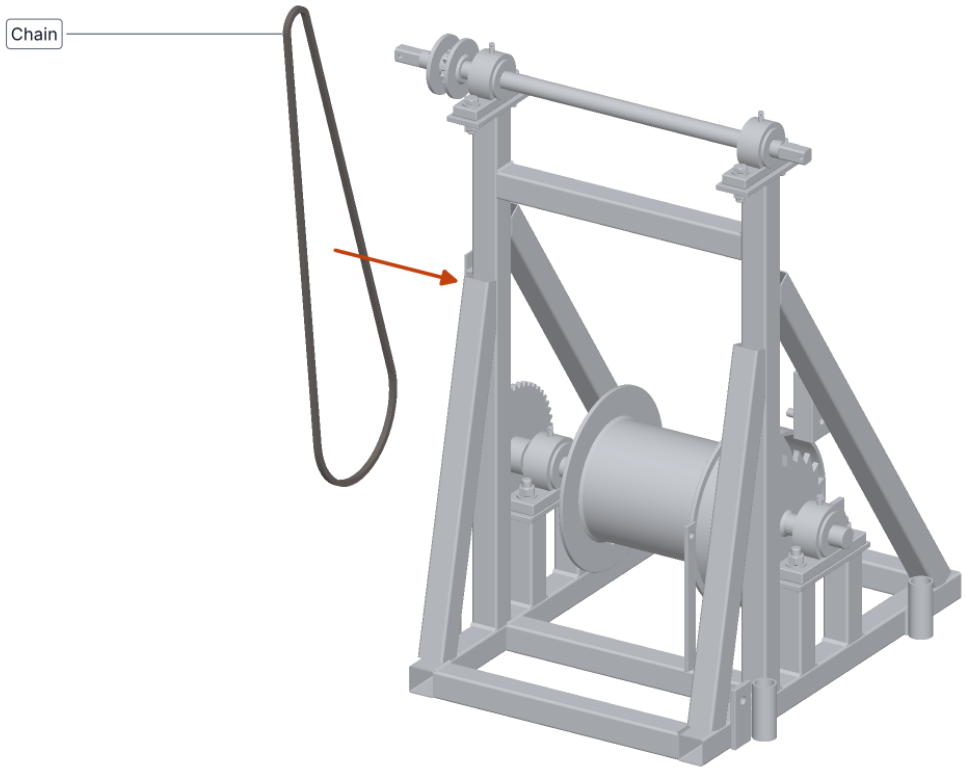


Slide the crank bearings onto the crank shaft with the sprocket hub on the -X end. Bolt the bearings to the top plates with M12 bolts, the small sprocket in line with the big one within 1 mm.

Step 4: fit the chain

Step 4: fit the chain

Over both sprockets; connecting link clip closed end leading

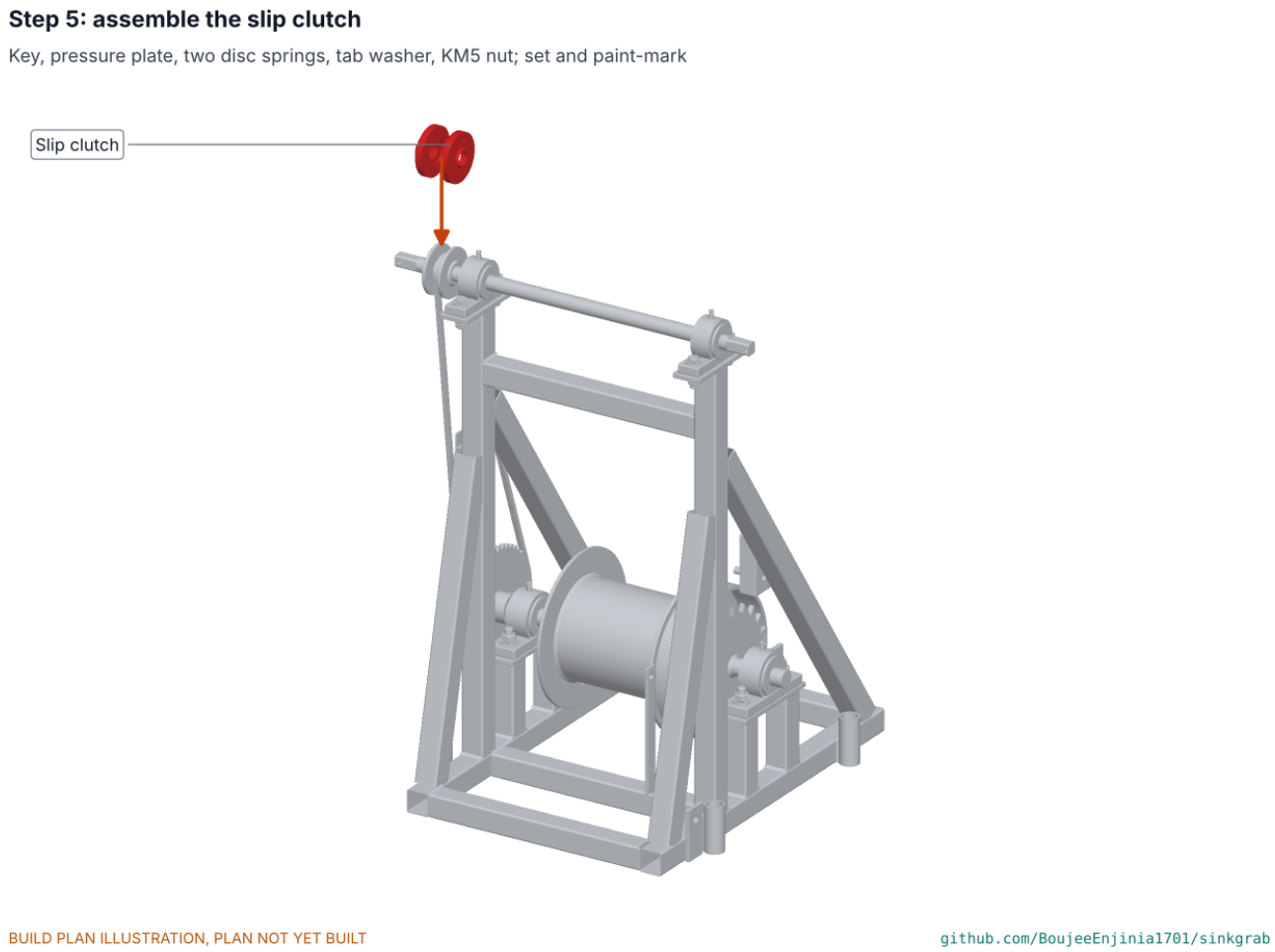


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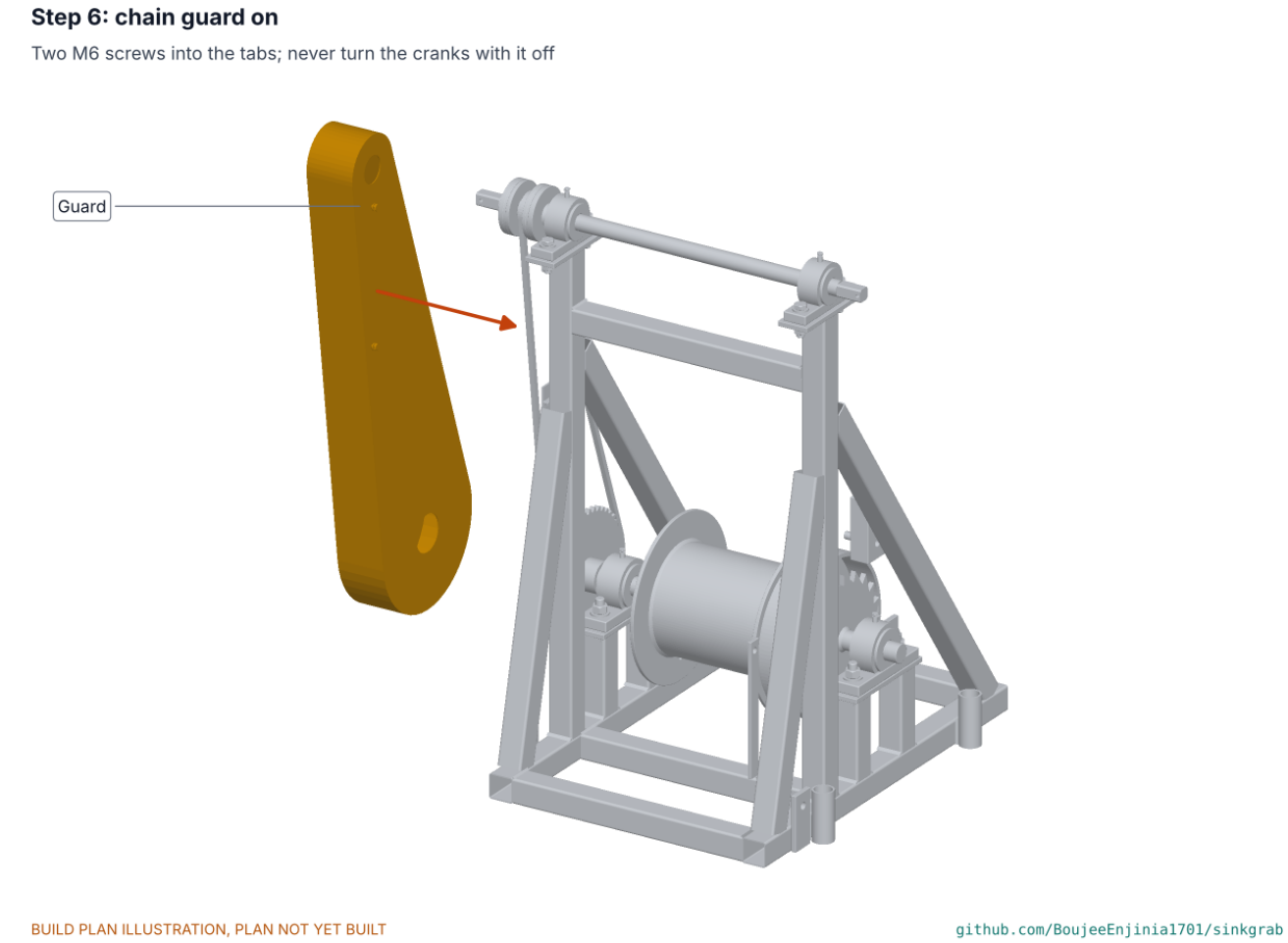
Lay the chain over both sprockets and join it with the connecting link, the clip's closed end leading when hoisting. It should lift about 10 mm at mid-span.

Step 5: assemble the slip clutch



With the driven sleeve on the shaft against the inner flange's facing, fit the key, slide on the pressure plate facing inward, then the two disc springs, the tab washer and the KM5 locknut. Bring the clutch to the setting found on CalRig (or set it there now), bend a tab of the washer into the locknut slot and paint-mark the locknut across the washer.

Step 6: chain guard on

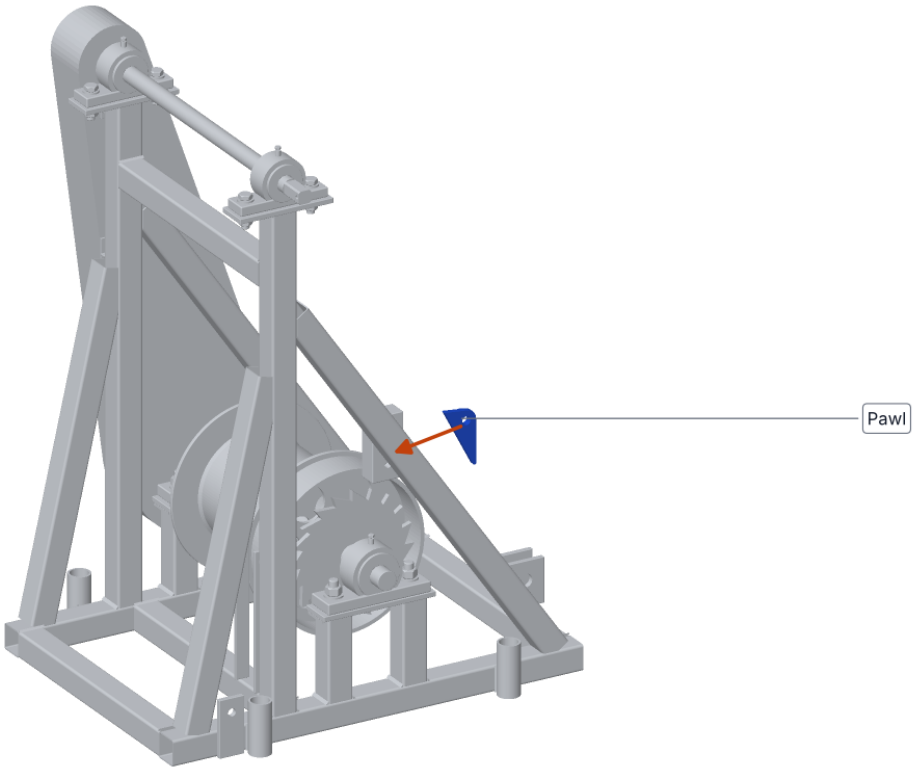


Fit the guard and fix it with two M6 screws. From now on the cranks are never turned with it off.

Step 7: pawl on its pin

Step 7: pawl on its pin

Washer and R-clip; it drops into the ratchet by its own weight



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

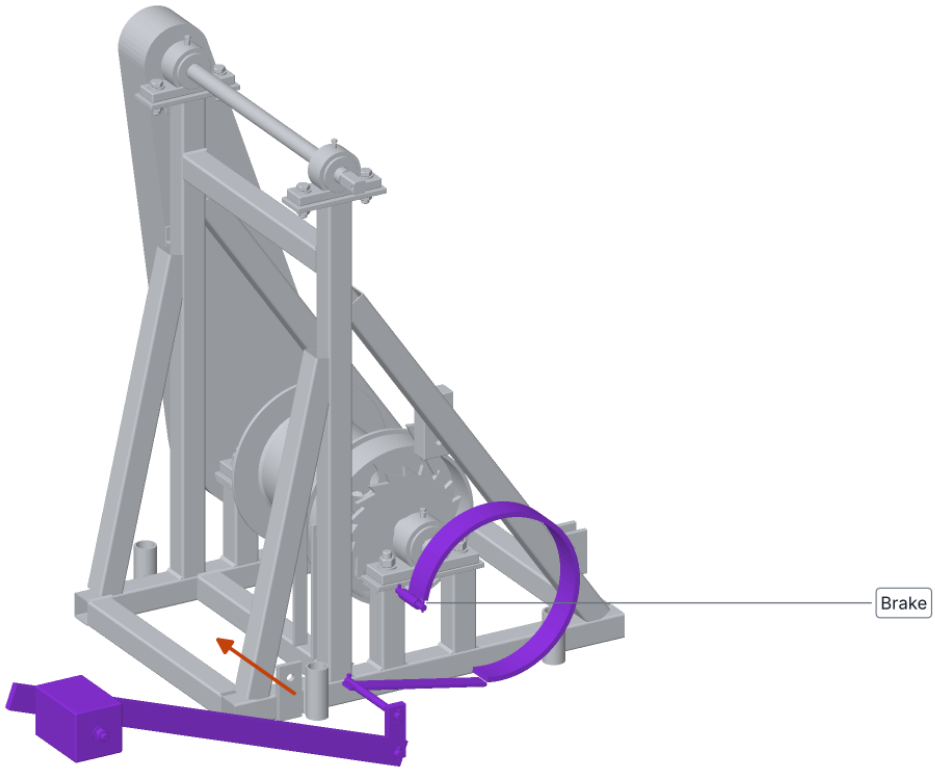
github.com/BoujeeEnjinia1701/sinkgrab

Hang the pawl on its pin with a washer and R-clip; check it drops into the ratchet.

Step 8: band brake and weighted lever

Step 8: band brake and weighted lever

Pin the anchor end, then the link and lever; weight on

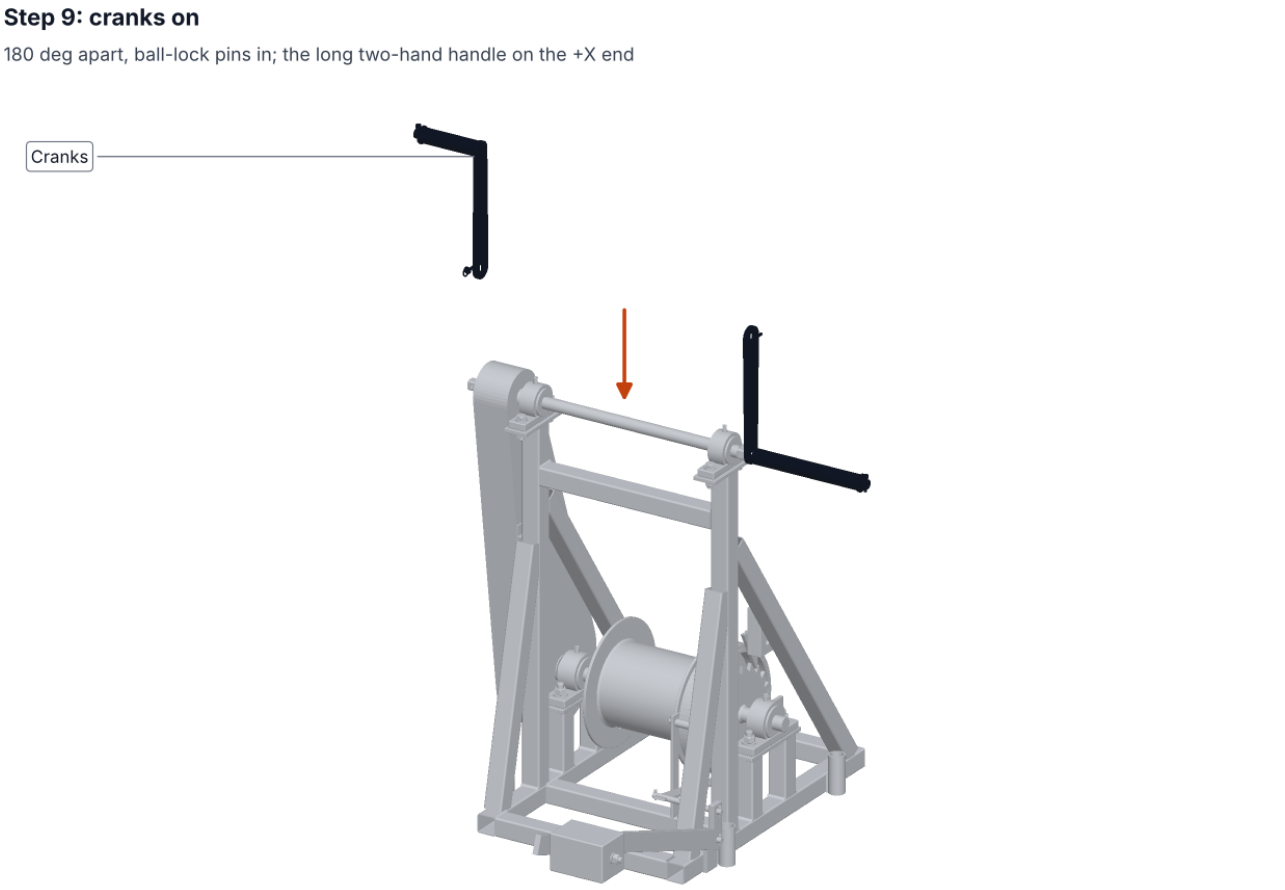


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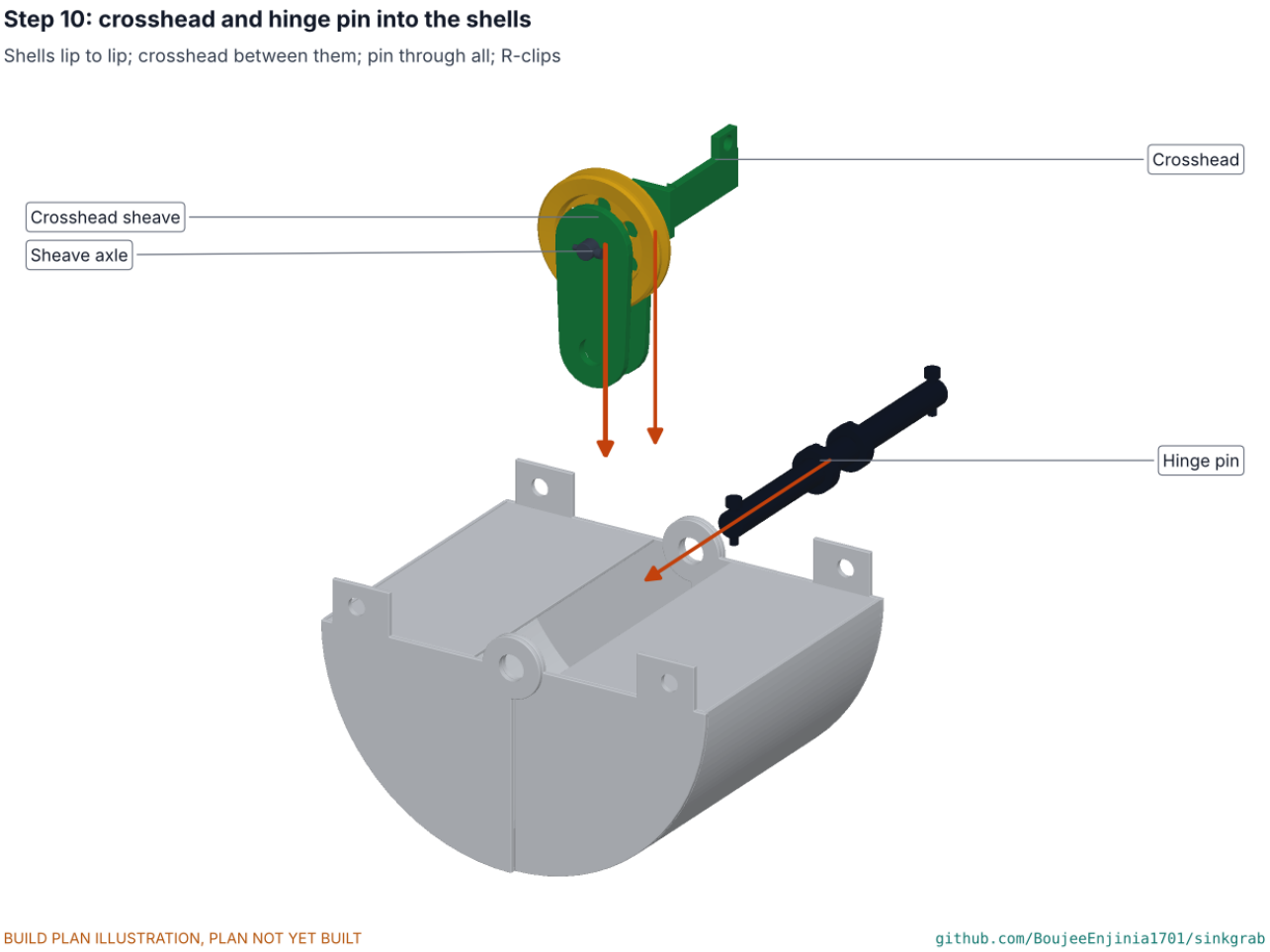
Wrap the band round the brake drum, pin the anchor end, then the link and lever; fit the weight. Check the drum is held.

Step 9: cranks on



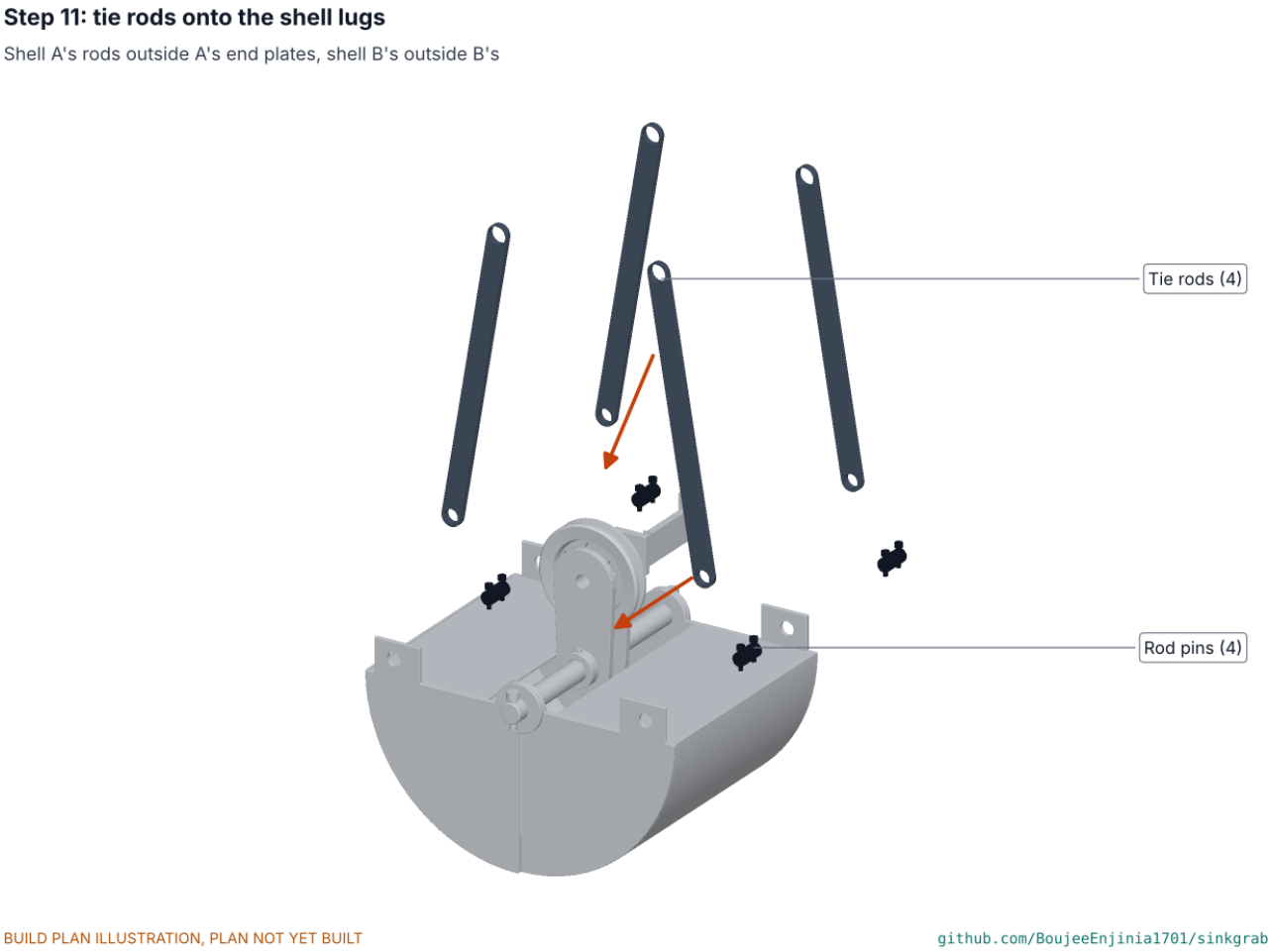
Slide each crank onto its square, 180 degrees apart, and push in the ball-lock pin. The crank with the long two-hand handle goes on the +X end, away from the chain guard; during the hoist two people work it and one the other crank.

Step 10: crosshead and hinge pin into the shells



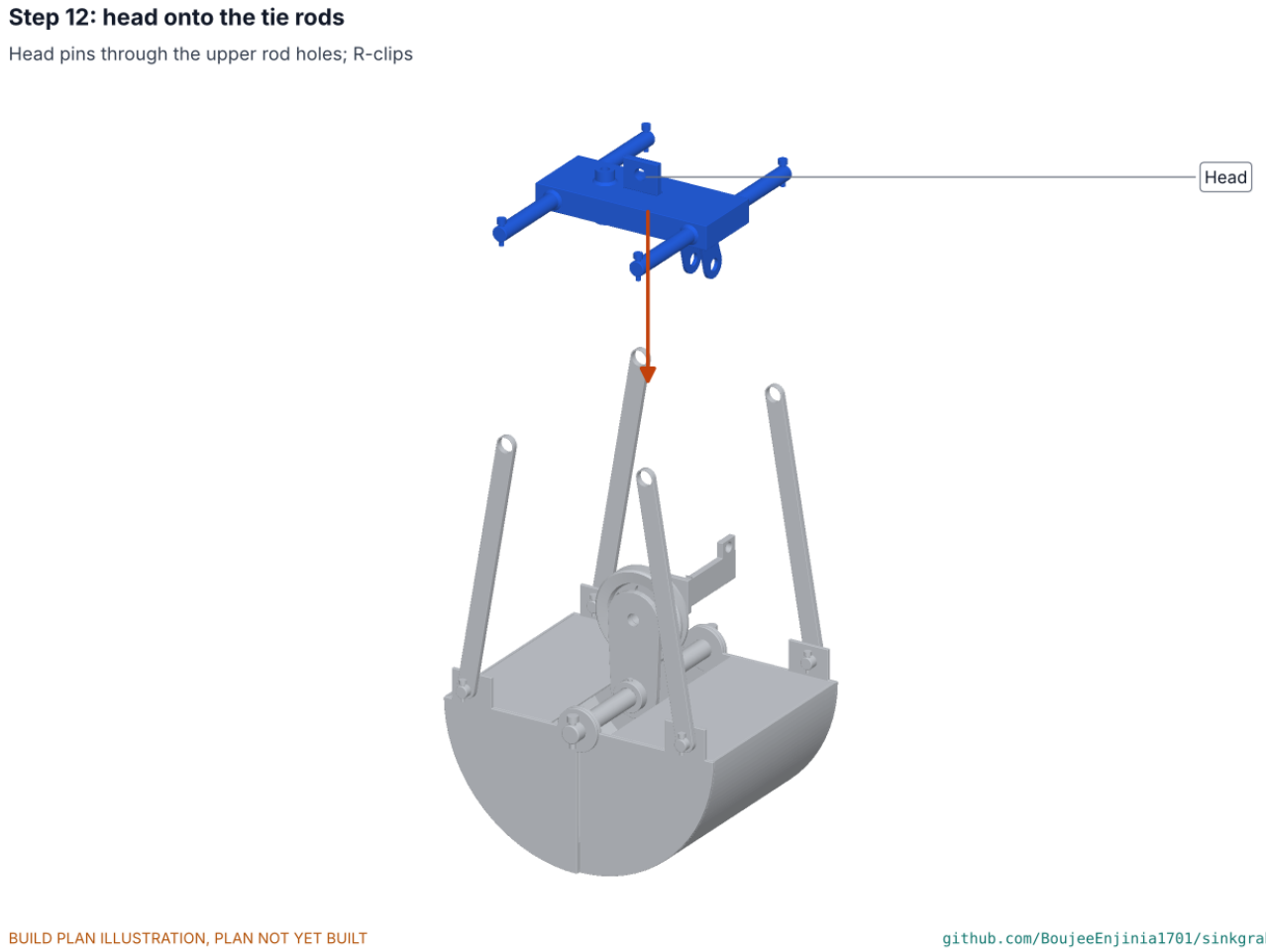
Lay the shells lip to lip. Fit the sheave between the crosshead plates on its axle, set the crosshead between the shells and push the hinge pin through all end plates, a set collar, both crosshead plates and the second collar; tighten the collars against the crosshead; R-clips. Work on a stand: the shells are 12 kg each.

Step 11: tie rods onto the shell lugs



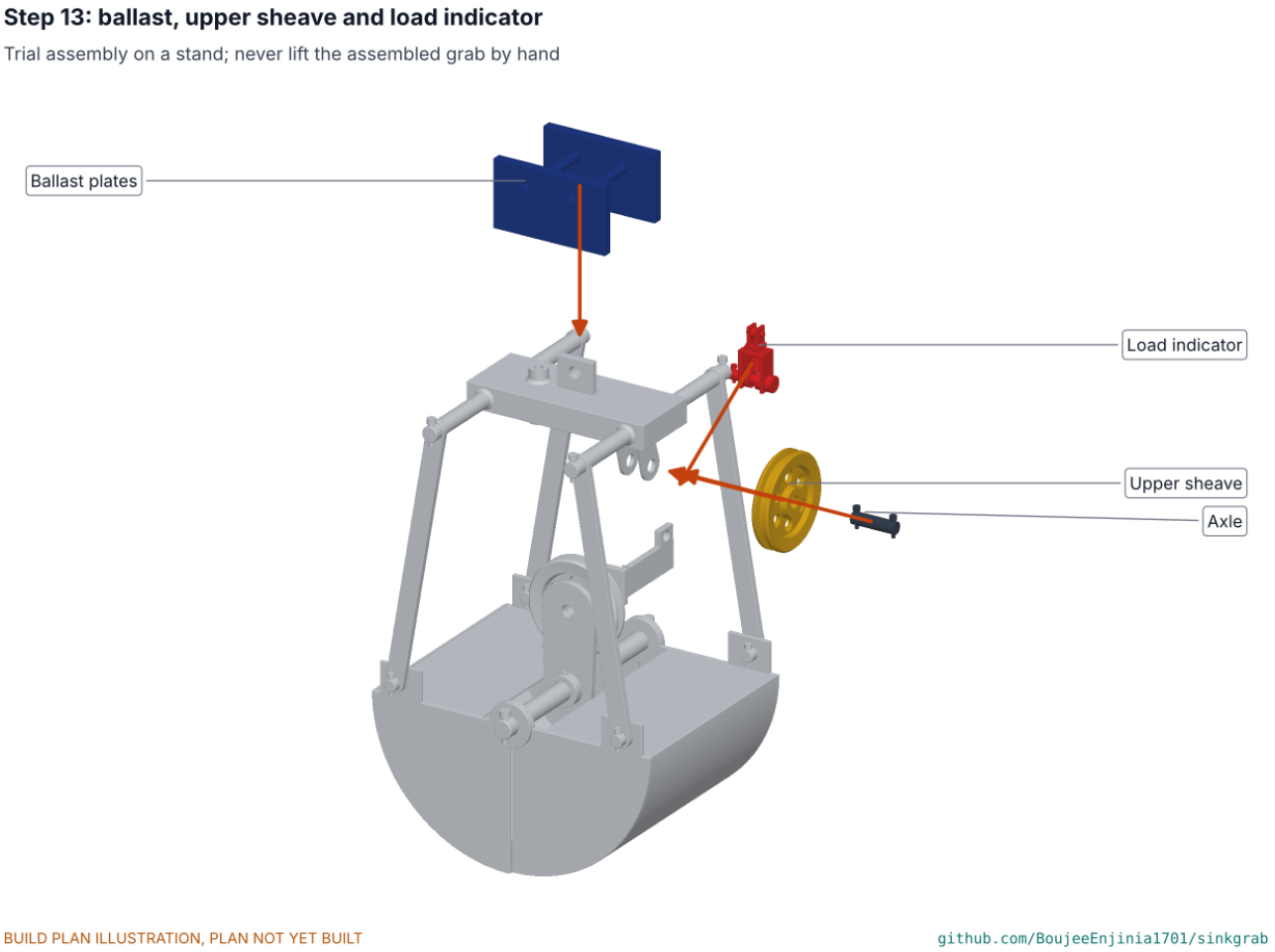
Pin two tie rods to each shell's lugs, outside that shell's end plates.

Step 12: head onto the tie rods



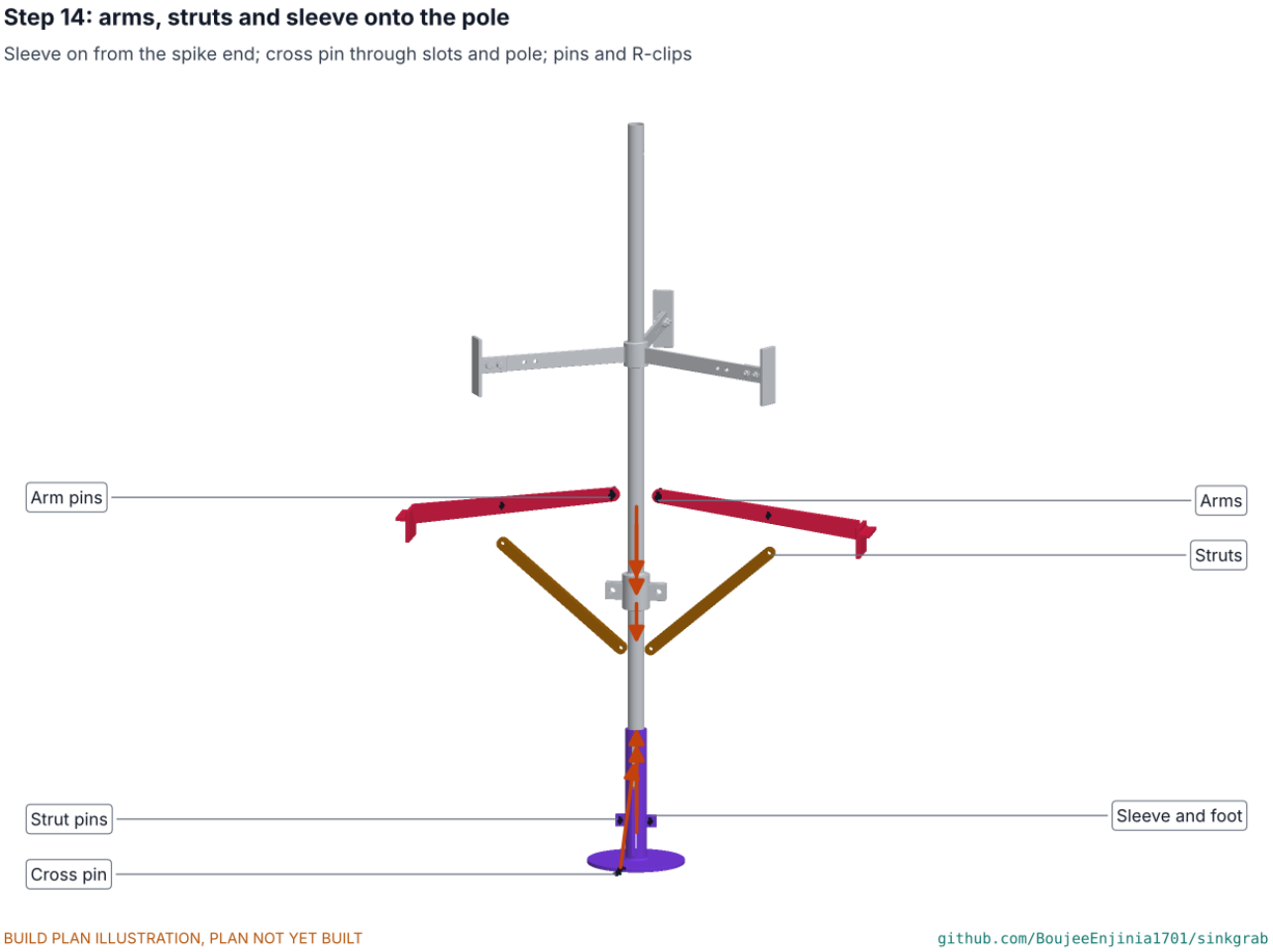
Lift the head over the rods and pass the head pins through their upper holes; R-clips. The grab now opens when the head is pushed down and closes when the crosshead is lifted toward it.

Step 13: ballast plates, upper sheave and load indicator



Bolt both ballast plates on with two M10 bolts through their spacer tubes. Fit the upper sheave between the head's cheeks on its axle; split pins. Pin the load indicator's straps to the crosshead's dead-end lug with the 16 mm clevis pin and its two spacers. Do this trial assembly on a stand or under a hoist: the assembled grab weighs about 52.7 kg, more than two or three people may lift together, so it is never lifted by hand. Take it apart again for carrying; the heaviest piece is the head with its plates, 14.3 kg.

Step 14: arms, struts and sleeve onto the pole

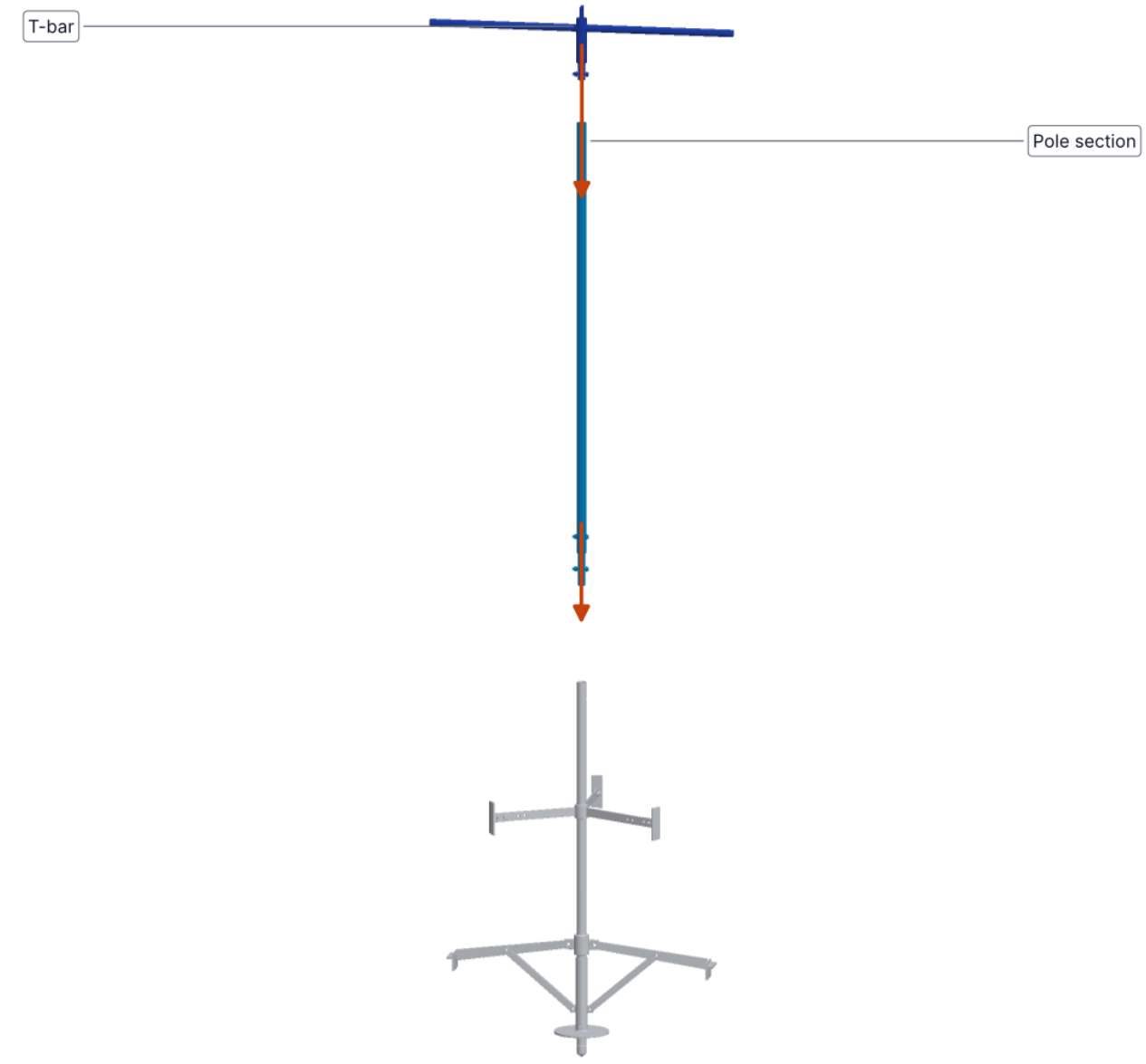


Slide the sleeve onto the pole from the spike end, slots facing the cross hole. Push the cross pin through one slot, the pole and the other slot, and fit its R-clip. Pin the arms to the hub lugs and the struts between the arms and the sleeve lugs.

Step 15: pole section and T-bar

Step 15: pole section and T-bar

Spigot into the pole, M12 bolt; T-bar socket over the top spigot



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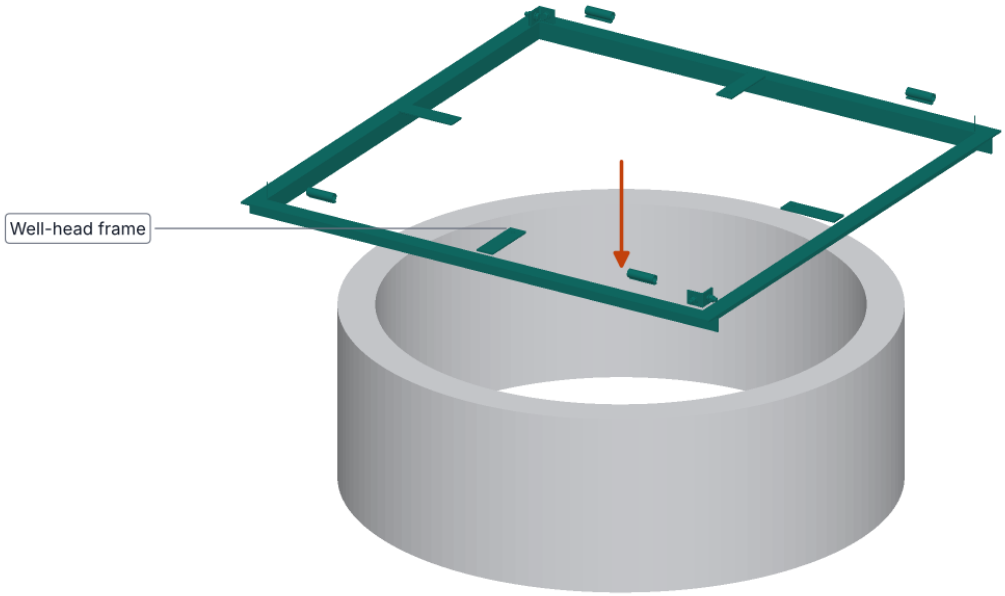
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Push a section's spigot into the scraper head and bolt it; fit the T-bar socket over the top spigot. At the well, sections are added one at a time at the doors. Four people turn the T-bar, two on each side (Amish's decision O5, 2026-10-25): with two the force is about 142 N each, over the about 100 N limit for repeated work. Keep the T-bar low: it clears the HatchSide legs only below about 1.27 m above ground.

Step 16: well-head frame on the collar

Step 16: well-head frame on the collar

Two halves, bolted at the corners; datum holes on the ring wall line

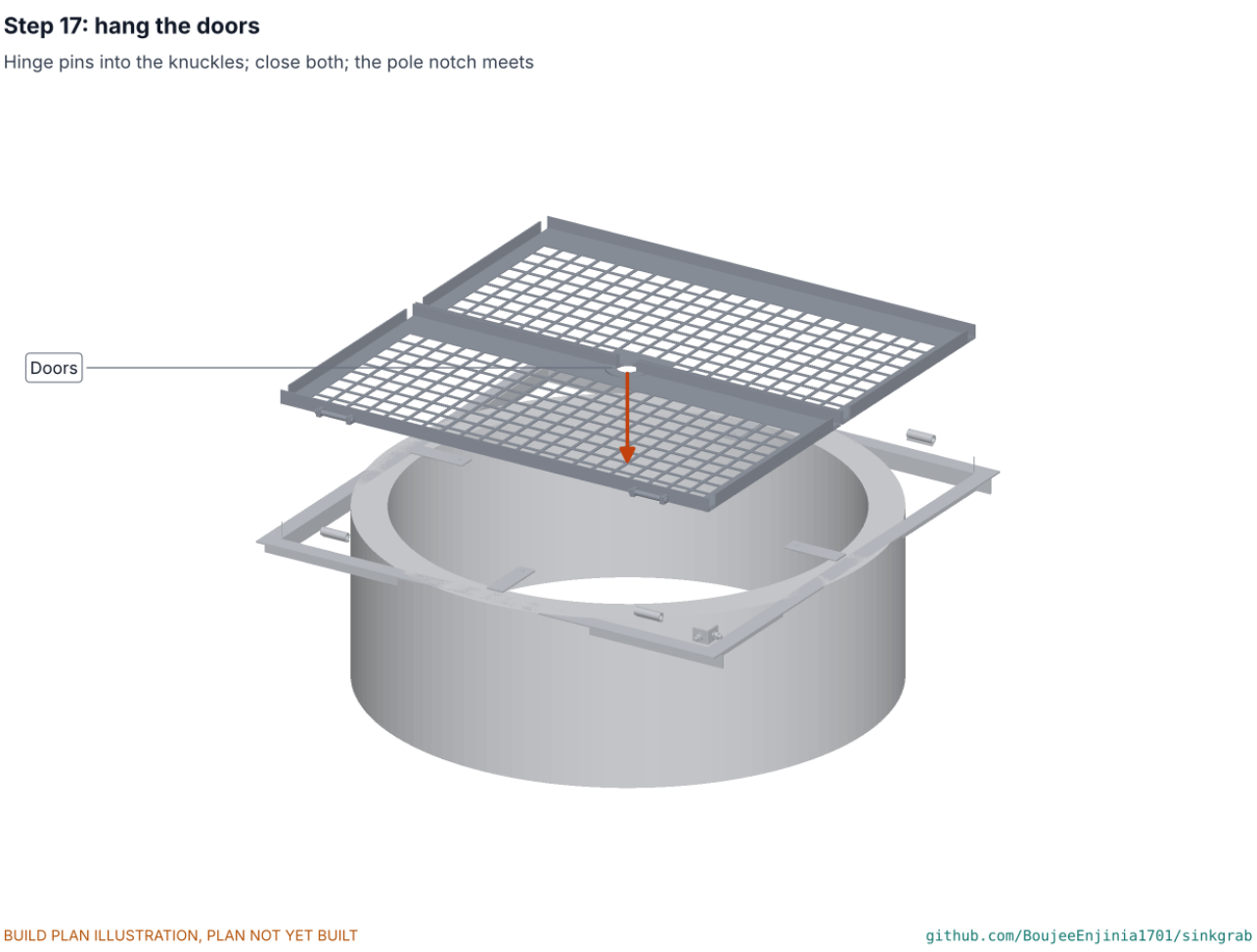


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Clear and level the collar top. Set the two frame halves on it, bolt the corners and turn the frame so its datum holes lie over the caisson ring wall.

Step 17: hang the doors

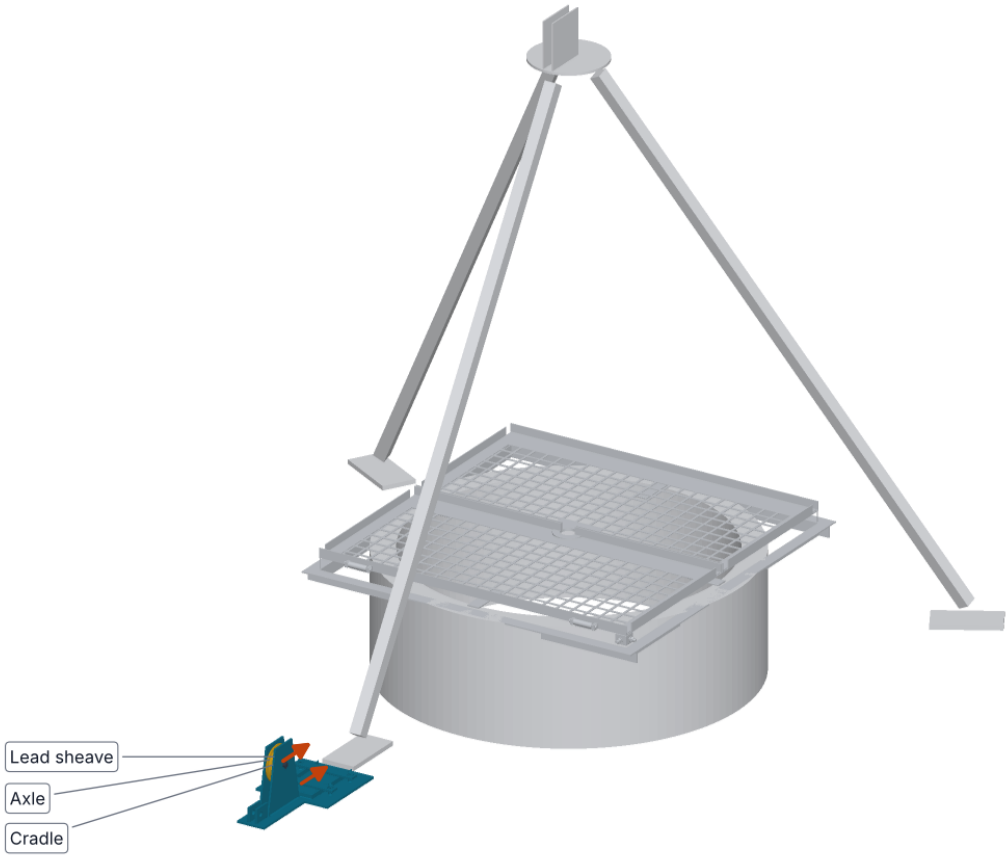


Drop each door's hinge pins into the knuckles and close both. The doors stay closed whenever nothing is passing through.

Step 18: cradle under the leg A foot

Step 18: cradle under the leg A foot

Lift the foot, slide the cradle under, clamp bars and M12 bolts



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

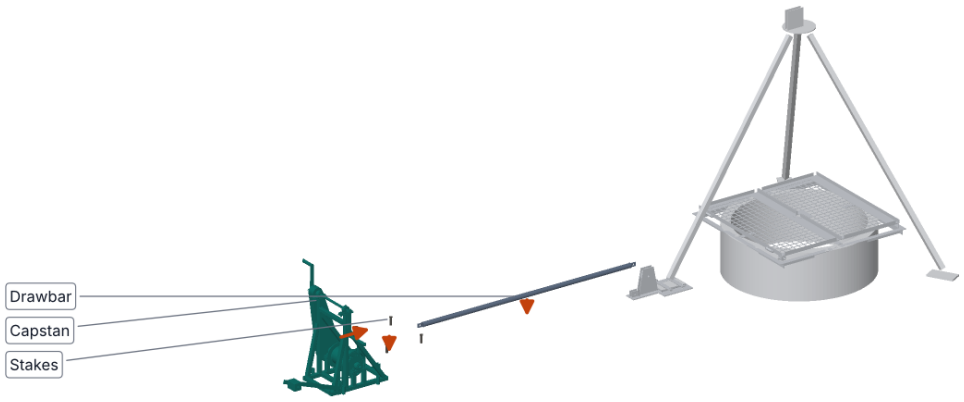
github.com/BoujeeEnjinia1701/sinkgrab

With the HatchSide tripod standing over the well and its winch removed, lift the leg A foot a few centimetres, slide the cradle under it with the sheave pointing away from the well, lower the foot inside the rim and bolt the two clamp bars down.

Step 19: capstan and drawbar

Step 19: capstan and drawbar

Drawbar pinned to both clevises, in line; stakes through the tubes

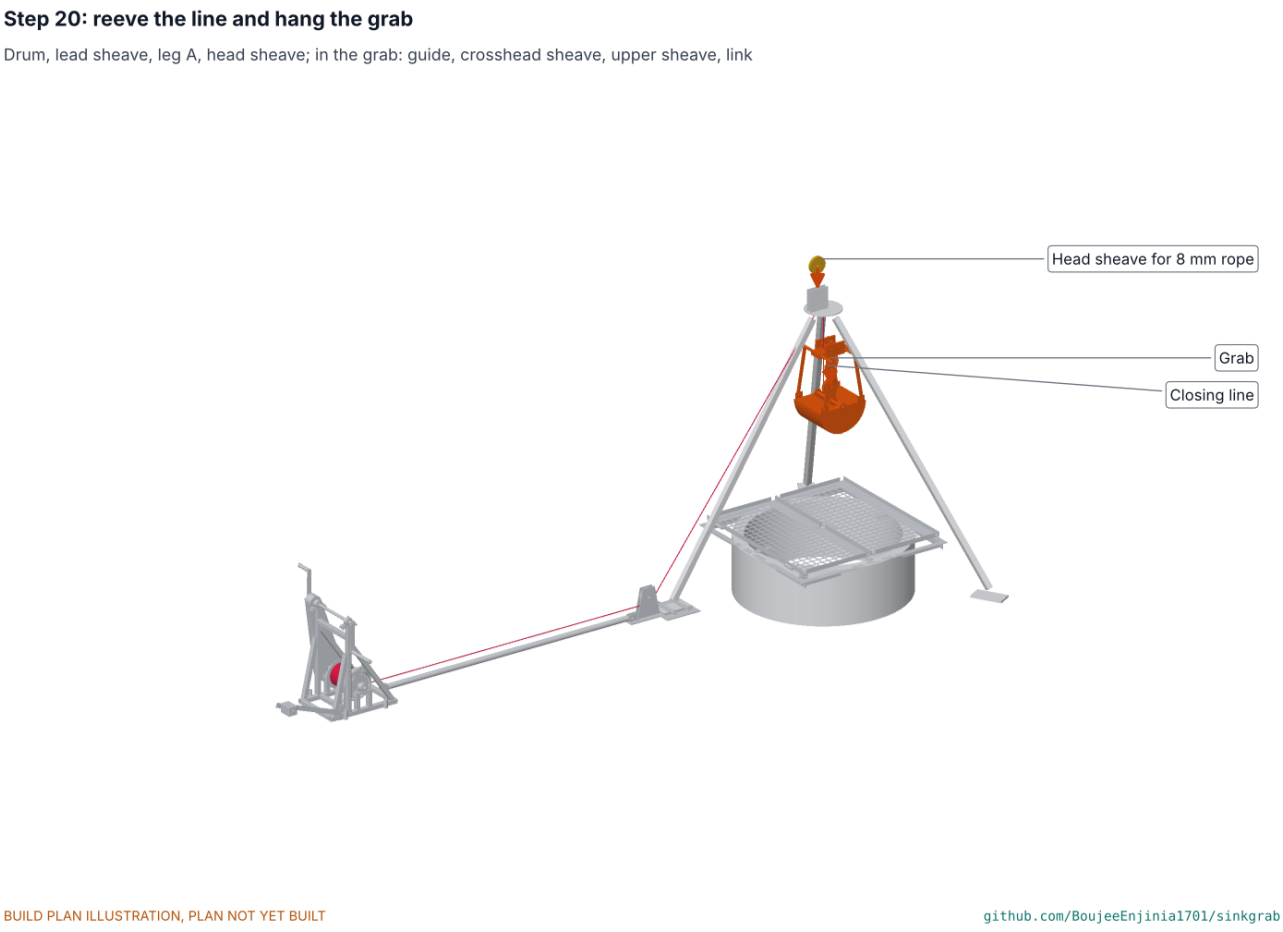


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Carry the capstan frame and the drum separately and reassemble them (steps 2, 4 to 9). Set the capstan in line with leg A, 3 m beyond the sheave. Join the drawbar halves and pin it to both clevises; drive the four stakes.

Step 20: reeve the line and hang the grab



Assemble the grab under the tripod, its shells resting closed on the closed doors: shells, crosshead and collars on the hinge pin, tie rods, head, then the ballast plates. Fit the 8 mm head sheave in the HatchSide head. Clamp the line on the drum, wind on three dead turns, lead it under the lead sheave, up beside leg A, over the head sheave and down the well axis. Thread it down through the grab head's guide, under the crosshead sheave, up over the upper sheave in the head and down to the load indicator on the crosshead; seat the thimble on the fork sleeve and fit the fork pin and its split pins. Tie the recovery line to the head's eye and make it off at the frame. From now on the grab is moved only on the line.

5. First checks

These are listed here and recorded in a TRL 4 test report, not in this plan.

Table 2. First checks.

Check	Requirement	How	Pass when
Fit in rings	R2	Pass the closed grab and the folded scraper through rings of 0.8, 1.0 and 1.3 m on the ground	Nothing binds; the arms open below the ring edge
Proof load	R7	CalRig or a load cell, held 1 min with the clutch locked out: line, tripod, cradle and capstan to 1.25 x the 150 kg rated load, 188 kg (1.84 kN)	No movement over 2 mm, no damage
Clutch setting	R7	On CalRig, set the locknut so the clutch slips at about 34 N m; pull through the capstan on each rope layer	Slips at or below 1.47 kN on the first layer and not below 1.13 kN on the third
Load indicator	R7	Pull the dead end through a load cell in 0.25 kN steps to 1.5 kN	The tell-tale reads within one ring of the load cell
Air test	PR-012	Bump-test the four-gas detector and the carbon dioxide monitor; lower both on a cord in a mesh bag and read at 2 m, mid depth and 1 m above the water (SKG-DEC-001 K22)	Both read and alarm; every level below the alarms
Clutch slip at the well	R7	Hold the grab against a fixed stop, crank slowly, at the start of each day	The clutch slips, the pawl holds, and the load indicator reads under 1.5 kN
Brake	R7	Lower the full grab on the brake, let go of the lever	The drum stops within one turn
Crank force	R5	Spring balance on a handle, full grab	150 N or less per person
Bite	R3	Test pit of saturated sand, 20 bites into a measured tub	Record the mean bite
Cycle time	R4	20 cycles at 10 m, three people at the cranks during the hoist (two otherwise); then 20 with two	3 min or less a cycle with three at the hoist; record both
Bentonite annulus	R1	Test pit first, then the partner well: bentonite slurry pumped down a feed pipe into the overcut annulus as the rings sink (decision O2)	Record the slurry used and whether the rings sink under their weight and saddles
Sinking and tilt	R1, R6	Partner well; dip tape every cycle	Record depth gained and tilt
Mass and packing	R9	Weigh each piece, the scraper head with its arms unpinned; load the kit	Each 25 kg or less; record the vehicle used

6. Safety stops

Work stops at each of these points until what is listed is true.

1. **Before anything is lifted over the well.** The barrier stands 2 m outside the tripod feet. The shaft air has been tested with the four-gas detector and the carbon dioxide monitor (PR-012), both lowered on a cord in a mesh bag and read at 2 m below the collar, at mid depth and 1 m above the water, each held for at least its response time (SKG-DEC-001 K22); a reading over range is a fail. It is tested again after every break and before anyone leans over the shaft. The chain guard is on and the brake weight is fitted. Everyone has been briefed with the card: one signaller, whistle and hand signals; one blast stops all work.
2. **Before the first lift on a site.** The line, cradle, drawbar, capstan and tripod have been proof-loaded for 1 min at 188 kg (1.25 x the rated load) with nobody near the well. The rope, splice, shackles and pins have been

inspected that day. The clutch's locknut carries its paint mark, the clutch has slipped against a fixed stop as set (Table 2), and the load indicator's rod is free and its tell-tale reset.

3. **Before each lift.** Nobody is under the grab, beside the line or on the doors. The doors are open only while the grab passes.
4. **Before lowering.** The cranks are off and the pawl is up; the person on the brake has both hands on the lever. During the hoist three people work the cranks; on the stop call all three let go together, the pawl holds the drum, and nobody stands between the capstan and the lead sheave.
5. **Before dumping.** The doors are closed under the grab and the tub is in place; nobody reaches under the grab. A full tub is never lifted: two people slide it off the doors on a board to the ground and tip it at the heap.
6. **When the clutch slips or the load indicator reads over 1.5 kN.** Stop, let the pawl hold, slack the line and find the snag; recover the grab on the recovery line if it cannot be freed. Never tighten the clutch to stop it slipping; if it slips on a steady full grab, take it back to CalRig.
7. **When tilt passes 1 in 160.** Scrape only the high side and move saddles to it. At 1 in 80, stop all undercutting until it is back under 1 in 160.
8. **If anyone must look into or enter the shaft.** SinkGrab work stops. Entry is a separate operation: gas test, harness on a separate line with rescue equipment, a person at the top throughout. SinkGrab's line is never used for a person.
9. **At the end of each day.** Close the doors, inspect the rope, pins, pawl, brake lining, clutch facings, load indicator and shackles, and replace anything damaged.

7. Tools, skills and workspace

- Stick welder with 2.5 mm electrodes or MIG, and a person who can weld 2 mm tube without burning through; welding screen, gloves and mask.
- Angle grinder with cutting and grinding discs; metal saw; pillar drill with bits to 30 mm.
- Plate rolls for the shell skins, or a fabricator who has them; a profile cutter for the ratchet, pawl and end plates.
- Spanners to M14, R-clip pliers, a splicing fid if the eye splices are made in house.
- A flat floor or welding table about 1.5 x 1.5 m; two people for the drum and frames; a stand or a hoist for the grab's trial assembly.
- Lead-free primer and paint for every part that goes into the well.
- At the well: the HatchSide tripod and its crew of two; a crew of four or five in all, and five or six on scraping days so four can turn the T-bar (decision O5).
- For sinking: a drum for mixing bentonite slurry, a hand pump and a feed pipe long enough to reach the toe through the overcut annulus, and bentonite for about 0.45 m3 of slurry over 3 m of sinking (decision O2); trial kit, not in the bill of materials.

8. Where the numbers come from

- `cad/src/model.py` : the parametric model, its constructability checks and the STEP and STL files in `cad/step` and `cad/stl` ; `cad/src/skg_detail.py` and `cad/src/skg_parts.py` : every part at detail level D3, with a STEP file per made or bought part in `cad/step/A/` ; `cad/fidelity.json` : the fit and fidelity check.
- `cad/drawings/SKG-DWG-001` and `SKG-DWG-002` : the well-head layout and the grab and scraper general arrangement; `SKG-DWG-101` to `SKG-DWG-117` : making sketches.

- `docs/04-calcs/01-sizing.md` and `docs/04-calcs/sizing.py` (SKG-CAL-001): loads, limits, forces, masses and cost.
- `bom/bom.csv` : parts, specifications and prices.
- `docs/decisions/0002-design-for-construction.md` (SKG-DDR-002): the changes in section 2; `docs/decisions/0003-round-2-requirement-decisions.md` (SKG-DDR-003): the third person at the cranks, the 3:1 tackle and the toe blades; `docs/decisions/0006-decisions-2026-10-25.md` (SKG-DDR-006): the slip clutch, load indicator, scraper crew and bentonite annulus.
- `cad/src/build_plan_media.py` : every picture in this plan.